

# Draft Integrated Urban Recovery Strategies

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Vol I: Main Report

Draft Integrated Urban  
Recovery Strategy

21 October 2022

# Draft Integrated Urban Recovery Strategy

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## Volume I: Main Report

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## EXECUTIVE SUMMARY

The pace of urbanization in Nepal is rapid and is likely to remain so in the future. It is also one of the fastest urbanizing countries in South Asia. Government of Nepal has classified urban municipalities as urban areas and rural municipalities as rural areas. In Nepal, three persons out of five now live in urban municipalities. The urban share in the national population has increased rapidly with the rural-to-urban migration and declaration of new municipalities as the major driving factors. The 2015 Gorkha earthquake of 25th April and the massive aftershocks brought a heavy loss to life and property. The 2015 Gorkha earthquake resulted in a loss of more than 8,000 lives and damaged around 500,000 private houses (PDRF, 2016). Rural areas were damaged more as compared to the urban areas due to the inferior construction quality. However, the reconstruction of urban areas stayed a challenge. The earthquake-affected 757 urban wards and 1796 rural wards across the urban and rural municipalities in the 32 earthquake-affected districts of the country. Around 40% of total housing damage from the earthquake were in urban areas (UEHRR, 2021).

After the 2015 Gorkha earthquake, Government of Nepal transferred the responsibility of reconstruction and recovery to National Reconstruction Authority (NRA). As the NRA has been phased out in December 2021, it is important to embed the NRA's lessons in the generic line agencies such as the Ministry of Urban Development (MoUD), NDRRMA and local governments. Hence, the main aim of this Integrated Urban Recovery Strategy is to organize learnings from the ongoing recovery efforts and prepare ground for integrating evidence-based urban strategies into specific government acts, policies, working procedures and regulations. The document focused on the following aspects:

- Identify and analyze the urban issues and challenges through available evidence and reports.
- Organize advocacy meetings and dissemination Workshops with key government (MoUD, MoFAGA, NDRRMA, DUDBC, KVDA, Provincial and local government officials, etc.) and urban stakeholders for the enhancement of the propositions.
- Review draft proposed amendments to the existing Acts, Policies, Regulations and Working Procedures
- Recommend the way forward towards integrating recommendations in relevant government legal and operational documents like the urban policies, heritage policies, etc.

The study team collected Proposed Urban Recovery Strategy documents prepared by NRA for peer review to draft Integrated Urban Recovery Strategy applicable to Local, Provincial and Federal government. The study team also collected a substantial body of international knowledge and information from the international literature review as well.

Proposed Urban Recovery Strategy documents is based on the “Urban Earthquake Housing Recovery” study conducted by Urban Technical Working Group (UR-TWG) of Housing Recovery and Reconstruction Platform-Nepal (HRRP). This report is also based on national level policies, documents from the NRA, CLPIU/Building, MOUD, MOFAGA etc. International reconstruction and recovery strategies and approaches are also reviewed.

This Integrated Urban Recovery Strategy is targeted to national, provincial and local governments and international professionals, academia, practitioners, policy makers, students, researchers, government ministries, donors, civil society members, development partners, reconstructions partners, UN agencies, national and international organizations who were involved in the post-earthquake reconstruction in Nepal and all other individuals/organizations who are interested in learning about the issues and its recommendations on post-earthquake recovery and reconstruction.

After 2015 Gorkha Earthquake, the recovery and reconstruction of buildings and infrastructure were taken care by the CLPIU (Building), CLPIU (Education) and Grant Management and Local Infrastructure (GMALI) and their respective units' set-up in the affected districts under NRA. The NRA authorized reconstruction projects, built policies, budgeting schemes and legislation, and disbursed them to the CLPIUs. The Project Directors of CLPIU delegated implementation tasks to District Level Project Implementation Units (DLPIU) Project Managers. The DLPIUs focused on implementing and checking the recovery and reconstruction projects in the districts. Several development partners, UN organizations and national and international NGOs supplied a range of technical, financial, and material

services during relief and early recovery, and many have been offering to provide substantial support to the recovery and reconstruction process.

Regarding the status of urban housing recovery, as of January 2021, out of the 373,104 households from urban municipalities who are eligible for receiving financial and technical support from NRA, 61.61% have completed their reconstruction. Further, data shows that urban recovery is not uniform across the 32 earthquake-affected districts; only 38% of earthquake-affected households in urban municipalities of Kathmandu Valley have completed reconstruction compared to 69.83% urban municipalities outside Kathmandu Valley. As of August 2022, tranche progress is as follows:

|       | Second Tranche Progress | Third Tranche Progress |
|-------|-------------------------|------------------------|
| Rural | 92%                     | 88%                    |
| Urban | 80%                     | 76%                    |

Source: CLPIU Building

Data shows that reconstruction in urban areas is slower as compared to reconstruction in rural areas. This discrepancy makes it important to find the issues associated with urban reconstruction so that they can be discussed in detail and addressed appropriately.

To complete the remaining tasks of housing recovery and reconstruction, the 15th Plan (Fiscal Year 2019/20- 2023/24) by the National Planning Commission proposes coordination with Provincial level and local level governments. Seven Provinces of Nepal were formed on September 20, 2015. Schedule 9 of the *Constitution* provides a basis for provinces to be active in disaster management. When the Gorkha earthquake hit Nepal, there was no system of Provincial government. The Four provinces namely: - Bagmati, Gandaki, Lumbini and Province no. 1 were affected and still have caseloads remaining. According to the municipalities affected by the Gorkha earthquake, there are 96 urban municipalities in earthquake affected districts. Data shows that Bhaktapur, Kathmandu, Kirtipur, Lalitpur, Dudhouli, Bhirkot urban municipalities have a high number of caseloads. Hence, a concentrated effort is needed for these municipalities.

Considering, reconstruction status (as of April 2021) by vulnerable beneficiaries in urban areas of earthquake affected districts, among 9256 total cases, 49% has completed the reconstruction; 40% are reconstructing their houses and 11 % have not yet started the reconstruction. There are 42 historic 'Newari' settlements within the Kathmandu Valley and 12 outside Kathmandu valley as declared by NRA. Data shows that among 794 heritage structures in the urban areas, 55% are reconstructed, 24% are under construction and 21% are yet to be constructed. Considering, reconstruction status of vulnerable beneficiaries in heritage settlements in urban areas of earthquake affected districts as identified by NRA, 678 beneficiaries have already obtained the first tranche, 244 beneficiaries have already obtained the second tranche, while only 203 beneficiaries have obtained the third tranche. Regarding economic recovery, various activities have been implemented by government and non-government agencies such as housing grant distribution; several types of concessional Loans; food for work or cash for work in the short term etc.

Reconstruction status (as of July 2022) by beneficiaries in urban and rural areas of earthquake affected districts is as follows:

| Tranche Progress |                         |                        |
|------------------|-------------------------|------------------------|
|                  | Second Tranche Progress | Third Tranche Progress |
| Rural            | 92%                     | 88%                    |
| Urban            | 80%                     | 76%                    |

Source: NHSRP

Studies on Urban Housing Reconstruction and Recovery have shown the issues on: Access to finance, land, compliance to heritage norms, communication, socio-technical assistance in reconstruction, retrofitting in urban areas, and aid to vulnerable. Finance issues include excessive cost of urban housing reconstruction, and less access to affordable finance/loan. Land issues include lack of ownership papers, multiple ownership of land, smaller plots below threshold, cadastral inconsistencies, geo-hazard risk/



vulnerable settlements and landlessness. Heritage issues include non-compliance with heritage norms, added cost for keeping vernacular architecture, lack of availability of traditional construction material, lack of skilled human resource and multi-ownership. Communication issues include lack of information flow, inadequate flow of information, early flow of information, inadequate dissemination of guidelines and decisions at local level, digitized communication system and misleading information. Socio-technical issues include Lack of provision of STA, ineffective part of STA in urban context, difficulty in obtaining building permits, expensive demolition and debris clearance and lack of Risk Sensitive Land Use Plan (RSLUP). Retrofitting issues include excessive cost of Retrofitting, low awareness and trust in retrofitting, complex system/technology, and dispute in retrofitting of some traditional buildings. Aid to vulnerable populations include neglected vulnerable population, lack of adequate document to apply loan and delisting of beneficiaries who are socio-economically vulnerable. To solve these issues existing policies, Acts, regulations, working procedures were studied and gaps in them were found. The gaps were mostly related to conservation of monuments and traditional settlements, invasion of Guthi land, ambiguous responsibilities between different government levels, and funding and implementation issues.

International experience in post-earthquake housing reconstruction and recovery and national experiences of urban planning projects were studied to understand the international practice of reconstruction and recovery and national practice of urban planning. This study suggests that urban recovery is a time-consuming process and effective strategies are needed to make the process efficient.

Suggestions for the efficient urban recovery strategies were also obtained from the Key informant interviews. Some of these suggestions include: Design and promote low cost housing; promote incremental/community housing; develop revolving fund/ CSR fund mechanisms to provide low interest loan, provide concessional loans, include microfinance institutions, adopt development mechanisms such as land/house pooling, provide technical drawings and designs, define heritage areas, train locals, adopt encouragement mechanisms for conservation, involve private sectors, provide a socio-technical assistance committee based at local level, clarify ambiguous roles and responsibilities, and update/amend policies, Acts and regulations.

To solve these issues, this study provides urban recovery strategies about funding, livelihood support, institutional arrangements, human resource planning, socio-technical support, and implementation for the post NRA phase. Potential projects for regeneration of three heritage settlements: Dolakha, Khokana, and Sankhu are planned based on the proposed strategies.

Regarding Institutional arrangement, reconstruction will be led by NDRRMA through its recovery and reconstruction directorate after NRA. Funding Modality of private buildings include NDDRMA grant, NDDRMA vulnerable grant, local government's heritage grant, Subsidized collateral free bank loan, regular loan, and subsidized loan through a company/cooperative of the beneficiaries. The subsidized loan and grants will be managed as a part of a "Housing Security Fund." Funding Modality of heritage structures and infrastructures include funding by concerned service supplying agencies such as Nepal Water Supply Corporation, Kathmandu Upatyaka Khanepani Limited (KUKL), Nepal Electricity Authority (NEA), Nepal Telecom (NTC) through loan eased by GoN; Central/Provincial/Local Government; Town Development Fund; bilateral/multilateral funding agencies; Public Private Partnership, private companies / private donations, and CSR. Livelihood support and economic recovery programs include specifically, designed projects; small and cottage industry development program with the support of prime minister Self-employment fund; and local micro-finance banks and cooperatives through identification of local trade, industry, tourism, training and technical support and start-up seed money.

## **PRELUDE**



## **ACKNOWLEDGEMENT**

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The Study Team

## LIST OF ABBREVIATIONS

|        |                                                                    |
|--------|--------------------------------------------------------------------|
| BDP    | Bhaktapur Development Project                                      |
| BECTP  | Bandipur Eco-Cultural Tourism Project                              |
| BPP    | Building Permit Process                                            |
| BPS    | Building Permit System                                             |
| CAO    | Chief Administrative Officer                                       |
| CBS    | Central Bureau of Statistics                                       |
| CEO    | Chief Executive Officer                                            |
| CLPIU  | Central Level Project Implementation Unit                          |
| CNI    | Confederation of Nepalese Industries                               |
| CRC    | Community Reconstruction Committees                                |
| CRS    | Catholic Relief Services                                           |
| CSR    | Corporate Social Responsibility                                    |
| CTEVT  | Council for Technical Education and Vocational Training            |
| CTDD   | Culture and Tourism Development Directorate                        |
| DAO    | District Administration Office                                     |
| DG     | Director General                                                   |
| DLPIU  | District Level Project Implementation Unit                         |
| DLRM   | Department of Land Resource Management                             |
| DoA    | Department of Archaeology                                          |
| DoS    | Department of Survey                                               |
| DRR    | Disaster Risk Reduction                                            |
| DTCO   | District Treasury Controller Office                                |
| DUDBC  | Department of Urban Development and Building Construction          |
| FGD    | Focus Group Discussion                                             |
| FNCCI  | Federation of Nepalese Chamber of Commerce & Industries            |
| GMALI  | Grant Management and Local Infrastructure                          |
| GoN    | Government of Nepal                                                |
| HRRP   | Housing Recovery and Reconstruction Platform                       |
| IDD    | Infrastructure Development Directorate                             |
| INGO   | International Non-Governmental Organization                        |
| KII    | Key Informant Interview                                            |
| KMC    | Kathmandu Metropolitan City                                        |
| KVDA   | Kathmandu Valley Development Authority                             |
| LGOA   | Local Government Operations Act                                    |
| LMC    | Lalitpur Metropolitan City                                         |
| MoFAGA | Ministry of Federal Affairs and General Administration             |
| MoHA   | Ministry of Home Affairs                                           |
| MoLPCA | Ministry of Land Management, Cooperatives, and Poverty Alleviation |
| MoTCA  | Ministry of Culture, Tourism and Civil Aviation                    |
| MoUD   | Ministry of Urban Development                                      |
| NBC    | National Building Code                                             |
| NDRRMA | National Disaster Risk Reduction & Management Authority            |



|          |                                                                           |
|----------|---------------------------------------------------------------------------|
| NGO      | Non-Governmental Organization                                             |
| NHSRP    | National Housing and Settlement Resilient Platform                        |
| NNBC     | Nepal National Building Code                                              |
| NPC      | National Planning Commission                                              |
| NRA      | National Reconstruction Authority                                         |
| NSET     | National Society for Earthquake Technology-Nepal                          |
| PDMA     | Provincial Disaster Management Authority                                  |
| PDRF     | Post Disaster Recovery Framework                                          |
| PMoIPD   | Provincial Ministry of Physical Infrastructure Development                |
| PMoIAL   | Provincial Ministry of Internal Affairs and Law                           |
| PMoLMCPA | Provincial Ministry of Land Management Cooperative and Poverty Allevation |
| PMoCT    | Provincial Ministry of Culture and Tourism                                |
| RSLUP    | Risk Sensitive Land Use Plan                                              |
| STA      | Socio-Technical Assistance                                                |
| UR-TWG   | Urban Recovery Technical Working Group                                    |
| UNESCO   | The United Nations Educational, Scientific and Cultural Organization      |

# 1 INTRODUCTION

## 1.1 Background

Today more than half of the world population live in cities. The pace of urbanization in Nepal is rapid and is likely to remain so in the future. It is also one of the fastest urbanizing countries in South Asia with a growth rate of 3.9% (CBS 2018). Government of Nepal (GoN) has classified urban municipalities as urban areas and rural municipalities as rural areas. In Nepal, three persons out of five now live in urban municipalities. The urban share in the national population has increased rapidly with the rural-to-urban migration and declaration of new municipalities as the major driving factors. In addition, the newly adopted federal administrative structure after 2015, has declared 293 municipalities where 62 percent people live.

According to the Urban Housing Sector Profile published by UN Habitat Nepal in 2010, urbanization in Nepal, as a process, is the outcome of location of new economic activities and population movement as opposed to the outcome of a planning effort. Haphazard growth, infrastructure deficit, heavy densification of the Kathmandu Valley, and an urban sprawl on agricultural land is the distinct character of the urbanization in Nepal (KVTDC, 2002). Local Government Operation Act, 2017 binds municipalities to prepare an integrated urban development plan. The Land Use Act, 2019 has a provision to set up a Land Use Council at the Local, Provincial and National level. These two legal provisions may be the instruments to develop risk sensitive land use plans for guiding urban development in a desirable way. Strengthening the ability of the local authorities to plan and enforce safe and resilient construction and reconstruction is needed.

Nepal is in a seismic zone 4 and 5. It has experienced damaging earthquakes throughout its history. The Gorkha earthquake of 25<sup>th</sup> April, 2015 measuring 7.8 magnitude and the massive aftershocks that followed a month later 12<sup>th</sup> May, 2015 measuring 7.3 magnitude, brought a heavy loss to life and property. The 2015 Gorkha earthquake resulted in a loss of more than 8,000 lives and damaged around 500,000 private houses (PDRF, 2016). Rural areas were damaged more as compared to the urban areas due to the inferior construction quality. However, the reconstruction of urban areas remained a challenge. The earthquake-affected 757 urban wards and 1796 rural wards across the urban and rural municipalities in the 32 earthquake-affected districts of the country. Around 40% of total housing damage from the earthquake were in urban areas (UEHRR, 2021).

## 1.2 Objective

As the National Reconstruction Authority (NRA) has been phased out in December 2021, it is important to embed NRA's lessons in the generic line agencies such as the federal, Provincial government ministries and entities and local governments. Hence, the main aim of this study is to focus on the following aspects:

- a) Map key stakeholders/authoritative implementers identified in “Proposed Urban Recovery Strategies”.
- b) Prepare a integrated version of “Proposed Urban Recovery Strategies” applicable to stakeholders/authoritative implementers identified in a) above.
- c) Conduct advocacy/briefing meetings with Ministry of Urban Development for the endorsement of “Proposed Urban Recovery Strategies” for its endorsement.
- d) Organize dissemination workshop to inform the officials of Ministry of Urban Development (MoUD), Department of Urban Development and Building Construction (DUDBC), National Disaster Risk Reduction and Management Authority (NDRRMA) and other relevant government stakeholders on the contents of the document.
- e) Organize learnings from the post Gorkha earthquake recovery efforts and prepare ground for integrating evidence-based urban strategies into specific government acts, policies, working procedures and regulations.

The proposed urban recovery strategy prepared by NRA in October 2021 focused on the following aspects:

- Identify and analyze the urban issues and challenges through available evidence and reports.
- Organize advocacy meetings and Dissemination Workshops with key government ministries and agencies, like MoUD, MoFAGA, MoLMCPA, MoCTCA, MoF, Nepal Rastra Bank, NDRRMA, Provincial and Municipal Government officials, etc. and urban stakeholders for the enhancement of the propositions.
- Review draft proposed amendments to the existing Policies, Acts, Regulations and Working Procedures.
- Recommend the way forward towards integrating recommendations in relevant government legal and operational documents like the urban policies, heritage policies, etc.

## 1.3 Guiding Principles

This study is based on the following guiding principles:

### a. Earlier urban studies as a base of urban strategy

This report is based on the “Urban Earthquake Housing Recovery” study conducted by Urban Technical Working Group (UR-TWG) of Housing Recovery and Reconstruction Platform-Nepal (HRRP), in October 2019, with Catholic Relief Services’ (CRS) financial support in coordination with NRA, Central level Project Implementation Unit-Building (CLPIU/Building), National Society for Earthquake Technology Nepal (NSET) and Lumanti. This report is also based on national level policies, documents from the NRA, CLPIU/Building, MOUD, MOFAGA, HRRP, bi-lateral & multilateral organizations and Non-governmental Organizations (NGOs). This study also did a detailed review on National Experiences of Historic City/Town Development projects and International Experience in Post-Earthquake Housing Reconstruction and Recovery.

### b. Urban characteristics as a complex scenario

Urban areas are composed of core urban, traditional settlements, inner fringe area, outer fringe area, and rural surroundings. Hence, a different approach is needed. These approaches could be radical or incremental and may consist of repair, retrofit or reconstruction.

### c. Housing and physical infrastructures as a major focus

Urban recovery is a complex issue making up of physical issues as well as livelihood aspects. This study focused on housing and physical infrastructures related to housing to be more specific.

### d. Issues of urban recovery and settlement approach as a major focus

As the reconstruction after Gorkha Earthquake 2015 could not be achieved in the past 7 years after the earthquake, this study focused on the issues of urban recovery to propose strategies, those targets, to solve these issues. There are 49 historic 'Newari' settlements within the Kathmandu Valley and 12 outside Kathmandu valley as recognized by then NRA. These settlements need to be conserved to conserve the urban essence. Hence, this study focuses on these heritage settlements.

### e. MoUD, NDRRMA, and Local Government as a key implementing agency

This study focuses on the institutional strategy led by MoUD, NDRRMA and Local Government as key implementing agencies. MoUD as a main government entity responsible for urban development, redevelopment, renewal, regeneration throughout the country will response to meet urban challenges in normal situation as well in post disaster reconstruction. NDRRMA will be activated only in post disaster urban reconstruction and actively manage and coordinate with concerned agencies especially with MoUD, provincial and Local government. Although technically weak in urban planning and reconstruction, local government will be responsible to implement local level urban planning, implementation, and post disaster urban recovery and reconstruction.

### f. Urban recovery as a long-term process

International experience in post-earthquake housing reconstruction and recovery and national experiences of urban planning projects were studied to understand the international practice of reconstruction and recovery and national practice of urban planning and development. This study suggests that urban recovery is a time-consuming process and effective strategies are needed to make the process efficient.

### g. Urban recovery strategy as a part of a continuous development process

The proposed urban recovery strategy targets not only the post disaster reconstruction but also the continuous overall development process.

## 1.4 Methodology

This Project of drafting integrated urban recovery strategy is based “Proposed Urban Recovery Strategies” prepared by NRA in October 2021.

- f) The study team collected urban housing recovery strategic issues, proposed urban recovery strategy and a substantial body of international knowledge and information from the international literature review as well. Then the team identified and mapped key stakeholders/authoritative implementers identified in “Proposed Urban Recovery Strategies”. The study team also prepared a integrated version of “Proposed Urban Recovery Strategies” applicable to stakeholders/authoritative implementers identified.

Series of advocacy meetings entitled “Integrated Urban Recovery Strategy” were conducted where the issues of urban development and post disaster recovery and reconstruction were discussed and valuable suggestions collected. CRS and National Housing and Settlements Resilience Platform (NHSRP)

Following it, a dissemination workshop with MOUD, DUDBC, KVDA and NDRRMA senior officials is planned to be conducted in near future where the issues of urban development, and post disaster urban recovery and reconstruction and role of federal government ministries, provincial government entities and local government discussed and the workable solutions will be identified.

## 1.5 Scope and Limitation

### 1.5.1 Scope

This Integrated Urban Recovery Strategy is based on Proposed Urban Recovery strategy prepared by NRA. The scope of the study is limited to map key stakeholders/authoritative implementers identified in “Proposed Urban Recovery Strategies” and prepare a integrated version of “Proposed Urban Recovery Strategies” applicable to stakeholders/authoritative implementers identified.

NRA’s Proposed Urban Recovery strategy has seven chapters in which the first chapter discusses the background, objective, method, scope, and limitation of the study. Chapter 2 presents the status of urban recovery. Chapter 3 presents the issues and challenges of urban recovery and reconstruction. Chapter 4 consists of existing policies, acts, regulations, and gaps in them. Chapter 5 consists of learnings from international experiences, national urban planning and issues and recommendations from KII . Chapter 6 consists of proposed urban recovery strategies: funding modality, livelihood support, institutional framework, and human resource plan and implementation strategies. Finally, chapter 7 consists of potential model projects.

NRA’s Proposed Urban Recovery strategy is based on the “Urban Earthquake Housing Recovery” study conducted by Urban Technical Working Group (UR-TWG) of Housing Recovery and Reconstruction Platform-Nepal (HRRP), in October 2019, with Catholic Relief Services’ (CRS) financial support in coordination with NRA, Central level Project Implementation Unit-Building (CLPIU/Building), National Society for Earthquake Technology Nepal (NSET) and Lumanti. This report is also based on national level policies, documents from the NRA, CLPIU/Building, MOUD, MOFAGA, then HRRP now NHSRP, bi-lateral & multilateral organizations and Nongovernmental Organizations (NGOs). International reconstruction and recovery strategies and approaches are also reviewed.

This Integrated Urban Recovery Strategy is targeted to national provincial, and local government, international professionals, academia, practitioners, policy makers, students, researchers, government ministries, donors, civil society members, development partners, reconstructions partners, UN agencies, national and international organizations who were involved in the post-earthquake reconstruction in Nepal and all other individuals/organizations who are interested in learning about the issues and its recommendations on post-earthquake recovery and reconstruction.

### 1.5.2 Limitation

- The drafting of integrated urban recovery strategy is based Proposed Urban Recovery Strategy prepared by NRA in October 2021 and suggestion and comments obtained from advocacy meetings and dissemination workshop.

## 2 OVERVIEW OF URBAN RECOVERY

This chapter presents the status of urban recovery on housing, heritage settlement and economic recovery.

### 2.1 Status of Urban Recovery or Recovery Progress

#### 2.1.1 Housing

When we discuss urban housing recovery, consideration is made towards recovery inside urban municipalities because the GoN recognizes urban municipalities as urban areas. It is noteworthy that urban housing recovery in Nepal has moved at a slower pace compared to the rural reconstruction. Reconstruction data from January 2021 states that 800,973 households all over Nepal have enrolled as reconstruction beneficiaries and 54,580 households have enrolled as retrofitting beneficiaries. The reconstruction beneficiaries receive NPR 300,000 (US \$ 3000) and retrofitting beneficiaries receive NPR 100,000 (US \$ 1000) along with the socio-technical aid packages. At present, there is no distinction between the grant amount and socio-technical aid that urban and rural households receive. As of January 2021, out of the 373,104 households from urban municipalities who are eligible for receiving financial and technical support from NRA, 61.61% have completed their reconstruction. In rural municipalities, the success rate is even higher where 77.6 % have completed their housing reconstruction. Further, data shows that urban recovery is not uniform across the 32 earthquake-affected districts; only 38% of earthquake-affected households in urban municipalities of Kathmandu Valley have completed reconstruction compared to 69.83% urban municipalities outside Kathmandu Valley (table 1). Table 2 shows that private housing recovery in Kathmandu valley is slowest in Lalitpur district as per the data of Jan 2021.

**Table 1: Comparison of final (third) tranche progress**

|                                     | Earthquake affected Rural areas | Earthquake affected Urban areas | Within Kathmandu valley | Outside Kathmandu valley |
|-------------------------------------|---------------------------------|---------------------------------|-------------------------|--------------------------|
| Beneficiaries completed 3rd tranche | 77.66%                          | 61.61%                          | 38.8%                   | 69.83%                   |

**Table 2: Recovery status of private housing in Kathmandu valley**

| Geographic area         | HHs eligible | HHs enrolled | 1st tranche | 2nd tranche | 3rd tranche | Progress achieved % |
|-------------------------|--------------|--------------|-------------|-------------|-------------|---------------------|
| Inside Kathmandu valley |              |              |             |             |             |                     |
| Kathmandu district      | 48,463       | 48,172       | 43,167      | 23,059      | 20,063      | 41.4%               |
| Lalitpur district       | 20,624       | 18,129       | 18,111      | 7,232       | 6,386       | 31.0%               |
| Bhaktapur district      | 28,619       | 25,089       | 25,089      | 11,876      | 11,444      | 40.0%               |
| Total                   | 97,706       | 91,390       | 86,367      | 42,167      | 37,893      | 38.8%               |

Table 3 shows that the status of urban recovery is better in areas outside Kathmandu valley where 262,469 Households (HH) are enrolled for support and 194,460 have completed their housing construction until January 2021. This is the achievement of 70.8% progress.

**Table 3: Housing recovery status in urban areas of earthquake affected districts**

| Geographic area          | HHs eligible | HHs enrolled | 1st tranche | 2nd tranche | 3rd tranche | Progress achieved % |
|--------------------------|--------------|--------------|-------------|-------------|-------------|---------------------|
| Inside Kathmandu valley  | 97,706       | 91,390       | 86,367      | 42,167      | 37,893      | 38.8%               |
| Outside Kathmandu valley | 274,613      | 262,469      | 259,749     | 217,767     | 194,460     | 70.8%               |
| Total                    | 372,319      | 353,859      | 346,116     | 256,154     | 232,353     | 62.4%               |

Data shows that reconstruction in urban areas is slower as compared to reconstruction in rural areas. This discrepancy makes it important to find the issues associated with urban reconstruction so that they can be discussed in detail and addressed appropriately.

To complete the remaining tasks of housing recovery and reconstruction, the 15th Plan (Fiscal Year 2019/20- 2023/24) by the National Planning Commission proposes coordination with Provincial level and local level governments. Seven Provinces of Nepal were formed on September 20, 2015. Schedule 9 of the *Constitution* produce a basis for provinces to be active in disaster management. When the Gorkha earthquake hit Nepal, there was no system of Provincial government. The Four provinces namely: - Bagmati, Gandaki, Lumbini and Province no. 1 were affected and still have caseloads being still. The following table 4 shows the recovery status in urban areas of these four Provinces.

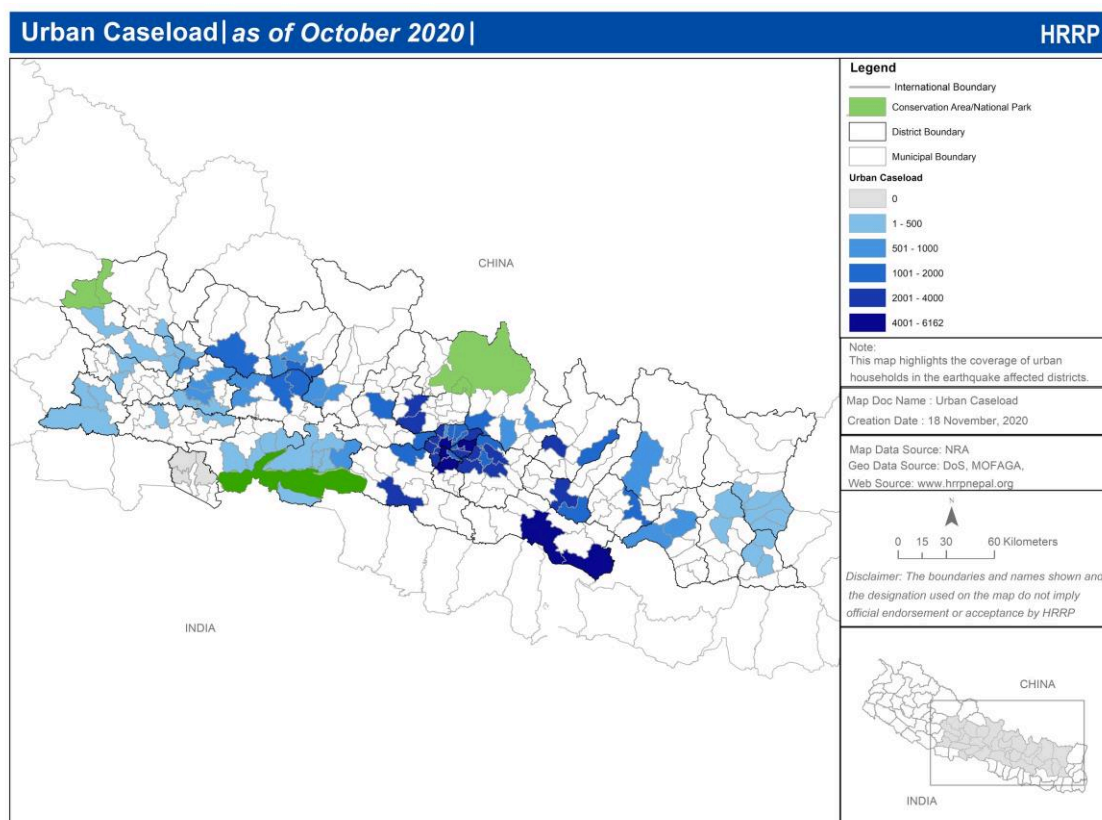
**Table 4: Recovery status of private housing in urban municipalities of earthquake affected Provinces**

| Administrative division | HHs eligible | HHs enrolled | 1st tranche | 2nd tranche | 3rd tranche | Progress achieved % |
|-------------------------|--------------|--------------|-------------|-------------|-------------|---------------------|
| Bagmati Province        | 304,952      | 288,106      | 281,376     | 208,945     | 184,499     | 60.5%               |
| Gandaki Province        | 50,955       | 48,871       | 48,262      | 38,434      | 34,115      | 66.95%              |
| Lumbini Province        | 2,149        | 2,090        | 2,124       | 1,459       | 1,165       | 54.21%              |
| Province no. 1          | 15,048       | 14,962       | 14,256      | 11,865      | 10,657      | 70.82%              |
| Total                   | 373,104      | 354,029      | 346,018     | 260,703     | 230,436     |                     |

**Table 5: Progress of various districts on private housing in urban areas**

| S.N. | Progress range      | Number of Districts | Name of Districts                                                                                                      |
|------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|
| 1    | 31% - 60% progress  | 12                  | Arghakhanchi, Bhaktapur, Kaski, Kathmandu, Kavrepalanchok, Lalitpur, Lamjung, Myagdi, Palpa, Parbat, Sindhuli, Syangja |
| 2    | 61% - 80% progress  | 12                  | Baglung, Bhojpur, Chitwan, Dhankuta, Dolakha, Gulmi, Khotang, Makwanpur, Nuwakot, Sankhuwasabha, Solukhumbu, Tanahun   |
| 3    | 81% - 100% progress | 6                   | Dhading, Gorkha, Nawalparasi, Okhaldhunga, Ramechhap, Sindhupalchowk                                                   |
|      | Total               | 30                  |                                                                                                                        |





**Map 1: Map showing District-wise urban caseload**

According to the municipalities affected by the Gorkha earthquake, there are 96 urban municipalities in earthquake affected districts. Data shows that Bhaktapur, Kathmandu, Kirtipur, Lalitpur, Dudhouli, Bhirkot urban municipalities have less progress rate than other municipalities. Hence, a concentrated effort is needed for these municipalities.

**Table 6: Progress of various urban municipalities on urban housing reconstruction**

| S.N. | Progress range      | Number of municipalities | Name of Urban Municipalities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|---------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 0% - 30% progress   | 6                        | <ul style="list-style-type: none"> <li>Bhaktapur, Kathmandu, Kirtipur, Lalitpur, Dudhouli, Bhirkot</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2    | 31% - 60% progress  | 46                       | <ul style="list-style-type: none"> <li>Bhumekasthan, Sitganga, Baglung, Dhorpatan, Galkot, Changunarayan, Madhyapur Thimi, Suryabinayak, Madi, Ratnanagar, Dhankuta, Pakhribas, Musikot, Pokhara Lekhnath, Budhanilkantha, Chandragiri, Dakshinkali, Gokarneshwor, Kageshwori Manohara, Nagarjun, Shankharapur, Tarakeshwor, Tokha, Banepa, Dhulikhel, Namobuddha, Panauti, Godawari, Mahalaxmi, Besishahar, MadhyaNepal, Sundarbazar, Hetauda, Beni, Rampur, Tansen, Kushma, Phalebas, Dharmadevi, Madi, Kamalamai, Chapakot, Putalibazar, Waling, Bhimad, Shuklagandaki</li> </ul> |
| 3    | 61% - 80% progress  | 30                       | <ul style="list-style-type: none"> <li>Sandhikharka, Jaimuni, Bhojpur, Shadananda, Bharatpur, Kalika, Khairahani, Rapti, Dhunibesi, Mahalaxmi, Bhimeshwor, Jiri, Resunga, Mandandeupur, Panchkhal, Halesi Tuwachung, Rupakot huwagadhi, Rainas, Thaha, Kawasoti, Belkotgadhi, Bidur, Ramechhap, Chainpur, Khandbari, Panchakhapan, Solududhakunda, Galyang, Bhanu, Byas</li> </ul>                                                                                                                                                                                                   |
| 4    | 81% - 100% progress | 14                       | <ul style="list-style-type: none"> <li>Nilakantha, Gorkha, Palungtar, Bardaghat, Devchuli, Gaidakot, Ramgram, Sunwal, Siddhicharan, Manthali, Barhabise, Chautara Sangachok Gadhi, Melamchi</li> </ul>                                                                                                                                                                                                                                                                                                                                                                               |
|      | Total               | 96                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



Considering, reconstruction status (as of April 2021) by vulnerable beneficiaries in urban areas of earthquake affected districts, among 9256 total cases, 49% has completed the reconstruction; 40% are reconstructing their houses and 11 % have not yet started the reconstruction.

## 2.1.2 Heritage Settlement

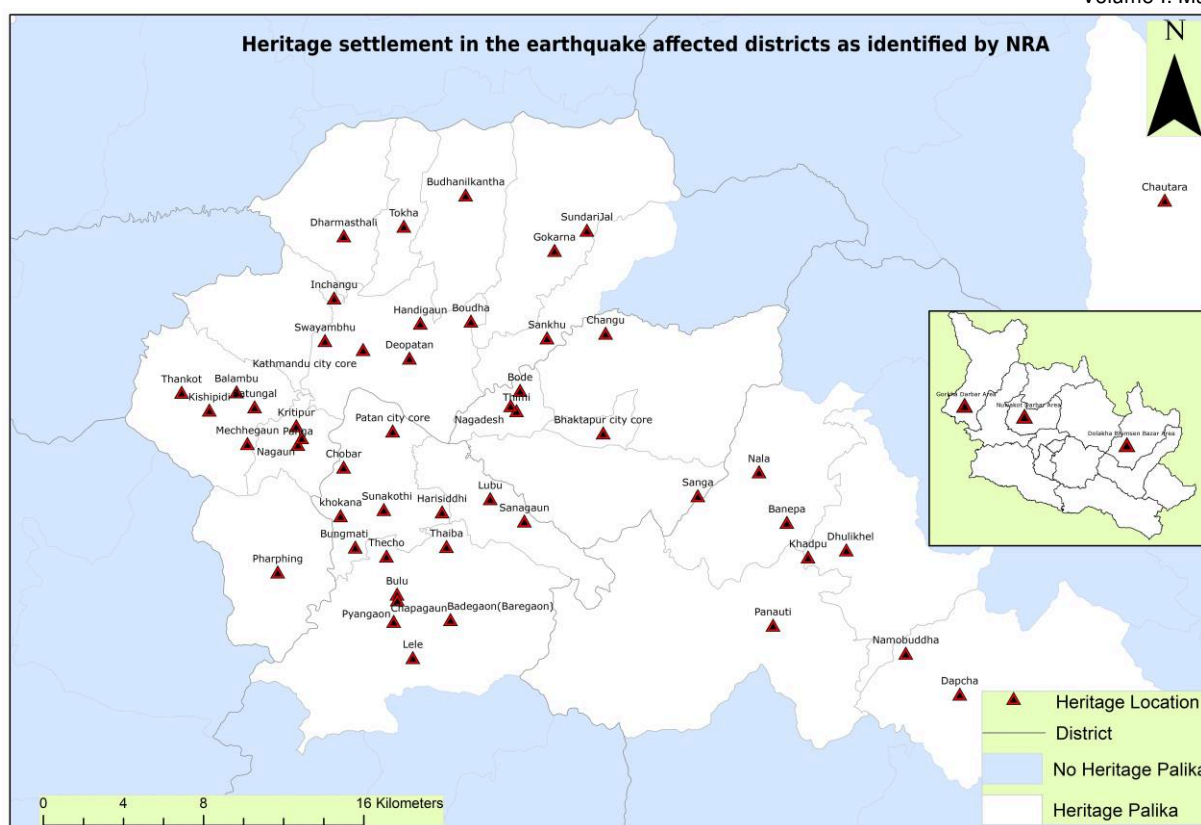
There are 42 historic '*Newari*' settlements within the Kathmandu Valley and 12 outside Kathmandu valley as declared by NRA. Kathmandu Valley historic settlements can be categorized into three groups.

1. The first group are the three historic durbar squares of Kathmandu, Lalitpur, and Bhaktapur city, two budhist monuments SwoyambhuNath and BoudhaNath Complexes and Two Hindu Monument complexes Pashupati Nath and Changu Narayan which are also chosen as the United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage Sites.
2. The second group of historic settlements are the four towns on the World Heritage Tentative List: Panauti, Khokana, Kirtipur, and Shankarapur.
3. The third group is composed of other towns in the valley, such as Bungamati, Panga, and Kishipidi, which are historic settlements with traditional values and satellite towns of the principal Cities of Kathmandu, Lalitpur, and Bhaktapur. For example: Khokana and Bungamati are the satellite towns of Lalitpur, while Kisipidi is that of Kathmandu.

Kathmandu valley has one World Heritage Site consisting of The Durbar Squares of Hanuman Dhoka (Kathmandu), Patan and Bhaktapur, the Buddhist stupas of Swayambhunath and Bauddhanath and the Hindu temples of Pashupatinath and Changu Narayan.

**Table 7: List of heritage settlements declared by NRA**

| A) Heritage Settlements in Kathmandu Valley      |                         |                                  |
|--------------------------------------------------|-------------------------|----------------------------------|
| 1. Kathmandu City Core                           | 15. Pharping            | 29. Tokha                        |
| 2. Thecho                                        | 16. Chobar              | 30. Budhanilkantha               |
| 3. Chapagaon                                     | 17. Nagaun              | 31. Sundarijal                   |
| 4. Bulu                                          | 18. Panga               | 32. Gokarna                      |
| 5. Lele                                          | 19. Kirtipur            | 33. Boudha                       |
| 6. Pyangaon                                      | 20. Machhegaun          | 34. Handigaun                    |
| 7. Bandegaon (Baregaon)                          | 21. Satungal            | 35. Deopatan                     |
| 8. Thaiba                                        | 22. Kishipidi           | 36. Patan City Core              |
| 9. Harisiddhi                                    | 23. Balambu             | 37. Thimi                        |
| 10. Sanagaun                                     | 24. Thankot             | 38. Nagadesh                     |
| 11. Lubhu                                        | 25. Ichangu             | 39. Bode                         |
| 12. Sunaguthi                                    | 26. Swoyambhu           | 40. Changu                       |
| 13. Bungamati                                    | 27. Balaju              | 41. Sankhu                       |
| 14. Khokana                                      | 28. Dharmasthali        | 42. Bhaktapur City Core+7 others |
| B) Heritage Settlements Outside Kathmandu Valley |                         |                                  |
| 43. Nala                                         | 47. Khadpu              | 51. Dolakha Bhimsen Bazar Area   |
| 44. Sanga                                        | 48. Panauti             | 52. Chautara                     |
| 45. Banepa                                       | 49. Gorkha Durbar Area  | 53. Namobudha                    |
| 46. Dhulikhel                                    | 50. Nuwakot Durbar Area | 54. Dapcha                       |



**Map 2: Map of Heritage Settlements in earthquake affected districts**

Data shows that among 794 heritage structures in the urban areas, 55% are reconstructed, 24% are under construction and 21% are yet to be constructed.

Considering, reconstruction status of vulnerable beneficiaries in heritage settlements in urban areas of earthquake affected districts as identified by NRA, 678 beneficiaries have already obtained the first tranche, 244 beneficiaries have already obtained the second tranche, while only 203 beneficiaries have obtained the third tranche.

Reconstruction status of vulnerable beneficiaries in heritage settlements in urban areas of earthquake affected districts as identified by NRA is picturized in **Annex 3, Volume II**.

### 2.1.3 Economic Recovery

The NRA has clearly set the vision on its Post Disaster Recovery Framework (PDRF) that is “*Well Planned, Resilient Settlements and a Prosperous Society*.” As per the recovery framework, there are five strategic objectives of the reconstruction process. Among them, one is an economic recovery related strategy “Develop and restore economic opportunities and livelihoods and re-establish a productive sector.”

It is urgent to restore economic opportunities and livelihoods. Regarding the on-farm economy, efforts will centre on the re-establishment of agricultural livelihoods of vulnerable earthquake-affected smallholder farmers, as well as meeting the most urgent recovery needs of the agriculture, livestock, and irrigation sector and of the families reliant on it. Another key consideration will be that all recovery programs should contribute to the restoration of basic operational systems and structures essential to restart sector growth.

The large owner-driven housing reconstruction program will also support local livelihoods, businesses, and products, thereby strengthening the local economy. Similarly, employment and livelihood related interventions will aim to enable households and workers to recover employment opportunities, engage in productive and income-generating activities, and ensure a stable and resilient asset base. Enhancing the role of environmental protection and forestry in improving community livelihoods will also be a key programmatic consideration. Labor intensive and local technologies will be used to ensure engagement and employment generation for local communities.

In the commerce and industry sector, the aim is to implement a comprehensive response involving the promotion of rights, decent employment, social protection, and social dialogue for ensuring representation of the informal sector. The focus will be on rebuilding the basic infrastructure essential to revive communication and information outreach. Additionally, there will be an effort to enhance labour market information and employment services and to set up better connections between skills development systems and training for productive work. Particular attention will be

given to enhancing youth employability and to meeting the government's aim of reducing youth unemployment, as well as to improving the productivity and abilities of young people in the workforce.

The following activities have been implemented by government and nongovernment agencies for economic recovery.

- Housing grant distribution
- Diverse types of concessional Loans
- Food for work or cash for work in the short term
- Banana cultivation
- Off-season vegetable production in a plastic house
- Goat raising.
- Free-range poultry
- Mushroom and ginger cultivation
- Bee keeping

#### **2.1.4 NRA's attempts for Urban Regeneration**

After Gorkha Earthquake 2015, NRA not only concentrated on the reconstruction but on its many aspects such as planning, policy formulation, and financing as discussed below:

**Planning:** After Gorkha Earthquake 2015, master plans of seven world heritage sites and six heritage settlements namely: Nuwakot, Gorkha, Dolkha, Bungmati, Sankhu and Khokana were formed by NRA. A separate deputy project director was assigned in CLPIU Buildings by NRA to guide and supervise the planning and implementation of the master plans. A budget of 2.5 billion was allocated for this and applied for loan from Exim Bank of India. However, Exim Bank's criteria could not be fulfilled and the fund is not realized.

NRA and Kathmandu Metropolitan city had signed MOU to initiate housepooling at Kilagal, Kathmandu based on the House Pooling Procedure which was being formulated simultaneously by the Ministry of Urban Development (MoUD). However, due to the lack of legal tools like Urban Regeneration Act, Regulation and Policy, the housepooling could not be implemented.

**Policies:** Including heritage byelaws NRA formulated twenty-two policies and guidelines for post earthquake recovery and reconstruction which local government could use after contextualizing them.

**Financing:** NRA grant of an amount (NPR 300,000) is segregated for all Gorkha earthquake victims to reconstruct their houses damaged by the earthquake. Additionally, heritage subsidy of NPR 50,000 is allocated by NRA for all Gorkha earthquake victims whose house lie in the heritage area. Similarly, NRA Vulnerable population subsidy of an amount NPR 50,000 is allocated by NRA for all vulnerable population affected by the earthquake. Additionally, subsidized loan at an interest of 2% is facilitated by GoN and NRA to help the Gorkha earthquake victims to reconstruct their house.

## **2.2 Stakeholder Mapping for Urban Housing Reconstruction and Recovery**

The recovery and reconstruction of buildings and infrastructure were taken care by the CLPIU (Building), CLPIU (Education) and Grant Management and Local Infrastructure (GMALI) and their respective units' set-up in the affected districts for the reconstruction of public buildings, educational institutions, local roads, and water supply services, respectively. The reconstruction of hospitals, major strategic roads, electricity services, telecommunication facilities, security buildings were taken care by the reconstruction unit at ministry of health, department of road reconstruction unit, Nepal electricity authority, Nepal Telecom, and the corresponding security organisation, respectively.

The CLPIUs are actively engaged in the recovery and reconstruction activities with continuous coordination, communication and collaboration with all the relevant government agencies, international and national non-state entities, and development partners. CLPIUs must follow the guiding policies, directives, and resources of the NRA as well as the line ministries. The CLPIUs setup District Level Project Implementation Units (DLPIU) according to their needs and workload. The NRA adopted policies for reconstruction from the corresponding sector Ministries. The NRA authorised reconstruction projects, built policies, budgeting schemes and legislation, and disbursed them to the CLPIUs. The Project Directors of CLPIU delegated implementation tasks to DLPIU Project Managers. DLPIUs have been set up in most of the affected districts under the CLPIUs. The DLPIUs focussed on implementing and monitoring the recovery and reconstruction projects in the districts.

Several development partners, UN organizations and national and international NGOs supplied a range of technical, financial, and material services during relief and early recovery, and many have been offering to supply substantial

support to the ongoing recovery and reconstruction process. The following stakeholder (Table 3) took part in reconstruction and recovery of Nepal in the post-earthquake scenario.

**Table 8: Stakeholders' Participation**

| S.No. | Urban Housing Reconstruction and Recovery Stakeholders |                                                                                                                              |
|-------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1.    | National Government                                    | NRA, CLPIU/Building, CLPIU/GMALI, MOUD, DUDBC, NDRRMA, KVDA, Guthi Sansthan                                                  |
| 2.    | Provincial Government                                  | MOPIT, MOEAP                                                                                                                 |
| 3.    | Local Government/ district level line agencies         | CRC, Municipal wards, Municipality, DLPIU GMALI, DLPIU/Building, DTCO, DCC, DAO, DLRM district office, DOS District office   |
| 4.    | Partner Organizations                                  | Donors, INGOs, NGOs,                                                                                                         |
| 5.    | Financial Service Provider                             | NRB, Banks, Cooperatives, MFIS, Remittance, Neighbors, Relatives and Friends, Insurance Board, and Insurance companies       |
| 6.    | Construction Professionals                             | NRA Technical Persons, Local Masons, Migrant Masons, Craftsman, contractors, consulting engineering firms, Social Mobilizers |
| 7.    | Construction Business                                  | Construction material suppliers, Construction material producers, FNCCI, CNI, Developers                                     |
| 8.    | Mass Media                                             | Social Media, Journalists                                                                                                    |
| 9.    | Academia and Research                                  | Researchers, Academic Institutions                                                                                           |
| 10.   | Community                                              | Ama Samuha (Mothers' Groups formed at community level), Youth Groups, Beneficiary                                            |

List of activities supported by Development Partners for Recovery and Reconstruction are attached to **Annex 4, Volume II.**

List of activities supported by UN Agencies for Recovery and Reconstruction are attached to **Annex 5, Volume II.**

### 3 ISSUES AND CHALLENGES

Studies on Urban Housing Reconstruction and Recovery have shown the following issues:

- Access to finance
- Land Issues
- Compliance to Heritage Norms
- Communication
- Socio-Technical Assistance (STA) in Reconstruction
- Retrofitting in Urban Areas
- Help to Vulnerable

In addition, international studies have shown that disaster recovery is a **time-consuming** task and Nepal is not an exception. Hence, the main issue of urban recovery was insufficient time.

#### 3.1 Access to Finance

Access to finance is one of the biggest challenges faced by the beneficiaries which is hindering the urban reconstruction to move forward. After the earthquake, though people had land and little assets, they did not have the required financial resources to start the immediate rebuilding of their houses. The construction cost in urban areas is far more than in rural areas. Finance issues includes High cost of urban housing reconstruction, less access to affordable finance/loan.

**Table 9: Summary of issues related to access to finance**

| Major Issues                                                                                                                                                                | Causes/<br>Problems                                                                                      | Past/Ongoing Efforts                                                                                                                                                                                         | Potential Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• High cost of urban housing reconstruction (Stone in cement:1000 sq. ft. NPR 1,450,000, RC: 1000 sq. Ft., NPR 3,400,000)</li> </ul> | <ul style="list-style-type: none"> <li>• High material costs</li> </ul>                                  | <ul style="list-style-type: none"> <li>• Community participation already exists in some cases</li> <li>• Tax exemption in some municipalities</li> <li>• Subsidized timber provided by government</li> </ul> | <ul style="list-style-type: none"> <li>• Promote incremental housing with development manuals with detailed process,</li> <li>• Amend by-laws accordingly.</li> <li>• Reform in existing policies / by-laws</li> <li>• Tax exemption provisions in approval process and materials cost</li> <li>• Supply free design and supervision from municipality.</li> <li>• Encourage in-kind labour (community participation)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• High building design &amp; supervision cost</li> </ul>          |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• High building permit tax</li> </ul>                             |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• High labour cost</li> </ul>                                     |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Less access to affordable finance/loan</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>• Collateral Requirements,</li> <li>• Bank's hesitance</li> </ul> | <ul style="list-style-type: none"> <li>• Some municipality started revolving fund</li> <li>• 'Unified Procedure for Interest Subsidy on Concessional Loan, NRA, 2018'</li> </ul>                             | <ul style="list-style-type: none"> <li>• Set up Revolving Funds with directives at municipality specially for marginalized Household</li> <li>• Mobilize community Based Cooperatives with seed money</li> <li>• Mobilize microfinance bank with seed capital</li> <li>• Develop comparatively flexible guidelines to supply loans</li> <li>• Increase deprived sector loan size. Facilitate the central, commercial and microfinance banks to bring change in policy.</li> <li>• Revise 'Unified Procedure for Interest Subsidy on Concessional Loan, 2018' to increase loan size.</li> <li>• Request NRB and concerned ministries of GON to priorities use of CSR fund to create revolving fund for providing housing loan for poor families</li> <li>• Develop risk transfer system through insurance</li> </ul> |

## 3.2 Land Issues

Land issues include lack of ownership papers, multiple ownership of land, smaller plots below threshold, cadastral inconsistencies, geo-hazard risk/ vulnerable settlements and landless.

**Table 10: Summary of issues related to land**

| Major Issues                                                                                                                 | Causes/Problems                                                                                                                                                                                             | Past/Ongoing Efforts                                                                                                                                                                                                                                                                                                                             | Potential Strategy                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of land ownership Documents ( <i>Lal Purja</i> – one of the prerequisites to sign grant agreement or to apply for loan) | <ul style="list-style-type: none"> <li>Guthi or public land dwellers (squatters)</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>Beneficiaries can build their house in Guthi land through 'abhilekikaran' by municipality and Guthi in few places</li> </ul>                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Policy review/guideline at national level by Guthi Sansthan</li> <li>Guideline for fast tracking remaining caseload of Guthi</li> </ul>                                                                                                                                                                                    |
|                                                                                                                              | Tenants (Mohi)                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Land Act 1964 (2021)</li> </ul>                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                              | <ul style="list-style-type: none"> <li>Ownership not transferred from generations even after death</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>Municipality has registration system (birth, death, and marriage),</li> <li>Ownership transfer system exists</li> </ul>                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Promote people to register family events at local level</li> <li>Maintain database of household at local level</li> </ul>                                                                                                                                                                                                  |
|                                                                                                                              | <ul style="list-style-type: none"> <li>Loss of land ownership due to some cases (disaster, theft etc.)</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>Can reapply for land ownership at Department of Land Reform and Management</li> </ul>                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Aware people</li> </ul>                                                                                                                                                                                                                                                                                                    |
| Multiple Ownership of Land                                                                                                   | <ul style="list-style-type: none"> <li>Division of property among family members</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>Procedure for reconstruction of single or Multiple Ownership in Heritage Settlement –NRA,2076</li> </ul>                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Amendment of the multiple ownership guideline for the longer-term to include               <ol style="list-style-type: none"> <li>1) core-urban areas besides designated heritage areas</li> <li>2) horizontal and vertical division</li> <li>3) include retrofitting beneficiaries</li> </ol> </li> </ul>                 |
|                                                                                                                              | <ul style="list-style-type: none"> <li>Disputes on ownership</li> <li>Litigation in court</li> </ul>                                                                                                        |                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Develop fast track decisive mechanism</li> <li>Can construct house on mutual understanding</li> <li>Compensation of expenditure by the person who gets the ownership</li> </ul>                                                                                                                                            |
| Smaller Plots below threshold (No building permit for Less than 2.2 Ana)                                                     | <ul style="list-style-type: none"> <li>Due to vertical subdivision of plots over generations</li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>Building permit has not restricted plot size for building construction in core cities (for pre-existing buildings)</li> </ul>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Mechanisms need to be developed (HP, LP, redevelopment etc.)</li> <li>Develop social awareness</li> <li>Review of smallest plot size for economically weak families (30-40 m<sup>2</sup> / family)</li> </ul>                                                                                                              |
|                                                                                                                              | <ul style="list-style-type: none"> <li>Part of the land within Right of Way (RoW)</li> </ul>                                                                                                                |                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Create a diversion so that the old settlement will not get affected</li> <li>Provide housing plot in planned land pooling site or govt. land of equivalent cost value (land Swap)</li> </ul>                                                                                                                               |
| Cadastral Inconsistencies                                                                                                    | <ul style="list-style-type: none"> <li>Obsolete cadastral map, errors in maps</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>Government land survey system exists</li> </ul>                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Develop verification mechanism</li> </ul>                                                                                                                                                                                                                                                                                  |
| Geo-hazard risk/ Vulnerable settlements                                                                                      | <ul style="list-style-type: none"> <li>Living there for generations due to linkage with livelihood</li> <li>Not aware of the potential risk but exposed by the earthquake</li> <li>No other land</li> </ul> | <ul style="list-style-type: none"> <li>Procedure for Relocation and rehabilitation of the vulnerable settlements–NRA, 2073</li> <li>Standards on habitable land purchase for earthquake affected beneficiaries, 2074</li> <li>Procedure for Development of Integrated Settlements, 2075: beneficiaries that come under risk category.</li> </ul> | <ul style="list-style-type: none"> <li>Identify hazard prone areas, settlements in hazardous areas and safer areas through risk sensitive land use planning.</li> <li>Adopt engineering solutions to mitigate/minimize disaster risk.</li> <li>If engineering solution is not possible, move vulnerable settlements in nearest safer settlement areas.</li> </ul> |



| Major Issues | Causes/Problems                                                                                 | Past/Ongoing Efforts                                                                           | Potential Strategy |
|--------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------|
| Landless     | <ul style="list-style-type: none"> <li>Poverty</li> <li>Loss of land due to disaster</li> </ul> | <ul style="list-style-type: none"> <li>Commission for solving land related problems</li> </ul> |                    |

### 3.3 Compliance to Heritage Norms

Heritage issues include Non-compliance to heritage norms, more cost for keeping vernacular architecture, lack of availability of traditional construction material, lack of skilled human resource and multi-ownership.

**Table 11: Summary of issues related to compliance with Heritage Norms**

| Major Issues                                              | Causes/Problems                                                           | Past/Ongoing Efforts                                                                                                                                                                                                                                                       | Potential Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-compliance to heritage norms                          | Lack of demarcation, identification, and nomination of heritage area      | <ul style="list-style-type: none"> <li>Ancient monument preservation Act 2013 and Basic bylaws on settlement development, urban planning and building construction 1<sup>st</sup> amendment, 2016</li> </ul>                                                               | <ul style="list-style-type: none"> <li>Revise and update ancient monument preservation Act 2013, and LGOA, 2074</li> <li>Identify, define, and nominate heritage area, update bylaws accordingly</li> </ul>                                                                                                                                                                                                                                                          |
| Additional cost for keeping vernacular architecture       | Expensive traditional material, labour intensive to meet compliance level | <ul style="list-style-type: none"> <li>NPR 50,000 more grants provided for reconstruction of house in heritage settlement in compliance with Basic bylaws on settlement development, urban planning, &amp; building construction, 1<sup>st</sup> amendment 2016</li> </ul> | <ul style="list-style-type: none"> <li>Update Inspection checklists prepared by NRA for technical verification.</li> <li>Develop design and technical construction guidelines for traditional houses in heritage settlements.</li> <li>Emphasize community engagement; and resident committees evaluate housing rehabilitation schemes</li> <li>Train and Involve community groups in housing repairs and retrofitting</li> </ul>                                    |
| Lack of availability of traditional construction material | Less production of material                                               |                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Motivate and mobilize private sector to manufacture traditional materials</li> <li>Bulk procurement by municipality: distribute to homeowners on par cost</li> </ul>                                                                                                                                                                                                                                                          |
| Lack of skilled human resource                            | Inadequate training                                                       |                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Encourage/promote skill transfer from generation to generation,</li> <li>Increase number of trainings, Provide incentives,</li> <li>Develop specific curriculum for mason training on traditional construction technology</li> <li>Train masons and contractors</li> </ul>                                                                                                                                                    |
| Multi-ownership                                           | Vertical sub-division, conflict in property division                      | <ul style="list-style-type: none"> <li>Procedure for reconstruction of single or Multiple Ownership in Heritage Settlement –NRA,2076</li> </ul>                                                                                                                            | <ul style="list-style-type: none"> <li>Finalize and overanalyse House pooling (urban renewal) Guideline,</li> <li>Enact Federal Urban Development Act that defines local area development projects like urban regeneration, urban renewal, slum and squatter area improvement and development, other public infrastructure construction, including land pooling, Guided land development and site and services and planning and implementation procedures</li> </ul> |

### 3.4 Communication Strategy

Issues related to communication strategy are lack of information flow, inadequate flow of information, advance flow of information, inadequate dissemination of guidelines and decisions at local level, digitized communication system and misleading information.



**Table 12: Summary of issues related to communication strategy**

| Major Issues                                                                                 | Causes/Problems                                                                                                                    | Past/Ongoing Efforts | Potential Strategy                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of flow of information (about grant, soft loan, and reconstruction procedure to public) | Lack of effective communication mechanism between local and central level and vice versa, Guidelines mostly available in soft copy |                      | <ul style="list-style-type: none"> <li>Federal and provincial government should help local government in formulation of communication strategy and plan</li> <li>Local bodies should pre-plan communication strategy</li> </ul> |
| Inadequate flow of information                                                               |                                                                                                                                    |                      |                                                                                                                                                                                                                                 |
| Untimely flow of information                                                                 |                                                                                                                                    |                      |                                                                                                                                                                                                                                 |
| Inadequate dissemination of Guidelines and decisions at local level                          |                                                                                                                                    |                      |                                                                                                                                                                                                                                 |
| Digitized communication system                                                               | Lack of stable internet/ communication network coverage                                                                            |                      | <ul style="list-style-type: none"> <li>Information dissemination system should be developed</li> <li>Ensure stable network</li> </ul>                                                                                           |
| Misleading Information                                                                       | Different interpretation of information by different stakeholder                                                                   |                      | <ul style="list-style-type: none"> <li>Strictly check information flow and correct misleading information</li> </ul>                                                                                                            |

### 3.5 Socio-Technical Assistance (STA)

Socio-Technical help is a people-based activity, requiring many roles, from defining building regulations to advising households on construction budgets, and from government, private sector, and aid backgrounds. Technical help is broadly defined as a range of social and technical support, for housing recovery that support to supply and apply proper standards and guidance for construction and ensure correct information reaches all communities and households in a prompt, consistent and effective manner. The issues related to STA are Lack of provision of STA, ineffective part of STA in urban context, difficulty in obtaining building permits, expensive demolition and debris clearance and lack of Risk Sensitive Land Use Plan (RSLUP).

**Table 13: Summary of issues related to socio-technical assistance**

| Major Issues                                  | Causes/Problems                                                                                                                                                                            | Past/Ongoing Efforts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Potential Strategy                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of provision of STA                      | <ul style="list-style-type: none"> <li>Insufficient social mobilizers and technical manpower</li> <li>Unaware communities about STA</li> </ul>                                             | <ul style="list-style-type: none"> <li>Socio-Technical Assistance Package designed by HRRP and endorsed by NRA which consists of seven components: <ul style="list-style-type: none"> <li>i. Community/ Household Orientations</li> <li>ii. Continuous door to door technical help</li> <li>iii. Short training for masons</li> <li>iv. On the job training for masons</li> <li>v. Helpdesk/Technical support centre</li> <li>vi. Demonstration construction</li> <li>vii. Community Reconstruction Committees</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>STA guidelines should be clear and concise</li> <li>Social mobilizers and technical person and masons should be trained on STA components</li> <li>Engineer and architects of local government should be trained in designing and building vernacular building and cultural heritage monuments</li> <li>Community groups should be trained and involved in housing repairs and retrofitting</li> </ul> |
| Ineffective component of STA in urban context | <ul style="list-style-type: none"> <li>STA adopted same format for urban and rural area</li> </ul>                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Difficulty in obtaining building permits      | <ul style="list-style-type: none"> <li>Lack of information on Building Permit Process (BPP)</li> <li>BPP was time-consuming</li> <li>Non-compliance of Bylaws and building code</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Local government should set up help desk for flow of information</li> <li>BPP should have clear and brief manual</li> <li>There should be trainings on bylaws and building code</li> </ul>                                                                                                                                                                                                             |

| Major Issues                              | Causes/Problems                                                                                                                                                                                                                  | Past/Ongoing Efforts | Potential Strategy                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Expensive demolition and debris clearance | <ul style="list-style-type: none"> <li>Demolition had a threat of damaging Neighbours' house in case of row housing</li> </ul>                                                                                                   |                      | <ul style="list-style-type: none"> <li>There should be guidelines for demolition and clearance of debris</li> </ul> |
| Lack of RSLUP                             | <ul style="list-style-type: none"> <li>Buildings were reconstructed in risk sensitive areas</li> <li>Risk Sensitive Land Use Plan (RSLUP) not formulated by municipalities</li> <li>People were not oriented on RSLUP</li> </ul> |                      | <ul style="list-style-type: none"> <li>Formulate RSLUP and orient people on it</li> </ul>                           |

### 3.6 Retrofitting in Urban Areas

Retrofitting issues include excessive cost of Retrofitting, low awareness and trust in retrofitting, complex system/technology, and dispute in retrofitting of some traditional buildings.

**Table 14: Summary of issues related to retrofitting in urban areas**

| Major Issues                                          | Causes/Problems                                                                                                                                                                                                                       | Past/Ongoing Efforts | Potential Strategy                                                                                                                                                                                                               |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excessive cost of Retrofitting                        | <ul style="list-style-type: none"> <li>Rate specifications does not exist for retrofitting</li> </ul>                                                                                                                                 | Retrofitting manual  | <ul style="list-style-type: none"> <li>Retrofitting manuals should be simplified</li> <li>Retrofitting guidelines should be prepared</li> <li>People need to be oriented on effectiveness of retrofitting and economy</li> </ul> |
| Low awareness and trust in retrofitting               | <ul style="list-style-type: none"> <li>Retrofitting is a new concept</li> <li>Some house owners are not convinced</li> <li>Some convinced ones could not get proper technical help</li> </ul>                                         |                      |                                                                                                                                                                                                                                  |
| Complex system/technology                             | <ul style="list-style-type: none"> <li>Retrofitting of adjoined buildings in case of row housing is complex and at times technically not possible</li> <li>Lack of adequate technical guidance and technical professionals</li> </ul> |                      |                                                                                                                                                                                                                                  |
| Dispute in retrofitting of some traditional buildings | <ul style="list-style-type: none"> <li>Traditional building needs to be compliant with byelaws of the municipality and Department of Archaeology</li> </ul>                                                                           |                      | <ul style="list-style-type: none"> <li>Prepare retrofitting guidelines for traditional buildings</li> </ul>                                                                                                                      |

### 3.7 Assistance to Vulnerable

Issues related to help to vulnerable populations include neglected vulnerable population, lack of adequate document to apply loan and delisting of beneficiaries who are socio-economically vulnerable.

**Table 15: Summary of issues related to help to vulnerable population**

| Major Issues                            | Causes/Problems                                                                                                                                                                                                          | Past/Ongoing Efforts                                                                                                        | Potential Strategy                                                                                                                                                              |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Neglected vulnerable population         | <ul style="list-style-type: none"> <li>Vulnerable population (poor, disabled, displaced) are often left behind</li> <li>No separate STA has been developed for vulnerable beneficiaries under NRA's criteria.</li> </ul> |                                                                                                                             | <ul style="list-style-type: none"> <li>NRA's vulnerability criteria should include all types of vulnerable population</li> </ul>                                                |
| Lack of adequate document to apply loan | <ul style="list-style-type: none"> <li>Bank needs collateral and pay back with interest</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li>More addition support of NPR 50,000 is available to vulnerable households</li> </ul> | <ul style="list-style-type: none"> <li>Government grant should be managed</li> <li>Building premix tax should be exempt</li> <li>Community housing should be planned</li> </ul> |

|                                                                  |                                                                                                                            |  |  |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--|--|
| Delisting of beneficiaries who are socio-economically vulnerable | <ul style="list-style-type: none"> <li>People do not fall into NRA's vulnerable category after tranche deadline</li> </ul> |  |  |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--|--|

Apart from the above-mentioned issues and challenges, the following highlights the areas and major issues in the Urban Development and Recovery Strategy (UDRS) in Local Governments of Nepal:

| S. No. | Areas                                | Main Issues in the UDRS                                                                                                                                                                                                                    |
|--------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Conservation and Protection          | CP1. Control hazard prone areas from development                                                                                                                                                                                           |
|        |                                      | CP2. Preservation and restoration of historical, cultural, and archaeological sites and settlements                                                                                                                                        |
| 2      | Municipal Land Use Planning          | MLUP1. Optimum use of Available Land Resources                                                                                                                                                                                             |
|        |                                      | MLUP 2. Infrastructure Development in built up areas                                                                                                                                                                                       |
|        |                                      | MLUP 3. Promote Studies for Urban/Built up Fringe Area Development                                                                                                                                                                         |
| 3      | Housing                              | H 1. Promotion of Housing Area Development in New Urban Area                                                                                                                                                                               |
|        |                                      | H 2. Promotion of affordable Housing for the Low-Income Groups and the Poor by Public Sector Housing Agencies                                                                                                                              |
| 4      | Economy and Employment               | EE 1. Promotion of local crafts and skills based small scale cottage industry                                                                                                                                                              |
|        |                                      | EE 2. Promotion of public private partnership in Heritage settlement regeneration strategy and development and implementation of urban regeneration plan                                                                                   |
| 5      | Infrastructure and Public Facilities | IPF 1. Incremental development and improvement of basic urban (municipal) infrastructure and services such as road, drainage, water supply and sanitation, solid and liquid waste management, power supply etc., in already built-up areas |
| 6      | Tourism and Recreation               | TR 1. Promotion of tourism through tourism infrastructure development and improvement                                                                                                                                                      |
|        |                                      | TR 2. Promotion of local heritage-based tourism                                                                                                                                                                                            |
| 7      | National Resources & Environment     | NRE. 1. Promotion of recreational open / green space                                                                                                                                                                                       |
|        |                                      | NRE. 2. Prevention of surface water from pollution                                                                                                                                                                                         |

## 4 EXISTING POLICIES, ACTS, REGULATIONS, WORKING PROCEDURE AND GAPS

This chapter consists of study and analysis of Government urban development acts, policies, guidelines and working procedures in housing, building construction, and urban development and study and analysis of reconstruction and recovery Acts, policies and procedures formulated by NRA.

### 4.1 Government Urban Development Policy gaps

Gaps (deficiencies/inadequacies) in relevant Acts, Policies, Guidelines and Working Procedures in Housing, Building Construction, and Urban Development Planning are as follows:

Details of amendments are attached to **Annex 1, Volume III**.

#### 4.1.1 Ancient Monument Preservation Act, 1956 (2013)

This act mentions the provisions for preserving ancient monuments by controlling the buying and selling of archaeological objects and excavation of the place of ancient monuments. This act provides authority to the Department of Archaeology to prepare, adopt and enforce through local authority the conservation Byelaws for the construction of buildings in the protected monumental zones.

Gaps: There is no provision for the protection of ancient human settlements, monument areas and traditional private buildings.

Analysis (Char Mahale) of this Act is attached to **Annex 1.1, Volume III**.

#### 4.1.2 Guthi Corporation Act, 1976 (2033)

This act was enacted to set up a Guthi Corporation to remove state trusts (Raj Guthi) from the authority of the Government of Nepal and place them under a corporation, and use the state trusts in a systematic manner, and it is useful, to make more effective and prompt provisions by amending and merging laws relating to Guthi with a view to keeping cordial relation between the people of various classes and economic interest and morality of the people.

Gaps: There is no provision of inventory of all Guthi land for safeguarding Guthi property and for controlling intrusion. Auctioning of Guthi land and conversion into Reiker are the major causes behind the loss of Guthi land.

Analysis (Char Mahale) of this Act is attached to **Annex 1.2, Volume III**.

#### 4.1.3 Town Development Act, 1988 (2045)

This Act provides the legal basis for the formation and operations of town development committees. It deals with the reconstruction, extension, and development of towns, including land-pooling projects.

Gaps: Physical planning is seen as a town development committee function, rather than a municipality task. It does not describe urban development terminologies.

Analysis (Char Mahale) of this Act is attached to **Annex 1.3, Volume III**.

#### 4.1.4 Kathmandu Valley Development Authority (KVDA) Act, 1988 (2045)

The KVDA handles preparing plans and executing them within Kathmandu valley.

Gaps: Overlapping mandates and duplication of functions among levels of governments in Kathmandu valley.

Analysis (Char Mahale) of this Act is attached to **Annex 1.4, Volume III**.

Note: - \* marked existing Policy, Act, Regulation and Working Procedure and Gaps are most relevant to local and provincial government for consideration

#### 4.1.5 The Ownership of Joint Housing Act 1997 (2054)

The Act puts forward the concept of group housing to promote the urban areas with higher densities.

Gaps: Approval process for joint housing is time-consuming including Initial Environmental Evaluation (IEE) / Environmental Impact Assessment (EIA) process. Approval is needed from multiple government agencies. This act does not address the demand for low-income housing. The act needs to be amended to include post-earthquake lessons.

Analysis (Char Mahale) of this Act is attached to **Annex 1.5, Volume III**.

#### 4.1.6 Building Act 2055 (1999)

This Act makes necessary arrangements for regularizing the construction work in order to keep the buildings safe from earthquakes, fires and other natural hazards.

Gaps: This act mentions that MoUD should monitor the implementation of building code in the constructed buildings permitted by local level. However, this provision does not seem to be implemented in real practice. For this a separate unit should be envisioned at MoUD to monitor the implementation of building code and to solve other issues regarding the lack of implementation of building code. This act should also add provisions to capacitate rural municipalities to implement building code and to make building permit process effective.

Analysis (Char Mahale) of this Act is attached to **Annex 1.6, Volume III.**

#### 4.1.7 National Urban Policy (NUP) 2007

The three aims of NUP are:

- i. To achieve a balanced national urban structure through proper guidance to development of and investment in the infrastructural facilities.
- ii. To raise the living standard of the urban residents through development of clean, secure, and economically vibrant urban environments.
- iii. To achieve effective urban management through institutional strengthening and legal empowerment of the local bodies, as well as through promotion of proper cooperation and coordination among the different institutions involved in urban development.

Gaps: Lack of clarity on how and when the aims will be achieved.

Analysis (Char Mahale) of this policy is attached to **Annex 1.7, Volume III.**

#### 4.1.8 Building Regulations 2066 (2010)

This regulation makes necessary arrangements for regularizing the construction work as per the Building Act 1999, to keep the buildings safe from earthquakes, fires, and other natural hazards.

Gaps: This act mentions that MoUD should monitor the implementation of building code in the constructed buildings permitted by local level. However, this provision does not seem to be implemented in real practice. For this a separate unit should be envisioned at MoUD to monitor the implementation of building code.

Analysis (Char Mahale) of this Building Regulations is attached to **Annex 1.8, Volume III.**

#### 4.1.9 National Adaptation Program of Action (NAPA, 2010)

It has found urban settlements and infrastructure as one of the six key themes and included promoting climate smart urban settlements and infrastructure as one of nine priority projects.

Gaps: Infrastructure investments in urban settlements are low.

Analysis (Char Mahale) of this NAPA is attached to **Annex 1.9, Volume III.**

#### 4.1.10 The National Shelter Policy, 2012 (2069)

For the first time, the new National Shelter Policy covered issues of squatter settlements and landless squatters. It also acknowledged that squatter settlements are increasing at a rapid pace.

Gaps: Implementation of the National Shelter Policy has been challenging.

Analysis (Char Mahale) of this policy is attached to **Annex 1.10, Volume III.**

#### 4.1.11 National Land Use Policy, 2015 (2072)

This policy was prepared with a vision to guide the proper management and use of land, along with the distribution of land in a fair manner, to achieve sustainable social, economic, and ecological development. The document outlines 15 policies. In terms of urban development, this policy has looked at supporting planned urbanization. The policy has also prioritized ensuring access by the urban poor to low-cost housing, housing finance, and income-generating activities.

Gaps: There is Lack of inter-departmental coordination, institutional fragmentation, and duplication of responsibility in government mechanism.

Analysis (Char Mahale) of this policy is attached to **Annex 1.11, Volume III.**

#### **4.1.12 Draft National Plan of Actions for Safer Building Construction, 2015 (2072)**

Draft NaPA on SBC proposes “vision 2035” to transform Nepal towards a disaster resilient nation in 20 years’ time, gives a strategic directive of different initiatives of Safer Building Construction across the nation. Draft NaPA on Safer Building Construction is not limited to National Building Code enforcement but aspires to mainstream all dimensions of Safer Building Construction in all levels of governance through internalization, adaptation and enforcement resulting into behavioral change in local community within 20 years timeframe. It has 4 policies and 10 strategies. It indicates 10 different streams of actions like:

1. Immediate Response to Recent Disasters
2. Policy and Regulation
3. Codes, Guidelines and Research
4. Risk Sensitive Land Use Planning and Building Bye Laws
5. Awareness, Education and Demand creation
6. Skill Development and Capacity Building
7. Physical Facility Support
8. Internalization, Adaptation and Enforcement
9. Retrofitting and Regeneration

Gaps: There are operational challenges and resources to implement the proposed plan of actions.

Analysis (Char Mahale) of this National Plan of Action is attached to **Annex 1.12, Volume III.**

#### **4.1.13 National Building Code (NBC) 203, 2015 (2072) Guideline**

It applies to helping earthquake resistant building construction especially for low strength masonry.

Gaps: Traditional masonry buildings with more than three stories are not allowed by the National Building Code (NBC). In historic settlements a common typology is three floors plus an attic which is a local building culture.

Analysis (Char Mahale) of this code is attached to **Annex 1.13, Volume III.** Note: - \* marked existing Policy, Act, Regulation and Working Procedure and Gaps are most relevant to local and provincial government for consideration

#### **4.1.14 National Reconstruction and Recovery Policy, 2015 (2072)**

The vision and guidelines for reconstruction and recovery are described in the government’s Reconstruction and Rehabilitation Policy. The recovery vision of the Government of Nepal is establishment of well-planned, resilient settlements and a prosperous society.

The guidelines for recovery and reconstruction set up the first goal of keeping the spirit of national unity, harmony and resilience experienced during rescue, relief, and search operations after the earthquake. Utilizing the ability and skill of the Nepalese people to a maximum degree in planning and implementing reconstruction is also part of this vision.

Gaps: Aims are not specific to use time proven traditional construction technology to preserve the identity of ancient settlements in sustainable way, and approaches for policy implementation are vague and not specific in promoting use of traditional construction technology, material, and skill.

Analysis (Char Mahale) of this policy is attached to **Annex 1.14, Volume III.**

#### **4.1.15 Third United Nations Conference on Housing and Sustainable Urban Development (Habitat III), A National Report, 2016**

The conference reinvigorated the global commitment to sustainable urbanization. It focuses on the implementation of a ‘New Urban Agenda,’ building on the Habitat Agenda of Istanbul in 1996. The six key areas the new report focuses on are urban demography; land and urban planning; environment and urbanization; urban governance and legislation; urban economy; and housing and basic services.

Gaps: Implementation challenges exist. There is limited technical ability to plan and manage cities. Slums and squatter resettlement is still a challenge.



Analysis (Char Mahale) of this national report is attached to **Annex 1.15, Volume III.**

#### **4.1.16 Post Disaster Recovery Framework (PDRF), 2016 (2073)**

PDRF documents the vision and strategic aims that guide recovery. The underlying visions of the GoN are well-planned, resilient settlements and a prosperous society through the recovery process, including safe structures, social cohesion, and access to services, livelihood support and ability building. PDRF has set five strategic recovery aims, which are a set of concrete, main recovery outcomes. Among the five strategic recovery aims, Strategic Recovery Objective 1 is to restore and improve disaster resilient housing, government buildings and cultural heritage, in rural areas and cities.

Gaps: The framework does not cover the restoration of ancient towns and settlements in a disaster resilient and sustainable manner.

Analysis (Char Mahale) of this framework is attached to **Annex 1.16, Volume III.**

#### **4.1.17 The Fundamental Construction Bylaws on Settlement Development, Urban Planning and Building Construction 2015 (2072)**

To be applied by all local government (Urban and Rural Municipalities); they will have to adjust the prevailing bylaws accordingly.

Gaps: Local government sees the Bylaws as an imposition by the central government rather than as guidance for proper design. There is a need to set up an effective system of planning and implementation guidelines for preserving local culture and traditional character of the historic townscape. 'Building bylaws' are only one of the many tools to implement urban conservation and development plans (including incentive and punishment for effective development control).

Analysis (Char Mahale) of this byelaw is attached to **Annex 1.17, Volume III.**

#### **4.1.18 National Urban Development Strategy, 2017 (2074)**

NUDS proposes 'Vision 2031: Balanced and prosperous national urban system'. It gives a strategic directive of urban development. NUDS is not limited to physical development but aspires to reach a qualitative vision for future cities that reflect the highest value of a society. It has 105 strategies. It acknowledges that provision of affordable, adequate, and safe housing is the aim of the urban housing sector.

Gaps: There are operational challenges and resources to implement the proposed strategies.

Analysis (Char Mahale) of this strategy is attached to **Annex 1.18, Volume III.**

#### **4.1.19 Local Government Operations Act (LGOA), 2017 (2074)**

The Act has paved a strong legal foundation towards institutionalizing legislative, executive, and quasi-judicial practice of the new local governments, restructured in 2017. The Act has stipulated several arrangements related to authorities, duties and responsibilities of local government, assembly meeting and working system, plan formulation and implementation, financial jurisdictions, and administrative structure. This Act describes the criteria to form urban municipalities or rural municipalities and their respective rights, duties, and responsibilities. It also details the list of rights of RM and UMs to form local laws and regulations as local governments (LGs).

Gaps: There are implementation issues. Also, there is a lack of human & financial resources to implement the Act in its entirety. Urban management abilities of local governments are limited, and their mandates are underfunded.

Analysis (Char Mahale) of this Act is attached to **Annex 1.19, Volume III.**

#### **4.1.20 Right to Housing Act, 2018 (2075)**

The Act empowers federal, provincial, and local governments to coordinate, prioritize and supply housing to those citizens and their families who do not own land and housing anywhere in Nepal and lack income and ability to manage it. Households may also include those displaced by natural hazards without any assets left. The Act enables Federal, Provincial and the Local governments to undertake "integrated settlement development" or if needed, to move settlements-which are remote to supply amenities or which are unsafe- to provision housing.

Gaps: There is less focus on safer settlements, livelihood issues and safety consideration in housing development.

Analysis (Char Mahale) of this Act is attached to **Annex 1.20, Volume III.**

#### 4.1.21 Land Use Act, 2019 (2076)

The Act's aim is to regulate land management and ensure sustainable use of land resources in the country. It stipulates formation of municipal land use bodies, acquisition of land use maps from the federal government, development of land use plans, and carrying out zoning process, land use change, land title update, land valuation and taxation. While developing the land use plan, the local governments will have to prepare separate plans for rural and urban areas.

Gaps: Land Use Council (LUC) vs. Municipalities. LGOA gives power to municipalities. The Act assigns all the three levels of governments to develop comprehensive land use plans and classifications based on purpose and other provisions. However, implementation by local governments is a challenge in terms of their ability and resources. It stays silent on cases of urban expansion areas. Despite the concurrent problem of land, no provisions for such scenarios have been suggested in the Act.

Analysis (Char Mahale) of this Act is attached to **Annex 1.21, Volume III.**

#### 4.1.22 Proposed National Urban Policy, 2019 (2076)

The three aims of the National Urban Policy are:

- i) To achieve a balanced national urban structure through proper guidance to development of and investment in the infrastructural facilities.
- ii) To raise the living standard of the urban residents through development of clean, secure, and economically vibrant urban environments.
- iii) To achieve effective urban management through institutional strengthening and legal empowerment of the local bodies, as well as through promotion of proper cooperation and coordination among the different institutions involved in urban development.

Gaps: Lack of clarity on how the aims are funded and executed.

Analysis (Char Mahale) of this policy is attached to **Annex 1.22, Volume III.**

#### 4.1.23 House Pooling Procedure, 2020 (2077)

The Ministry of Urban Development has formulated and implemented this procedure to manage uncontrolled urbanization and supply infrastructure and services in the ancient settlement and urbanizing areas.

Gaps: It includes land pooling provisions, which is very difficult to implement in already builtup areas with many small parcels. Engagement of private developer as practiced in many other countries and practices in urban renewal is missing in this procedure. It does not encourage local business and miss some of the detail parameters of house pooling.

Analysis (Char Mahale) of this working Procedure is attached to **Annex 1.23, Volume III.**

#### 4.1.24 Urban Development Bill, 2021(2078)

Ministry of Urban Development drafted this Urban Development Bill as an umbrella Act in Urban Development in Nepal. The bill is focused more on physical development planning of an urban area, not in comprehensive integrated urban development planning.

Gaps: There is no provision for gradual improvement of existing cities and planning and implementation system of new cities and determination of decision-making process. There is a lack of planning and implementation systems in physical, social, economic, and environmental aspects for urban development in the proposed bill.

Detailed revision of the Bill is attached to **Annex 1.24, Volume III.**

Policy recommendation are as follows:

| S.N. | Policies                                       | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Ancient Monument Preservation Act, 1956 (2013) | <ul style="list-style-type: none"> <li>Local government shall declare monument areas as heritage and conserved areas.</li> <li>Local government shall define such an area and publish public notice.</li> <li>People within the heritage area shall ask for permission to install public infrastructure, to park vehicles, to stick posters, to shoot movies, to organize festivals (other than cultural festivals) etc. with local government.</li> <li>Construction and addition in the structure shall be within the premise of heritage bylaws or else local government can ban or demolish such construction.</li> <li>Local government shall make inventory of heritage buildings (more than 100 years old), define its area, and publish public notice.</li> <li>The owner of the heritage buildings shall handle conserving the property. For this the local government shall supply 20-30% of the repair and maintenance cost and exempt property tax.</li> <li>Local government shall form a technical expert committee to conserve heritage buildings.</li> <li>Government of Nepal shall hand over the responsibility of preservation of ancient monuments and heritage settlements at province level to province government and at local level to local government.</li> <li>Local government shall buy the private heritage property after its evaluation if it is not conserved by the owner.</li> <li>Necessary arrangements shall be made by the concerned local government to prevent the misuse of the archeological or historically important monuments or temples under the care of the Government of Nepal.</li> <li>Local government shall preserve unique types of artefacts and monuments that show the importance of a certain period if the Department of Archeology fails or delays the preservation.</li> </ul> |
| 2    | Guthi Corporation Act, 1976 (2033)             | <ul style="list-style-type: none"> <li>Construction of houses and shops shall only be allowed in places where the premise of the temple is not adversely affected and the Jatra or festivals are not conducted.</li> <li>Guthi land shall be leased for a fixed period for agriculture or forestry.</li> <li>The inventory of Guthi land shall be managed and land use shall restrict its intrusion.</li> <li>Only those who have built a house on the Guthi land with their own capital for more than 10 years can have the land registered in their own name.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3    | Town Development Act, 1988 (2045)              | <ul style="list-style-type: none"> <li>Town development plan shall be defined as “A plan having well identified land use areas, integrated infrastructure development, environmental protection plan, disaster management and resilience plan, institutional capacity, and urban areas and urban expansion areas. This plan shall have key action plans, targets, expected achievements, implementation period and list of responsible bodies. The plan shall be approved by the local, provincial, and central government as per its requirement.</li> <li>Local planning shall be defined as a local area development plan implemented by the local level within its area. For example, land pooling project, bus Park and bus terminal construction plan, industrial park development plan, public infrastructure development plan, squatter settlement or slum improvement project, local disaster, and climate resilience plan etc.</li> <li>Integrated Urban Development Plan shall be defined as a clear identification of land use, integrated infrastructure development, environment protection, local disaster and climate resilience plan, social, economic, financial, and institutional ability development, etc.</li> <li>Integrated Infrastructure Development Plan shall be defined as a plan for the integrated development of basic infrastructures such as: transport, communication, drinking water, sewerage treatment and electricity distribution in a coordinated manner.</li> <li>The local level town development plan shall be approved by the Executive Officer of the local level concerned.</li> </ul>                                                                                                                                                                                                        |

| S.N. | Policies                                                                    | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | Kathmandu Valley Development Authority (KVDA) Act, 1988 (2045)              | <ul style="list-style-type: none"> <li>Keeping in mind the current need to face the growing population and the challenge of urbanization in the Kathmandu Valley, the role and responsibilities of the federal, provincial, and local levels, and participation of the valley dwellers, and cooperative sectors shall be clarified to ensure development of integrated infrastructure and services.</li> </ul>                                                                             |
| 5    | Ownership of Joint Housing Act 1997 (2054)                                  | <ul style="list-style-type: none"> <li>Include post-earthquake lessons by creating a provision of social housing planning and implementation to cater the need of urban poor.</li> </ul> <p>Approval process for joint housing shall be reorganized and systemized to step up the Initial Environmental Evaluation (IEE) / Environmental Impact Assessment (EIA) process.</p>                                                                                                              |
| 6    | Building Act 1998 (2055)                                                    | <ul style="list-style-type: none"> <li>Ammend this Building Act in consonant with the latest legal provision and federal system of administration</li> </ul> <p>To add provision to create a permanent committee for updating NBC</p>                                                                                                                                                                                                                                                      |
| 7    | National Urban Policy (NUP) 2007                                            | <ul style="list-style-type: none"> <li>Clarity on how and when the aims can be achieved must be spelled out.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                    |
| 8    | Building Regulations 2066 (2010)                                            | <ul style="list-style-type: none"> <li>Ammend this regulation in consonant with the latest legal provision and federal system of administration</li> <li>Include ToR of Building Code implementation monitoring unit and formats of monitoring and reporting of the building code. implementation in local government</li> </ul>                                                                                                                                                           |
| 9    | National Adaptation Program of Action (NAPA, 2010)                          | <ul style="list-style-type: none"> <li>GoN shall consider formulation of Value Capture Policy for infrastructure development.</li> <li>MoUD shall plan and implement the National Urban Development Strategy by mobilizing local resources and funds from bilateral and multilateral donor agencies.</li> </ul>                                                                                                                                                                            |
| 10   | The National Shelter Policy, 2012 (2069)                                    | <ul style="list-style-type: none"> <li>Central government shall develop implementation strategy, action plan, program, and projects.</li> <li>Provincial and local governments shall develop their own housing policy.</li> <li>Central and Provincial governments shall help local government to formulate plans for solving the issues of squatter settlements and landless squatters.</li> </ul>                                                                                        |
| 11   | National Land Use Policy, 2015 (2072)                                       | <ul style="list-style-type: none"> <li>Establish inter-departmental coordination mechanism and create conducive environment to function properly</li> <li>Institutional fragmentation, and duplication of responsibility in government mechanism shall be eliminated by defining role, responsibility, and obligation of each agency responsible by amending Land Use Policy.</li> </ul>                                                                                                   |
| 12   | Draft National Plan of Actions for Safer Building Construction, 2015 (2072) | <ul style="list-style-type: none"> <li>Rewrite this Draft National Plan of Action in consonant with the latest legal provision and federal system of administration and submit for approval from GON</li> <li>MoUD with support from MoFA (finance ministry) shall plan and implement National Plan of Action for Safer Building Construction by mobilizing local resources and fund from bilateral and multilateral donor agencies</li> </ul>                                             |
| 13   | National Building Code (NBC) 203, 2015 (2072) Guideline                     | <ul style="list-style-type: none"> <li>Number of Storey shall be up to three stories and an added attic floor. The maximum height (floor to floor) of a building shall not exceed 12 times the wall thickness at the superstructure.</li> <li>Shape shall be rectangular or square.</li> </ul>                                                                                                                                                                                             |
| 14   | National Reconstruction and Recovery Policy, 2015 (2072)                    | <ul style="list-style-type: none"> <li>In the reconstruction of the earthquake-affected ancient settlements, local materials, knowledge, skills, labor, and architecture shall be used to preserve the ancient settlements.</li> <li>Completely and partially damaged ancient heritage settlements, residential buildings, community and government buildings, ancient monuments and other structures shall be repaired, retrofitted, and reconstructed in an integrated manner</li> </ul> |

| S.N. | Policies                                                                                                            | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                     | by adopting right technology as per the local needs to be resistant to natural calamities.                                                                                                                                                                                                                                                                                                                                                           |
| 15   | Third United Nations Conference on Housing and Sustainable Urban Development (Habitat III), A National Report, 2016 | <ul style="list-style-type: none"> <li>Efforts must be taken to tackle slums and squatter problems by developing and implementing slums and squatter's area improvement plans by local government with full technical and financial support from central and provincial governments.</li> </ul>                                                                                                                                                      |
| 16   | Post Disaster Recovery Framework (PDRF), 2016 (2073)                                                                | <ul style="list-style-type: none"> <li>Financial help for housing recovery shall be provided through the banking system in instalments, which will be linked to the progress of construction. In general, house owners will need to conform to government-approved seismic standards. The house owners of heritage settlement will need to conform to government approved seismic standards and building byelaws for heritage settlement.</li> </ul> |
| 17   | The Fundamental Construction Bylaws on Settlement Development, Urban Planning and Building Construction 2017 (2074) | <ul style="list-style-type: none"> <li>Federal government shall develop guidelines for engagement of urban stakeholders in preparation of municipal development control and building construction byelaws.</li> <li>Local government shall prepare regulation of building in areas of urban development and cultural and natural scenic beauty and in rural areas.</li> </ul>                                                                        |
| 18   | National Urban Development Strategy, 2017 (2074)                                                                    | <ul style="list-style-type: none"> <li>MOUD with support from MoFA (finance ministry) shall plan and implement the National Urban Development Strategy by mobilizing local resources and funds from bilateral and multilateral donor agencies.</li> </ul>                                                                                                                                                                                            |

| S.N. | Policies                                            | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 19   | Local Government Operations Act (LGOA), 2017 (2074) | <ul style="list-style-type: none"> <li>Heritage area means the place where an ancient monument or ruins of an ancient settlement exists or human settlement which is inhabited more than 100 years ago.</li> <li>Tourist area means an area that attracts tourists with naturally beautiful places, mountains, hills, plains, forests, forests, lakes, rivers, waterfalls, and places with their own originality and culture.</li> <li>Urban area means an area with a population of at least 5,000, population density of 10 persons per hectare and 50% of the population above 15 years of age engaged in occupations other than agriculture. Similarly, dense areas, areas with houses on both sides of roads, and areas having basic urban infrastructures are termed as urban areas.</li> <li>Additionally, in the definition of urban area, different land use areas of the city, various levels of urban planning, urban infrastructure and service facilities, overall plan, local level planning, as well as rural areas within the village, rural development plan, rural infrastructure and service facilities shall be added.</li> <li>Evaluate urban infrastructure and services of Municipalities, Sub-Metropolitan City, and Metropolitan City and add the required urban infrastructure and services within 2 years by GoN shall be added in the Act.</li> <li>From the date of issuance of this Act, the province government shall identify the cultural or tourist areas within five years and declare them in the gazette of the province government.</li> <li>Definition of heritage or tourist area shall be added.</li> <li>Requirements for defining heritage areas are: <ul style="list-style-type: none"> <li>The location of temples, monuments and other structures of archeological importance or human settlements which are more than one hundred years old, which are important from the point of view of history, art, science, and architecture.</li> </ul> </li> <li>Requirements for defining tourist areas are: <ul style="list-style-type: none"> <li>Areas that attract or can attract tourists such as: mountains, hills, Terai forests, farms, rivers, cliffs, waterfalls, lakes, natural landscapes, cultural sites and settlements etc.</li> </ul> </li> <li>Demarcation of boundary of heritage or tourist area: <ul style="list-style-type: none"> <li>The demarcation of heritage or tourist area shall be done by the local level in collaboration with the concerned expert group</li> <li>The perimeter occupied by the cultural or tourist area shall be considered as the border.</li> <li>To keep such boundaries, aerial maps, satellite images and historical documents prepared in 1970 or earlier shall be used as the basis.</li> </ul> </li> <li>Declaration of heritage or tourist area: <ul style="list-style-type: none"> <li>Subject to the grounds mentioned above, on the recommendation of the concerned expert group, the executive of the concerned municipality may declare the cultural and tourist areas as preserved areas and publish public notice.</li> <li>A person who is not satisfied with the declaration may lodge a complaint with the Province Government within thirty-five days of the posting of such notice.</li> <li>The State Government shall make the final decision on the complaint and publish the demarcation of such sites in the gazette.</li> <li>People within the heritage area shall ask for permission to install public infrastructure, to park vehicles, to stick posters, to shoot movies, to organize festivals (other than cultural festivals) etc. with local government.</li> <li>People within the heritage area shall ask for permission to construct a house or to alter the house with local government. As an incentive, the municipality shall make provision for free design and drawings and exempt tax.</li> <li>Decisions of the local government on conservation, promotion, development and management plan, disaster risk reduction and management plan and other development plans should be published in the local gazette.</li> </ul> </li> <li>Within five years of the commencement of this Act, GoN shall prepare detailed guidelines for formulating and implementing policies, laws, norms, provincial and local level development projects related to urban development, settlement development and rural development.</li> </ul> |



| S.N. | Policies                                    | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|      |                                             | <ul style="list-style-type: none"> <li>Local level shall formulate policy, law, criteria guidelines within 2 years of preparation of all above-mentioned documents.</li> <li>Based on the risk sensitive land use plan, disaster risk areas and safe areas are found and the settlements in the disaster risk areas are protected by engineering work to minimize the risk if possible. Where it is not possible, resettlement is done to a safer location.</li> <li>Definition of periodic and annual plans as well as the medium- and long-term plans shall be added.</li> <li>Within 5 years from the commencement of this law by Central government and within 3 years of formulation of this law by Province government, implementation shall be done by local government within 2 years.</li> <li>If the local level does not formulate or does not formulate policies, laws, and norms as per the policies, laws and norms implemented by the Central and the Province government, the provincial government shall formulate local level policies, laws, and norms by following the policies and laws of the Central governments.</li> <li>Rural municipality and municipality should make long-term, medium-term, periodical, and annual plans for local level development by making at least 2 and at most 3 options and consider the most suitable plan as possible.</li> <li>Integrated property tax shall be levied on privately owned houses built in the local architecture reflecting originality within the rural municipality or municipality.</li> </ul> |
| 20   | Right to Housing Act, 2018 (2075)           | <ul style="list-style-type: none"> <li>The current land and land title distribution program launched by high level land distribution commission shall continue by mobilizing support from local governments and other concerned government agencies.</li> <li>While undertaking "integrated settlement development" or moving settlements- which are remote to supply amenities or which are unsafe- to supply housing, Federal, Provincial and the Local governments action may be based on NRA's Lessons learned in planning and developing integrated settlement and relocation of unsafe settlements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 21   | Land Use Act, 2019 (2076)                   | <ul style="list-style-type: none"> <li>Land use planning and best use of available land shall be a part of the municipal development plan which shall be clearly defined.</li> <li>Central and provincial governments shall assist local government in municipal development of land use planning and implementation by formulating land use planning implementation guidelines and building ability of the municipal staff and institutions in specified period.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 22   | Proposed National Urban Policy, 2019 (2076) | <ul style="list-style-type: none"> <li>Article 6) Vision: "Prosperous, vibrant, beautiful and inclusive city" shall be replaced by "balanced, inclusive and prosperous urban system at the federal, provincial and local levels".</li> <li>The productive of subsection 8.3 of section 8 shall be replaced by "uplifting".</li> <li>Strategy 10.4.4: shall be changed as follows: To run urban development programs to minimize the damage caused by natural calamities and human induced disasters and climate change.</li> <li>11.1.3 E) Hazard shall be replaced by "disaster".</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 23   | House Pooling Procedure, 2020 (2077)        | <ul style="list-style-type: none"> <li>The "public use within the planning area" of section 2c shall be removed and replaced with "collective purpose within the construction planning area".</li> <li>Article 2e) after the ownership, "deducting certain contribution part" shall be removed.</li> <li>The commercial ground floor of Article 2 f) shall be further changed by "or other purposes similar to residential purposes".</li> <li>Article 2 g) integrated shall be replaced by "joint housing".</li> <li>To develop the unorganized and unplanned settlements as a well-planned and managed settlement, all existing scattered buildings, and structures except monuments within the project site should be demolished and reconstructed. These</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| S.N. | Policies                           | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                    | <p>buildings shall be evaluated, and equivalent property shall be returned to the former owner.</p> <ul style="list-style-type: none"> <li>• To recover the cost of the project, the remaining housing unit and any other developed infrastructure or facility shall be sold or leased.</li> <li>• Transfer of ownership of various housing units and facilities of developed settlements and buildings to the former owners or to the new buyer selected from the competition shall be done as per the principle of house pooling.</li> <li>• When returning a new residential unit to the former property owner, one or more flats of reasonable size based on the area of the earlier house and land, number of rooms, connected facilities, number of family members, etc. shall be returned</li> <li>• If a family is unable to get back even a one-room housing unit (flat), a collective housing unit with kitchen, toilet, and common room to be used collectively shall be planned to include other such families. However, in such a case, if the collective use is not possible, the householder will not be removed from his / her earlier place of residence. In such a case, the house owner can pay for the apartment in installments.</li> <li>• Some property to be acquired through negotiation following the prevailing law if the owner is dissatisfied and obstructs the operation of the project.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 24   | Urban Development Bill, 2021(2078) | <ul style="list-style-type: none"> <li>• No definition of urban area, urban plan, and urban infrastructure, urban regeneration, regeneration area, urban regeneration plan etc. exists in Urban Development Bill. However, some definitions exist in Urban Development Policy, 2007. Hence, some definitions are proposed based on the Urban Development Policy:</li> <li>• Urban planning" is defined as any physical, social, economic, and environmental program, activity or project conducted to provide physical, social and economic basic urban infrastructure and services in the urban area to achieve social, economic, environmental or other objectives.</li> <li>• Basic urban infrastructure and service facilities" is defined as physical, social, and economic infrastructure such as drinking water, sewerage, rainwater drainage, roads, waste management, electricity distribution, communication system, etc.</li> <li>• Urban Development Bill is more focused on physical infrastructure rather than overall urban development. It misses the linkage between development and physical infrastructure.</li> <li>• In the Bill, local area development projects are limited to guided land development, land pooling, site, and services and house pooling. It misses to include urban renewal, heritage settlement regeneration, industrial park, City Park, bus terminal, river front development, squatter and slum area improvement and development and community infrastructure development projects etc.</li> <li>• Gazette publication for updated plan is mentioned. However, the Bill does not mention the publication of approved plan for public.</li> <li>• Details are missed in Annex.</li> <li>• Bill proposes to define and prioritize the project planning components through regulation which shall be defined in the bill.</li> <li>• Land use plan shall be according to land use Act.</li> </ul> |
| 25   | Draft Urban Regeneration Act 2021  | <ul style="list-style-type: none"> <li>• The proposed urban development Bill, 2021 does not cover urban regenerations as a local area plan. Hence, an Urban Regeneration Act is proposed to support urban regeneration projects in Nepal.</li> <li>• The purpose of this draft Act is to contribute to enhancement of the quality of life of the people, such as the expansion of the base for sustainable growth of historic cities, towns and settlements, improvement of competitiveness of those cities/towns, and recovery of local community, by strengthening the public role and support for the economic, social, and cultural revitalization of historic cities, towns, and settlements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| S.N. | Policies                                | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26   | Draft Urban Regeneration Guideline 2021 | <ul style="list-style-type: none"> <li>The proposed urban development Bill, 2021 does not cover urban regenerations as a local area plan. The settlement development and building construction guideline covers the monument conservation but it requires legal tools to implement properly. Urban regeneration should be a part of an umbrella Act. Hence, a guideline for urban regeneration is developed which mentions the vision, goals, aim, and guiding principle together with the overall process of urban regeneration project implementation.</li> <li>The proposed Urban Regeneration guideline supplies guidance on implementation of urban regeneration in an integrated manner. It supports to revitalize traditional settlements of Nepal damaged by the earthquake in a planned and participating manner keeping the urban fabric intact. It supports the rebuilding of traditional houses to make them safe and convenient while conserving local heritage. This guideline promotes conservation and use of public spaces for community services. It stimulates local economy by promoting entrepreneurship, tourism and retail commercial activities that are compatible to heritage conservation and eventually strengthens local ability of municipality and community groups in managing urban growth.</li> </ul> |

## 4.2 Reconstruction: NRA policy gaps

NRA has developed 22 Guidelines/standards/Documents over the years to address the various issues that appeared during reconstruction process in addition to 13 technical standards. Out of 22 Guidelines/Documents, 5 are related to land, 3 are related to grant /finance, 1 is related to heritage and the rest of documents are related more to administrative or managerial aspect.

### 4.2.1 Reconstruction of Earthquake Affected Structures Act, 2072 (2015)

This Act sets up/empowers National Reconstruction Authority (NRA) to complete the reconstruction work.

Gaps: It does not separately define rural and urban context and does not include the components of DRRM.

Note: - \* marked existing Policy, Act, Regulation and Working Procedure and Gaps are most relevant to local and provincial government for consideration

### 4.2.2 Reconstruction Rules and policies for earthquake affected Structures, 2072 (2015)

This policy defines policies and rules to implement Reconstruction of Earthquake Affected Structures Act, 2072.

Gaps: It does not separately define rural and urban context and does not consider urban housing needs. It does not clarify the terminologies as well.

### 4.2.3 Procedure for the landownership registration in the name of earthquake-affected persons, 2072 (2015)

This procedure simplifies the process and rules for registration of land in the name of EQ affected person including traditionally inhabited land (without ownership).

Gaps: It does not clarify the process of six categories of lands to be registered including Guthi (on unpaid revenue, building permit and ownership certificate).

### 4.2.4 Land Acquisition procedure for reconstruction of the earthquake affected structure, 2072 (2015)

This procedure eases the acquisition of land for integrated settlement, re-settlement, and re-location projects.

Gaps: It allows flexibility to change the use of bought land which leaves room for the misuse of its original aim.

### 4.2.5 Environmental impact assessment procedure for the reconstruction of earthquake affected structure, 2072 (2015)

This procedure supplies provision for conducting Initial Environmental Examination (IEE) and Environment Impact Assessment (EIA).

Gaps: It does not cover environmental assessment in general for core-city/ dense traditional areas and small plots.

CRS and National Housing and Settlements Resilience Platform (NHSRP)

#### **4.2.6 Public Procurement Procedure for Reconstruction, 2072 (2015)**

This procedure supplies a faster transparent, systematic, economical procurement process for reconstruction.

Gaps: It mentions that the procurement needs to be made by agencies listed in VAT and PAN but does not make it mandatory.

#### **4.2.7 I/NGOs mobilization procedures for rehabilitation and reconstruction, 2072 (2015)**

This procedure makes the mobilization, coordination, support, and roles of I/NGOs transparent, simple, faster, and integrated.

Gaps: It does not differentiate between urban and rural context and does not have clear or strong provision to make I/NGOS more accountable.

#### **4.2.8 Private housing reconstruction grant distribution procedure, 2073 (2016)**

This procedure makes the reconstruction and retrofitting grant process smooth, easier, and systematic for affected beneficiaries.

Gaps: It consists of a complex procedure of detailing the ownership process (Ansa Banda).

#### **4.2.9 Procedure for Relocation and rehabilitation of the vulnerable settlements, 2073 (2016)**

This procedure makes necessary arrangements to the vulnerable settlements and households for relocation and rehabilitation.

Gaps: It lacks criteria for prompt geo-hazard assessment to move and does not clarify relocation criteria in urban context.

#### **4.2.10 NRA Procedure for the legal appeal, 2073 (2016)**

This procedure eases hearing grievances, complaints and appeals filed related to reconstruction.

Gaps: It does not simply the issues such as: complex documentation procedure, lack of people's awareness about legal appeal procedure etc.

#### **4.2.11 Procedure for School reconstruction, 2073 (2016)**

This procedure eases the planning and completion of school infrastructures damaged by the 2015 earthquake.

Gaps: This procedure focuses on reconstruction through the School Management Committee (SMC) but lacks provision to assess and capacitate SMC for overall management. It relies on DLPIU (Education) engineers. It does not consider DRR related activities such as retrofitting.

#### **4.2.12 Guideline on training operation and management, 2073 (2016)**

This guideline supplies provision to conduct training in a planned and organized manner to develop the required workforce for reconstruction and rehabilitation of earthquake affected structures.

Gaps: It does not consider post-training monitoring of employment status of trainees; managing roster of trainers in municipality; skill certification by Council for Technical Education and Vocational Training (CTEVT) and complete training approach.

#### **4.2.13 Guidelines on Community Reconstruction Committees (CRC), 2073 (2016)**

This guideline enhances local community participation and ownership in private housing reconstruction.

Gaps: This guideline was developed when there was no local government so there is no clear guidance, authority, and awareness to Palika on formation of CRC and its role. Ward and palika representatives are not clear about their own and CRC member's role which is the reason CRC vision could not be effectively implemented.

#### **4.2.14 Standards on habitable land purchase for earthquake affected beneficiaries, 2074 (2017)**

To define procedures to buy habitable land for landless beneficiaries.

Gaps: These standard lacks information about the process of land purchase at municipality and ward office. This standard applies to the rural context, not the urban/ peri-urban context where land cost is remarkably high. Completely moves some of the landless beneficiaries to new settings.

#### **4.2.15 Integrated Procedure for the interest on the subsidized loan, 2074 (2017)**

This procedure supplies several types of subsidized loans to eligible EQ-affected families for housing reconstruction.

Gaps: It does not cover/simplify the issues such as: complex loan application paperwork, high interest rate to apply loan for an amount more than 5 lakhs etc.

#### **4.2.16 Reconstruction Grievance management procedure, 2074 (2017)**

This procedure addresses the private house reconstruction grievances of listed beneficiaries after inspection of house.

Gaps: No units are dedicated to redressing the grievance.

#### **4.2.17 Procedure for finding beneficiaries under vulnerable category, 2074 (2017)**

This procedure defines the four categories for vulnerable beneficiaries' identification and support.

Gaps: Urban poor and marginalized communities are not considered for vulnerable categorization. Excessive cost of construction in urban areas adds to the vulnerability, which is not considered in the guideline.

#### **4.2.18 Procedure for development of integrated settlement, 2075 (2018)**

This procedure eases development of land and settlement, as per need, in an integrated way in the earthquake affected districts with regards to long-term housing provisions.

Gaps: It duplicates procedure with 'Resettlement of vulnerable settlement and integrated settlement development related procedure 2075' from MoUD. It does not consider environmental impact assessment and justification; livelihood; integrated housing / apartment housing concept for urban areas; and gender and protection issues in design and implementation.

#### **4.2.19 Procedure for the reconstruction of Gumba, Stupa, 2075 (2018)**

This procedure eases the process of conservation, restoration and reconstruction of Gumbas/Bihars/Stupas affected by earthquake.

Gaps: No gaps related to urban issues.

#### **4.2.20 Procedure for the reverification and resurvey, 2075 (2018)**

This procedure eases the grievance address process through reverification and resurvey by NRA technical team.

Gaps: It does not mention criteria of registering actual data on the number of grievances. Once the survey team are on the field, the number of grievances will increase.

#### **4.2.21 Procedure for supplying the grant to the structure constructed before the mobilization of Technical Team, 2076 (2019)**

This procedure makes provision of supplying the remaining grant amount to structures constructed before mobilization of technical team.

Gaps: It considers only those houses who received building completion certificate from municipality before the mobilization of technical team. It does not consider the ones in receiving the completion certificate or those who have not applied for the building permit.

#### **4.2.22 Procedure for reconstruction of single or Multiple Ownership in Heritage Settlement, 2076 (2019)**

This procedure eases housing reconstruction in heritage settlements while conserving traditional identity and architecture through grant support.

Gaps: Conflicting provision of municipality to define boundary from Local Level Operations Act 2072 and UNESCO guideline. No provision of vertical division of house. Lengthy process to receive more grant that demotivate beneficiaries to reconstruct house as per conservation rules and regulations. Grant amount is not sufficient for heritage conservation.

Analysis of NRA policies is attached to **Annex 2, Volume III**.



## 5 LEARNINGS

International experience in post-earthquake housing reconstruction and recovery and national experiences of urban planning projects were studied to understand the international practice of reconstruction and recovery and national practice of urban planning.

### 5.1 International Experience in Post-Earthquake Housing Reconstruction and Recovery

Many countries across the world have faced disasters due to multi-hazards. They have adopted various approaches to address the reconstruction and recovery issues. Some of them are as follows:

- After Kobe Earthquake 1995, Japan arranged interest free long-term loan. It managed the cost of reconstruction through the sale of more floors. Private house reconstruction was the responsibility of individuals. Japan linked reconstruction with their long-term planning system which made the post-earthquake operation more sustainable.
- After Lijiang Earthquake 1996, China applied owner driven and community engagement reconstruction approach. Local Government supplied grants covering 25 to 30% of total cost of reconstruction according to the level of damage.
- After Marmara earthquake 1999, Turkey adopted government subsidy-based reconstruction system. Low-cost apartments, housing and employment opportunities were provided. The Turkish Government created a Turkish Catastrophe Insurance Pool to transfer risk from the public to the private sector.
- After Gujarat earthquake 2001, India adopted owner driven reconstruction with partial relocation intervention. The government adopted a policy of supplying minimal housing. In rural areas a largest grant of Indian Rupees 90,000 (US\$2,117) and in urban areas, a largest grant of Indian Rupees 175,000 (US\$4,117) per destroyed house was provided by the government.
- After Aceh Tsunami and Earthquake 2004/2005, Indonesia adopted community-based implementation. The ministerial-level Reconstruction and Rehabilitation Agency implemented reconstruction through the contractors. It supplied grants based on the level of damage.
- After East Japan Earthquake 2011, Japan adopted public-private partnership in supplying housing plots for reconstruction of damaged houses. It applied land readjustment approach.
- After Christchurch Earthquake 2010/2011, New Zealand adopted insurance led model.

The details of International Experience in Post-Earthquake Housing Reconstruction and Recovery is attached to **Annex 6, Volume II**.

### 5.2 National Experiences of Historic City/Town Development projects

Some national urban development projects in Nepal are also studied. They have adopted various approaches to address urban development issues. Some of them are as follows:

#### 5.2.1 Bhaktapur Development Project (BDP) (1974 -1986), Bhaktapur:

Bhaktapur is the smallest town of Kathmandu Valley. It lies in a former trade route between India and Tibet. Most buildings in Bhaktapur are built in vernacular Newari style. BDP targeted to conserve and restore the historic environment of Bhaktapur keeping its urban fabric intact (Parajuli, 1992) (Dorji, 2019).

#### 5.2.2 Bandipur Eco-Cultural Tourism Project (BECTP) 2005-2006 AD, BANDIPUR, Tanahun.

Bandipur is a hilltop settlement and a municipality in Tanahun District, (Gandaki Zone) of Nepal. According to the National Census 2011, it had a population of 15591 people living in 3750 individual households. The BECTP focused on one aspect of urban renewal i.e., tourism promotion, to revive the town from great decline in the commercial trade after 1970, when it termed to have turned into a semi-ghost town (Dorji, 2019).

#### 5.2.3 Pro-poor Urban Regeneration Pilot Project of Lalitpur Metropolitan City

Pro-poor Urban Regeneration Pilot Project is a project funded by Japan Social Development Fund (JSDF) and administered by the World Bank. The aim of the project was to contribute to improving the living conditions of poor and vulnerable households in selected wards of the historic core of Lalitpur Metropolitan City through urban regeneration. The urban regeneration activities included improvement of infrastructure and services, conservation of local cultural heritage, and improvement of livelihood (LMC, 2017).

## 5.2.4 Pilanchhe Regeneration Project (2016-still ongoing)

Pilanchhe is a neighbourhood in Patan, Lalitpur Metropolitan city. Pilanchhe lies in a ten minutes' distance from Tribhuvan International Airport. It consists of 7 chowks including 110 houses out of which 82 were totally damaged by the 2015 Gorkha Earthquake. Pilanchhe project was started by the Maya Foundation and the locals.

Inventory of Past Urban Development Projects in Nepal is attached to Error! Reference source not found. Details of some Urban Development Projects in Nepal are attached to **Annex 8, Volume II**.

## 5.3 Issues and Recommendations of Urban Recovery

According to Proposed Urban Recovery Strategy, then conducted KII have shown the following issues:

- Access to finance
- Land Issues
- Compliance to Heritage Norms
- Communication
- Socio-Technical Assistance (STA) in Reconstruction
- Retrofitting in Urban Areas
- Help to Vulnerable

### 5.3.1 Access to Finance

The key issues related to access to finance are- lack of affordability, difficulty accessing grant from NRA, less access to soft loan/loans and lack of ability to pay the loan back.

**Table 16: Summary of issues and recommendations related to access to finance**

| Major Issues                         | Causes/Problems                                                 | Recommendation                                                              | Remarks                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of affordability                | High cost of urban housing reconstruction                       | Design and Promote Low-Cost Housing                                         | Examples from Nepal: Janta Awas<br>Examples from India: Jawaharlal Nehru Awas, Indira Gandhi Awas                                                                                                                                                                                                          |
|                                      |                                                                 | Promote Community housing                                                   |                                                                                                                                                                                                                                                                                                            |
|                                      |                                                                 | Promote Incremental Housing                                                 |                                                                                                                                                                                                                                                                                                            |
| Difficult to access grant from NRA   | Insufficient Grant Amount                                       | It should be calculated practically                                         |                                                                                                                                                                                                                                                                                                            |
|                                      |                                                                 | Involve Private Sectors                                                     |                                                                                                                                                                                                                                                                                                            |
|                                      | Difficulty in accessing the grant/delayed distribution of grant | Establish assisting committee at local level                                |                                                                                                                                                                                                                                                                                                            |
| Less access to soft loan/loans       | High interest rate                                              | Develop revolving fund/ CSR fund mechanisms to supply low interest loan     | The profit of A, B, C and D class banks is around 1 kharab (100 billion). Last year this fund was used for COVID. We can plan to create a revolving fund for reconstruction based on it. (LMC Mayor)<br>GoN has its own CSR fund use policy and focus on health and education rather than housing for poor |
|                                      |                                                                 | Ministry of finance and Nepal Rastra Bank should ease low interest loan     |                                                                                                                                                                                                                                                                                                            |
|                                      |                                                                 | NRB should take low interest loan from World bank or ADB backed up by GoN   |                                                                                                                                                                                                                                                                                                            |
|                                      | Short term loans                                                | GoN should ease long term loans                                             |                                                                                                                                                                                                                                                                                                            |
|                                      | Collateral requirement cannot be met                            | NRB should revisit collateral criteria                                      |                                                                                                                                                                                                                                                                                                            |
|                                      |                                                                 | Micro finance supplies collateral free loan up to 5 lakhs                   |                                                                                                                                                                                                                                                                                                            |
| Lack of ability to pay the loan back | No saving                                                       | GoN should ease to build low-cost house for (brick in mud masonry) for poor | The soft loan with 3% interest loan was approached by rich people rather than urban poor in LMC                                                                                                                                                                                                            |
|                                      |                                                                 | Supply long term loan (20-30 years)                                         |                                                                                                                                                                                                                                                                                                            |
|                                      | High income source required for EMI                             | Build the ability of local, so that they can earn and pay back              |                                                                                                                                                                                                                                                                                                            |

### 5.3.2 Land Issues

Land issues include- lack of land ownership certificate, multiple-ownership/small plots and landless population. CRS and National Housing and Settlements Resilience Platform (NHSRP)

**Table 17: Summary of issues and recommendations related to land**

| Major Issues                       | Causes/Problems                                                                                                                           | Recommendations                                                                                                                           | Remarks                                                                                                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Lack of land ownership certificate | Difficulty in obtaining building permit and/or loan                                                                                       | Develop policies to support                                                                                                               | LMC supplied 1 lakh to reconstruct house in heritage area; but people could not get it due to the lack of land certificate |
|                                    |                                                                                                                                           | Abhilekhikaran by municipalities is needed                                                                                                |                                                                                                                            |
|                                    | Guthi land occupied by landless                                                                                                           | Mohi's right should be reserved, land used for private purpose should be privatized and landless occupying Guthi land should be relocated | As per MuAN After 2052, Mohiyani right has been ended                                                                      |
| Multiple-Ownership/small plots     | Conflict in property distribution along siblings                                                                                          | Apartment System/Community Building Concept                                                                                               | Ms. Sangita Shrestha's family house is an example of owners as shareholders                                                |
|                                    |                                                                                                                                           | Owners as share holders                                                                                                                   |                                                                                                                            |
|                                    |                                                                                                                                           | Single ownership policy is needed                                                                                                         |                                                                                                                            |
|                                    | Replotting of a land (less than 2 Ana 2 Paisa)                                                                                            | Distribution of different property (land, building) among siblings                                                                        |                                                                                                                            |
|                                    | Vertical division of land                                                                                                                 | House Pooling to combine smaller houses                                                                                                   |                                                                                                                            |
| Landless                           | No land ever/ landless due to widening of road                                                                                            | Supply support from Land Related Problem-Solving Commission                                                                               | Dhulikhel municipality has segregated 1 crore 40 lakhs budget for landless.                                                |
|                                    | Government lack land for development or resettlement                                                                                      | Land bank concept should be incorporated where everyone should deposit the unused land in the bank and GoN should manage its use          |                                                                                                                            |
|                                    | The budget segregated by Commission for solving land related problems (2 lakhs per household) is not adequate to buy a plot in urban area | The amount should be calculated and contextually                                                                                          |                                                                                                                            |

### 5.3.3 Compliance to Heritage Norms

Heritage issues include Non-compliance to heritage norms and bylaws, more cost for keeping vernacular architecture, lack of skilled human resource, lot of resource needed to conserve a heritage settlement and heritage conservation threatened by development.

**Table 18: Summary of issues and recommendations related to compliance with Heritage Norms**

| Major Issues                     | Causes/Problems                                                                                                                                      | Recommendations                                                                                                                    | Remarks                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-compliance to heritage norms | Heritage bylaws are different than other urban bylaws<br>Difficult to conserve a heritage settlement                                                 | Supply technical drawings to keep slope roof and cater water tank and solar panels in roof (seen slope + hidden flat roof concept) | Khokana is in a verge of being enlisted in the UNESCO world heritage site, the slope roofs should be kept as needed.                                                     |
|                                  | Reconstruction in heritage area is a problem<br>Bylaws restrict building height & height allowed by bylaws could not fulfil the owners' requirements | Supply design and drawings from municipality                                                                                       | Design and Drawings by LMC comply to bylaws, but the requirement of house owners was not fulfilled and hence, they did not come to the ward for free design and drawings |
|                                  | People are unaware of the importance of conservation                                                                                                 | Develop vision/ overall plan to aware supported by policies from provincial & central Government                                   |                                                                                                                                                                          |
|                                  | Listing as UNESCO world                                                                                                                              | Zone/ define heritage area                                                                                                         |                                                                                                                                                                          |

| Major Issues                                               | Causes/Problems                                                                | Recommendations                                                                                                                    | Remarks                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | heritage is a complex process                                                  | There should be different National policies for UNESCO world heritages and local level heritages                                   |                                                                                                                                                                                                                                                                                                     |
| Non-compliance to the heritage bylaws                      | Difficult to conserve a heritage settlement                                    | Supply technical drawings to keep slope roof and cater water tank and solar panels in roof (seen slope + hidden flat roof concept) | Khokana is in a verge of being enlisted in the UNESCO world heritage site, the slope roofs should be maintained as needed.                                                                                                                                                                          |
| Additional cost for maintaining vernacular architecture    | Traditional materials are expensive                                            | Exempt tax to encourage conservation                                                                                               | In KMC, NPR 500 per square feet based on the façade area is being planned (which is NPR 2 lakhs to 2 lakhs 50 thousand<br>In Bhaktapur 35% of reconstruction fund is provided as grant<br>In Sankhu, NPR 1 lakhs is supplied irrespective of the façade size<br>In LMC the grant is not yet decided |
|                                                            |                                                                                | Grant is needed for conservation; however, Restoration should be prioritized over reconstruction                                   |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Grant system should also support restoration more than reconstruction or transformation                                            |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Grant should be supplied based on the percentage of the reconstruction cost                                                        |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | trading of traditional building material should be managed (use local material rather than importing them)                         |                                                                                                                                                                                                                                                                                                     |
| Lack of skilled human resource                             | Transfer of skill, technology and terminologies has been significantly reduced | Train local people for sustainability                                                                                              |                                                                                                                                                                                                                                                                                                     |
| Lot of resource required to conserve a heritage settlement |                                                                                | Involve Private Sector                                                                                                             |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Supply vision to locals                                                                                                            |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Supply an overall plan                                                                                                             |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | engage relevant students to promote conservation                                                                                   |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Utilize innovative technologies.                                                                                                   |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Educate on the difference between terminologies (restoration, regeneration, reconstruction etc.)                                   |                                                                                                                                                                                                                                                                                                     |
|                                                            |                                                                                | Give recognition to heritage buildings                                                                                             | e.g., In Europe and US, over 50 years old buildings are given a recognition of heritage building                                                                                                                                                                                                    |
| Heritage conservation threatened by development            | Widening of road calls for demolition of heritage buildings                    | Alternative roads should be proposed                                                                                               | UN's Sustainable Development Goal prioritizes conservation                                                                                                                                                                                                                                          |

### 5.3.4 Communication Strategy

Issues related to communication strategy are Lack, inadequate, early flow of information.

**Table 19: Summary of issues and recommendations related to communication strategy**

| Major Issues                                   | Causes/Problems              | Recommendations                                            | Remarks |
|------------------------------------------------|------------------------------|------------------------------------------------------------|---------|
| Lack, Inadequate, Untimely flow of information | Gap in conveying information | Local level should give information                        |         |
|                                                |                              | A working team/committee should be set up at a local level |         |
|                                                |                              | Clarity of responsibilities of 3 tiers of GoN              |         |

### 5.3.5 Socio-Technical Assistance (STA)

Issues related to socio-technical help are difficulty in obtaining a building permit and lack of provision of STA.

**Table 20: Summary of issues and recommendations related to socio-technical help**

| Major Issues                            | Causes/Problems                                                               | Recommendations                                                                                                                     | Remarks                                           |
|-----------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Difficulty in obtaining building permit | Lack of information on Building Permit Process (BPP), BPP was time- consuming | Review and update building code and bylaws as per context                                                                           | Pilanchhe took 6 months to obtain building permit |
|                                         | Non-compliance of Bylaws and building code                                    | Supply socio technical help                                                                                                         |                                                   |
| Lack of provision of STA                | Insufficient technical workforce                                              | Supply a socio-technical help committee based at local level, ward level is more preferred<br>Supply specific STA for heritage area |                                                   |
|                                         | Communities unaware about STA                                                 |                                                                                                                                     |                                                   |

### 5.3.6 Retrofitting in Urban Areas

Issues related to retrofitting are: Low knowledge, awareness, and trust in retrofitting.

**Table 21: Summary of issues and recommendations related to retrofitting in urban areas**

| Major Issues                                        | Causes/Problems           | Recommendations                                                | Remarks |
|-----------------------------------------------------|---------------------------|----------------------------------------------------------------|---------|
| Low knowledge, awareness, and trust in retrofitting | Complex system/technology | Increase awareness on importance & feasibility of retrofitting |         |
|                                                     |                           | Supply different retrofitting guidelines as per typologies     |         |

### 5.3.7 Assistance to Vulnerable

Issues related to assistance to vulnerable populations include neglected vulnerable population.

**Table 22: Summary of issues and recommendations related to assistance to vulnerable population**

| Major Issues                    | Causes/Problems                                                         | Recommendations                                                                            | Remarks |
|---------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------|
| Neglected vulnerable population | Vulnerable population (poor, disabled, displaced) are often left behind | Supply a socio-technical help committee based at local level, ward level is more preferred |         |

### 5.3.8 National Policies

Issues related to National Policies include Unclear roles of 3 level of government, more central level focused existing policies, less prioritized Micro-Finance Institutions (MFI) and policy does not support conservation of vernacular architecture and traditional buildings.

**Table 23: Summary of issues and recommendations related to National Policies**

| Major Issues                                                                         | Causes/Problems                                                                                                | Recommendations                                                                                                                                         | Remarks |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Unclear roles of 3 level of government, More central level focused existing policies | DRRM Act and LGOA Act assigns roles for all 3 levels of GoN, but specific roles/responsibilities are not clear | Clarify responsibilities so that there are no ambiguous roles<br>Integrated and independent tasks of different level of government should be segregated |         |
|                                                                                      | DRRM Act and LGOA Act assigns roles for all 3 levels of GoN, but local level lacks resources                   | better equip local government with budget and human resources<br>Special unit in Police and Army to be created for disaster response                    |         |

|                                                               |                                                                                                                                                   |                                                                                                                     |  |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--|
|                                                               | Policies supports the development but could not provide vision to people                                                                          | Long term urban development plans (master plans) should be made for proper long-term urban development              |  |
| Less prioritized Micro-finance Institutions (MFI)             |                                                                                                                                                   | MFI are close to community; they should be backup up by GoN so that they can supply low interest loan to the people |  |
| Policy does not support conservation of traditional buildings | Traditional building consists of Chidi, Matan, Chota, and Buigal; building code does not support 4 storey constructions of a stone in mud masonry | Update building code and bylaws to support and strengthen the conservation of traditional buildings                 |  |
|                                                               |                                                                                                                                                   | Policy should make people believe that stone construction can be earthquake resistant                               |  |
|                                                               |                                                                                                                                                   | Conserve all traditional styles of Nepal (Newari, Maithili, Tharu etc.)                                             |  |

### 5.3.9 NRA Policies

Issues related to NRA Policies include Reconstruction grant eligibility (one door policy), wide ranged policy and post-NRA exit issues.

**Table 24: Summary of issues and recommendations related to NRA Policies**

| Major Issues                                       | Causes/Problems                                                    | Recommendations                                                                                                                                                                                                                                                                                                                                               | Remarks                                                                                     |
|----------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Reconstruction grant eligibility (one door policy) | Policy mentions that grant should be received from only one source | NRA/GoN should reconsider policy                                                                                                                                                                                                                                                                                                                              | In Pilanchhe, reconstruction grant is not received but being supported from Maya foundation |
| Wide ranged policy                                 | NRA policies are targeted for whole nation                         | Need to change/update as per local context/area                                                                                                                                                                                                                                                                                                               |                                                                                             |
| Post-NRA exit                                      | Smooth running of reconstruction tasks                             | Role of successor organization should be clear and brief<br>Post-NRA, KVDA should work for Kathmandu Valley<br>Skills/ability of local level should be increased<br>To speed up the remaining reconstruction, local level should coordinate with the successor agency of NRA<br>The successor of NRA should not be burdened by the NRA's remaining case loads | LMC advisor Khokana                                                                         |

## 6 PROPOSED URBAN RECOVERY STRATEGIES

All the earlier studies and research on urban recovery and reconstruction and KIIs conducted in this process also reveal that the major barriers hindering urban reconstruction is the limited access to finance followed by land related issues, stringent compliance to heritage norms, inadequate communication, limited Socio-Technical Assistance (STA) in urban areas including retrofitting and inadequate STA to vulnerable household.

The reasons for limited access to finance include lack of affordability, difficulty accessing grant from NRA, less access to soft loan/loans and lack of ability to pay the loan back. Similarly, lack of land ownership certificate resulted from landlessness and people living in public land or Guthi land, multiple-ownership, small plot size than the threshold, and landlessness are the major problems associated with land issues. Likewise, non-compliance to heritage norms and bylaws, added cost for keeping vernacular architecture, lack of skilled human resource, huge resource requirement to conserve a heritage settlement and heritage conservation threatened by development are the underlying causes for delay in recovery in heritage sites. The major problems related to communication issues are inadequate information, late dissemination, misinterpretation, and lack of information. The issue of STA includes a lengthy process for building permits, and limited aid in preparing designs and drawings. Low knowledge, awareness, and trust in retrofitting are the major problems associated with retrofitting issues.



The nature of urban recovery issues is quite diverse in terms of simplicity, genesis, and level of interventions needed, level of funding requirement, stakeholder engagement and the types of socio- technical help need.

Some issues are simple and straight forward, but some are complex and often interlinked. For example, some of the multiple ownership issues other than heritage sites can be solved simply by amendment of the existing multiple ownership guidelines for heritage settlements. However, some issues need bigger program such as Land Pooling and House Pooling schemes.

Most of the issues are not only the issues of recovery and reconstruction but they are related to housing and urban development which were prevalent before the Gorkha Earthquake. For example, lack of land ownership certificate relates to the absence of registration system for family events at municipality level and database of households at local level for a long time. This suggests for setting up registration system and keeping updated database.

Some of the urban issues can be addressed at local level, but some need national level interventions in policies and institutional mechanisms. Municipality can revise by-laws to address some of the urban recovery issues. Local government can also supply more support to vulnerable household by creating a revolving fund at local level but ownership issues regarding Guthi land can be resolved only by the national level interventions in Guthi policy. Similarly, amendments in the existing acts and landless issues need national level interventions through Commission like institution. Furthermore, development of new urban policies and guidelines also calls for national interventions.

Access to finance issues also have manifolds. There exist various funding mechanisms of NRA namely NRA grant, heritage subsidy, vulnerable subsidy, subsidized loan, grant/loan from national agencies and grant/loan from international agencies etc. which further needs to be strengthened as this has not been able to address the urban recovery in whole. Access to a loan with a concessional interest rate may be sufficient to address some urban recovery issues while loans without collateral and further financial support is needed in other cases. Some further need support in housing loan repayment scheme through livelihood support. Establishment of Housing Security Fund can resolve most of the access to finances relates issues. Mobilizing corporate social responsibility fund in housing recovery can address some issues. Separate fund is also needed for non-revenue generating infrastructure development such as road, open space, heritage structures and revenue generating infrastructure such as water supply, electricity, telecommunication while dealing in urban settlements.

Urban has different Social Technical Assistance (STA) issues than the rural one. Specific support is needed particularly in retrofitting, heritage settlements recovery and vulnerable population to resolve these issues through orientation at various levels and training to construction technicians, development of technical guidelines, support in preparing designs and drawings for building permit process, setting of help desk for easy access to information, building tax exemption to vulnerable households and heritage buildings.

The in-depth analysis of the nature of the urban recovery issues shows that all issues are related either to policy, finance, or institutions or STA or in combination and need to be addressed through these four key areas policies interventions, funding modality, institutional mechanisms, and socio-technical help. This calls for a need of an urban recovery strategy that revolves around these four clusters:

- Policies Interventions
- Institutional Mechanism
- Funding Modality
- Socio-Technical Assistance (STA)

Hence, an urban recovery strategic framework has been proposed to set a long-term road map of post-earthquake reconstruction in a planned and coordinated way to achieve vision, mission, goals, and objectives of the reconstruction integrating with urban housing development eventually leading to sustainable resilient urban community with national ownership.

## 6.1 Policy Interventions

Urban recovery issues that have appeared in recovery and reconstruction are not the issues of recovery and reconstruction alone. It is also linked with the housing development issue. Experiences in similar post disaster situation across the globe clearly shows that urban recovery is a long-term process. Hence policy interventions in the existing housing and urban development policies, acts, guidelines are needed to address urban recovery issues in a long run. Policy intervention is also needed in the existing NRA policies developed during recovery and reconstruction as they have not been able to resolve urban recovery issues entirety and urban recovery will continue for few more years. Some of the provisions of policies and acts are also in contradictory with other existing policies and some provisions mean duplication. Furthermore, CRS and National Housing and Settlements Resilience Platform (NHSRP)

### Policy Intervention

Policy Rforms/Amendments ien  
the existing polices and Acts  
Development of new policies  
and acts

development of the new policies is also imperative to address some of the urban development, and post disaster urban recovery and reconstruction issues, as even amendment cannot resolve them.

### 6.1.1 Policy Reforms through Amendments in the Existing Policies and Acts

Some of the urban recovery issues can be addressed through amendments in the existing policies and acts and guidelines. There exist many policies acts, guidelines related to urban housing and development. Many more policies, acts, guidelines have been developed to address the recovery and reconstruction including urban housing recovery. After reviewing the existing policies and acts related to housing and urban development, more than 18 have been proposed for amendments. Amendments requiring policies is briefed below:

#### a. Ancient Monument Preservation Act, 1956 (2013)

The following provisions are proposed to include or to amend in this act or this policy:

- Local government **shall declare monument areas** as heritage and conserved areas.
- Local government **shall define such an area** and publish public notice.
- People within the heritage area **shall ask for permission** to install public infrastructure, to park vehicles, to stick posters, to shoot movies, to organize festivals (other than cultural festivals) etc. with local government.
- Construction and addition in the structure **shall be within the premise of heritage bylaws** or else local government can ban or demolish such construction.
- Local government **shall make inventory of heritage buildings** (more than 100 years old), define its area, and publish public notice.
- The owner of the heritage buildings **shall handle conserving** the property. For this the local government shall supply 20-30% of the repair and maintenance cost and exempt property tax.
- Local government **shall form a technical expert committee** to conserve heritage buildings.
- Government of Nepal **shall hand over the responsibility of preservation** of ancient monuments and heritage settlements at province level to province government and at local level to local government.
- Local government **shall buy the private heritage property** after its evaluation if it is not conserved by the owner.
- **Necessary arrangements shall be made** by the concerned local government to prevent the misuse of the archeological or historically important monuments or temples under the care of the Government of Nepal.
- Local government **shall preserve** unique types of artefacts and monuments that show the importance of a certain period if the Department of Archeology fails or delays the preservation.

Details of review and recommendations for amendment of Ancient Monument Preservation Act are presented in **Annex 1.1, Volume III**.

#### b. Guthi Corporation Act, 1976 (2033)

The following provisions are proposed to include or to amend in this act or this policy:

- Construction of houses and shops **shall only be allowed** in places where the premise of the temple is not adversely affected and the Jatra or festivals are not conducted.
- Guthi land **shall be leased** for a fixed period for agriculture or forestry.
- **The inventory of Guthi land shall be managed** and land use shall restrict its intrusion.
- Only those who have built a house on the Guthi land with their own capital for more than 10 years can have the **land registered in their own name**.

Details of review and recommendations for amendment of Guthi Corporation Act are presented in **Annex 1.2, Volume III**.

#### c. Town Development Act, 1988 (2045)

The following provisions are proposed to include or to amend in this act or this policy:

- **Town development plan shall be defined as** “A plan having well identified land use areas, integrated infrastructure development, environmental protection plan, disaster management and resilience plan, institutional capacity, and urban areas and urban expansion areas. This plan shall have key action plans, targets, expected achievements, implementation period and list of responsible bodies. The plan shall be approved by the local, provincial and central government as per its requirement.
- **Local planning shall be defined as** a local area development plan implemented by the local level within its area. For example, land pooling project, bus Park and bus terminal construction plan, industrial park

development plan, public infrastructure development plan, squatter settlement or slum improvement project, local disaster, and climate resilience plan etc.

- **Integrated Urban Development Plan shall be defined as** a clear identification of land use, integrated infrastructure development, environment protection, local disaster and climate resilience plan, social, economic, financial, and institutional ability development, etc.
- **Integrated Infrastructure Development Plan shall be defined as** a plan for the integrated development of basic infrastructures such as: transport, communication, drinking water, sewerage treatment and electricity distribution in a coordinated manner.
- The local level town development plan **shall be approved by** the Executive Officer of the local level concerned.

Details of review and recommendations for amendment of Town Development Act are presented in **Annex 1.3, Volume III.**

#### **d. Kathmandu Valley Development Authority (KVDA) Act, 1988 (2045)**

The following provisions are proposed to include or to amend in this act or this policy:

- Keeping in mind the current need to face the growing population and the challenge of urbanization in the Kathmandu Valley, **the role and responsibilities** of the federal, provincial, and local levels, and participation of the valley dwellers, and cooperative sectors **shall be clarified to ensure development** of integrated infrastructure and services.

Details of review and recommendations for amendment of Kathmandu Valley Development Authority Act are presented in **Annex 1.4, Volume III.**

#### **e. The Ownership of Joint Housing Act 1997 (2054)**

The following provisions are proposed to include or to amend in this act or this policy:

- Include post-earthquake lessons by creating a provision of social housing planning and implementation to cater the need of urban poor.
- Approval process for joint housing shall be reorganized and systemized to step up the Initial Environmental Evaluation (IEE) / Environmental Impact Assessment (EIA) process.

Details of review and recommendations for amendment of Apartment Ownership Act are presented in **Annex 1.5, Volume III.**

#### **f. Building Act 1998 (2055)**

The following provisions are proposed to include or to amend in this act:

- Amend this Building Act in consonant with the latest legal provision and federal system of administration
- To add provision to create a permanent committee for updating NBC

Details of review and recommendations for amendment of Building Act are presented in **Annex 1.6, Volume III.**

#### **g. National Urban Policy (NUP) 2007**

The following provisions are proposed to include or to amend in this act or this policy:

- **Clarity on how and when the aims can be achieved** must be spelled out.

Details of review and recommendations for amendment of National Urban Policy are presented in **Annex 1.7, Volume III.**

#### **h. National Adaptation Program of Action (NAPA, 2010)**

The following provisions are proposed to include or to amend in this act or this policy:

- **GoN shall consider formulation of Value Capture Policy** for infrastructure development.
- **MoUD shall plan and implement** the National Urban Development Strategy by mobilizing local resources and funds from bilateral and multilateral donor agencies.

Details of review and recommendations for amendment of National Adaptation Program of Action are presented in **Annex 1.8, Volume III.**

#### **i. Building Regulations 2010 (2066)**

The following provisions are proposed to include or to amend in this regulation:

CRS and National Housing and Settlements Resilience Platform (NHSRP)

- Amend this regulation in consonant with the latest legal provision and federal system of administration
- Include ToR of Building Code implementation monitoring unit and formats of monitoring and reporting of the building code. implementation in local government

Details of review and recommendations for amendment of Building Regulation are presented in **Annex 1.9, Volume III.**

#### **j. The National Shelter Policy, 2012 (2069)**

The following provisions are proposed to include or to amend in this act or this policy:

- Central government **shall develop implementation strategy, action plan, program, and projects.**
- Provincial and local governments **shall develop their own housing policy.**
- Central and Provincial governments shall help local government to **formulate plans for solving the issues of squatter settlements and landless squatters.**

Details of review and recommendations for amendment of The National Shelter Policy are presented in **Annex 1.10, Volume III.**

#### **k. National Land Use Policy, 2015 (2072)**

The following provisions are proposed to include or to amend in this act or this policy:

- **Establish inter-departmental coordination mechanism** and create conducive environment to function properly
- Institutional fragmentation, and **duplication of responsibility in government mechanism shall be eliminated** by defining role, responsibility, and obligation of each agency responsible by amending Land Use Policy.

Details of review and recommendations for amendment of National Land Use Policy are presented in **Annex 1.11, Volume III.**

#### **l. Draft National Plan of Actions for Safer Building Construction, 2015 (2072)**

The following provisions are proposed to include or to amend in this action plan:

- Rewrite this Draft National Plan of Action in consonant with the latest legal provision and federal system of administration and submit for approval from GON
- MoUD with support from MoFA (finance ministry) shall plan and implement National Plan of Action for Safer Building Construction by mobilizing local resources and fund from bilateral and multilateral donor agencies

Details of review and recommendations for amendment of Draft National Plan of Action are presented in **Annex 1.12, Volume III.**

#### **m. National Building Code (NBC) 203, 2015 (2072) Guideline**

The following provisions are proposed to include or to amend in this act or this policy:

- Number of Storey shall be up to three stories and an added attic floor. The maximum height (floor to floor) of a building shall not exceed 12 times the wall thickness at the superstructure.
- Shape shall be rectangular or square.

Details of review and recommendations for amendment of National Building Code are presented in **Annex 1.13, Volume III.**

#### **n. National Reconstruction and Recovery Policy, 2015 (2072)**

The following provisions are proposed to include or to amend in this act or this policy:

- In the reconstruction of the earthquake-affected ancient settlements, local materials, knowledge, skills, labor, and architecture shall be used to preserve the ancient settlements.
- Completely and partially damaged ancient heritage settlements, residential buildings, community and government buildings, ancient monuments and other structures shall be repaired, retrofitted, and reconstructed in an integrated manner by adopting right technology as per the local needs to be resistant to natural calamities.

Details of review and recommendations for amendment of National Reconstruction and Recovery Policy are presented in **Annex 1.14, Volume III.**

**o. Third United Nations Conference on Housing and Sustainable Urban Development (Habitat III), A National Report, 2016**

The following provisions are proposed to include or to amend in this act or this policy:

- Efforts must be taken to tackle slums and squatter problems by developing and implementing slums and squatter's area improvement plans by local government with full technical and financial support from central and provincial governments.

Details of review and recommendations for amendment of Third United Nations Conference on Housing and Sustainable Urban Development are presented in **Annex 1.15, Volume III.**

**p. Post Disaster Recovery Framework (PDRF), 2016 (2073)**

The following provisions are proposed to include or to amend in this act or this policy:

- Financial help for housing recovery shall be provided through the banking system in instalments, which will be linked to the progress of construction. In general, house owners will need to conform to government-approved seismic standards. The house owners of heritage settlement will need to conform to government approved seismic standards and building byelaws for heritage settlement.

Details of review and recommendations for amendment of Post Disaster Recovery Framework are presented in **Annex 1.16, Volume III.**

**q. The Fundamental Construction Bylaws on Settlement Development, Urban Planning and Building Construction 2017 (2074)**

The following provisions are proposed to include or to amend in this act or this policy:

- Federal government shall develop guidelines for engagement of urban stakeholders in preparation of municipal development control and building construction byelaws.
- Local government shall prepare regulation of building in areas of urban development and cultural and natural scenic beauty and in rural areas.

Details of review and recommendations for amendment of The Fundamental Construction Bylaws on Settlement Development, Urban Planning and Building Construction are presented in **Annex 1.17, Volume III.**

**r. National Urban Development Strategy, 2017 (2074)**

The following provisions are proposed to include or to amend in this act or this policy:

- MOUD with support from MoFA (finance ministry) shall plan and implement the National Urban Development Strategy by mobilizing local resources and funds from bilateral and multilateral donor agencies.

Details of review and recommendations for amendment of National Urban Development Strategy are presented in **Annex 1.18, Volume III.**

**s. Local Government Operations Act (LGOA), 2017 (2074)**

The following provisions are proposed to include or to amend in this act or this policy:

- Heritage area means the place where an ancient monument or ruins of an ancient settlement exists or human settlement which is inhabited more than 100 years ago.
- Tourist area means an area that attracts tourists with naturally beautiful places, mountains, hills, plains, forests, forests, lakes, rivers, waterfalls, and places with their own originality and culture.
- Urban area means an area with a population of at least 5,000, population density of 10 persons per hectare and 50% of the population above 15 years of age engaged in occupations other than agriculture. Similarly, dense areas, areas with houses on both sides of roads, and areas having basic urban infrastructures are termed as urban areas.
- Additionally, in the definition of urban area, different land use areas of the city, various levels of urban planning, urban infrastructure and service facilities, overall plan, local level planning, as well as rural areas within the village, rural development plan, rural infrastructure and service facilities shall be added.
- Evaluate urban infrastructure and services of Municipalities, Sub-Metropolitan City, and Metropolitan City and add the required urban infrastructure and services within 2 years by GoN shall be added in the Act.
- From the date of issuance of this Act, the province government shall identify the cultural or tourist areas within five years and declare them in the gazette of the province government.
- Definition of heritage or tourist area shall be added.



- Requirements for defining heritage areas are:
  - The location of temples, monuments and other structures of archeological importance or human settlements which are more than one hundred years old, which are important from the point of view of history, art, science, and architecture.
- Requirements for defining tourist areas are:
  - Areas that attract or can attract tourists such as: mountains, hills, Terai forests, farms, rivers, cliffs, waterfalls, lakes, natural landscapes, cultural sites and settlements etc.
- Demarcation of boundary of heritage or tourist area:
  - The demarcation of heritage or tourist area shall be done by the local level in collaboration with the concerned expert group
  - The perimeter occupied by the cultural or tourist area shall be considered as the border.
  - To keep such boundaries, aerial maps, satellite images and historical documents prepared in 1970 or earlier shall be used as the basis.
- Declaration of heritage or tourist area:
  - Subject to the grounds mentioned above, on the recommendation of the concerned expert group, the executive of the concerned municipality may declare the cultural and tourist areas as preserved areas and publish public notice.
  - A person who is not satisfied with the declaration may lodge a complaint with the Province Government within thirty-five days of the posting of such notice.
  - The State Government shall make the final decision on the complaint and publish the demarcation of such sites in the gazette.
  - People within the heritage area shall ask for permission to install public infrastructure, to park vehicles, to stick posters, to shoot movies, to organize festivals (other than cultural festivals) etc. with local government.
  - People within the heritage area shall ask for permission to construct a house or to alter the house with local government. As an incentive, the municipality shall make provision for free design and drawings and exempt tax.
  - Decisions of the local government on conservation, promotion, development and management plan, disaster risk reduction and management plan and other development plans should be published in the local gazette.
- Within five years of the commencement of this Act, GoN shall prepare detailed guidelines for formulating and implementing policies, laws, norms, provincial and local level development projects related to urban development, settlement development and rural development.
- Local level shall formulate policy, law, criteria guidelines within 2 years of preparation of all above-mentioned documents.
- Based on the risk sensitive land use plan, disaster risk areas and safe areas are found and the settlements in the disaster risk areas are protected by engineering work to minimize the risk if possible. Where it is not possible, resettlement is done to a safer location.
- Definition of periodic and annual plans as well as the medium- and long-term plans shall be added.
- Within 5 years from the commencement of this law by Central government and within 3 years of formulation of this law by Province government, implementation shall be done by local government within 2 years.
- If the local level does not formulate or does not formulate policies, laws, and norms as per the policies, laws and norms implemented by the Central and the Province government, the provincial government shall formulate local level policies, laws, and norms by following the policies and laws of the Central governments.
- Rural municipality and municipality should make long-term, medium-term, periodical, and annual plans for local level development by making at least 2 and at most 3 options and consider the most suitable plan as possible.
- Integrated property tax shall be levied on privately owned houses built in the local architecture reflecting originality within the rural municipality or municipality.

Details of review and recommendations for amendment of Local Government Operation Act are presented in **Annex 1.19, Volume III**.

#### **t. Right to Housing Act, 2018 (2075)**

The following provisions are proposed to include or to amend in this act or this policy:

- The current land and land title distribution program launched by high level land distribution commission shall continue by mobilizing support from local governments and other concerned government agencies.



- While undertaking "integrated settlement development" or moving settlements-which are remote to supply amenities or which are unsafe- to supply housing, Federal, Provincial and the Local governments action may be based on NRA's Lessons learned in planning and developing integrated settlement and relocation of unsafe settlements.

Details of review and recommendations for amendment of Right to Housing Act are presented in **Annex 1.20, Volume III.**

#### **u. Land Use Act, 2019 (2076)**

The following provisions are proposed to include or to amend in this act or this policy:

- Land use planning and best use of available land shall be a part of the municipal development plan which shall be clearly defined.
- Central and provincial governments shall assist local government in municipal development of land use planning and implementation by formulating land use planning implementation guidelines and building ability of the municipal staff and institutions in specified period.

Details of review and recommendations for amendment of Land Use Act are presented in **Annex 1.21, Volume III.**

### **6.1.2 Review of newly Developed draft Policies and Acts and Development of New Policies and Acts**

Recently Nepal government has developed few urban and housing related policies, guidelines, procedures, and acts which is yet to be approved. These includes National Urban Policy, House Pooling Procedure, urban regeneration Act. Amendment in these have also proposed after reviewing those policies and acts to address some of the urban issues. Few new policies and guideline also have been developed and proposed in this process itself as there is still a gap in some area. They are as follows:

#### **6.1.2.1 Proposed National Urban Policy, 2019 (2076)**

The following provisions are proposed to include or to amend in this act or this policy:

- Article 6) Vision: "Prosperous, vibrant, beautiful and inclusive city" shall be replaced by "balanced, inclusive and prosperous urban system at the federal, provincial and local levels".
- The productive of subsection 8.3 of section 8 shall be replaced by "uplifting".
- Strategy 10.4.4: shall be changed as follows: To run urban development programs to minimize the damage caused by natural calamities and human induced disasters and climate change.
- 11.1.3 E) Hazard shall be replaced by "disaster".

Details of review and recommendations for amendment of Proposed National Urban Policy are presented in **Annex 1.22, Volume III.**

#### **6.1.2.2 House Pooling Procedure, 2020 (2077)**

The following provisions are proposed to include or to amend in this act or this policy:

- The "public use within the planning area" of section 2c shall be removed and replaced with "collective purpose within the construction planning area".
- Article 2e) after the ownership, "deducting certain contribution part" shall be removed.
- The commercial ground floor of Article 2 f) shall be further changed by "or other purposes similar to residential purposes".
- Article 2 g) integrated shall be replaced by "joint housing".
- To develop the unorganized and unplanned settlements as a well-planned and managed settlement, all existing scattered buildings, and structures except monuments within the project site should be demolished and reconstructed. These buildings shall be evaluated, and equivalent property shall be returned to the former owner.
- To recover the cost of the project, the remaining housing unit and any other developed infrastructure or facility shall be sold or leased.
- Transfer of ownership of various housing units and facilities of developed settlements and buildings to the former owners or to the new buyer selected from the competition shall be done as per the principle of house pooling.

- When returning a new residential unit to the former property owner, one or more flats of reasonable size based on the area of the earlier house and land, number of rooms, connected facilities, number of family members, etc. shall be returned
- If a family is unable to get back even a one-room housing unit (flat), a collective housing unit with kitchen, toilet, and common room to be used collectively shall be planned to include other such families. However, in such a case, if the collective use is not possible, the householder will not be removed from his / her earlier place of residence. In such a case, the house owner can pay for the apartment in installments.
- Some property to be acquired through negotiation following the prevailing law if the owner is dissatisfied and obstructs the operation of the project.

Details of review and recommendations for amendment of House Pooling Procedure are presented in **Annex 1.23, Volume III.**

#### 6.1.2.3 Urban Development Bill, 2021 (2078)

The following provisions are proposed to include or to amend in this act or this policy:

- No definition of urban area, urban plan, and urban infrastructure, urban regeneration, regeneration area, urban regeneration plan etc. exists in Urban Development Bill. However, some definitions exist in Urban Development Policy, 2007. Hence, some definitions are proposed based on the Urban Development Policy:
  - Urban planning" is defined as any physical, social, economic and environmental program, activity or project conducted to provide physical, social and economic basic urban infrastructure and services in the urban area to achieve social, economic, environmental or other objectives.
  - Basic urban infrastructure and service facilities" is defined as physical, social, and economic infrastructure such as drinking water, sewerage, rainwater drainage, roads, waste management, electricity distribution, communication system, etc.
- Urban Development Bill is more focused on physical infrastructure rather than overall urban development. It misses the linkage between development and physical infrastructure.
- In the Bill, local area development projects are limited to guided land development, land pooling, site, and services and house pooling. It misses to include urban renewal, heritage settlement regeneration, industrial park, City Park, bus terminal, river front development, squatter and slum area improvement and development and community infrastructure development projects etc.
- Gazette publication for updated plan is mentioned. However, the Bill does not mention the publication of approved plan for public.
- Details are missed in Annex.
- Bill proposes to define and prioritize the project planning components through regulation which shall be defined in the bill.
- Land use plan shall be according to land use Act.  
Note: - \* marked existing Policy, Act, Regulation and Working Procedure and Gaps) are most relevant to local and provincial government for consideration

Details of review and recommendations for amendment of Urban Development Bill are presented in **Annex 1.24, Volume III.**

#### 6.1.2.4 Draft Urban Regeneration Act

The proposed urban development Bill, 2021 does not cover urban regenerations as a local area plan. Hence, an Urban Regeneration Act is proposed to support urban regeneration projects in N

The purpose of this draft Act is to contribute to enhancement of the quality of life of the people, such as the expansion of the base for sustainable growth of historic cities, towns and settlements, improvement of competitiveness of those cities/towns, and recovery of local community, by strengthening the public role and support for the economic, social, and cultural revitalization of historic cities, towns, and settlements.

The proposed Urban Regeneration Act is presented in **Annex 1, Volume IV.**

#### 6.1.2.5 Draft Urban Regeneration Guideline

The proposed urban development Bill, 2021 does not cover urban regenerations as a local area plan. The settlement development and building construction guideline covers the monument conservation but it requires legal tools to

implement properly. Urban regeneration should be a part of an umbrella Act. Hence, a guideline for urban regeneration is developed which mentions the vision, goals, aim, and guiding principle together with the overall process of urban regeneration project implementation.

The proposed Urban Regeneration guideline supplies guidance on implementation of urban regeneration in an integrated manner. It supports to revitalize traditional settlements of Nepal damaged by the earthquake in a planned and participating manner keeping the urban fabric intact. It supports the rebuilding of traditional houses to make them safe and convenient while conserving local heritage. This guideline promotes conservation and use of public spaces for community services. It stimulates local economy by promoting entrepreneurship, tourism and retail commercial activities that are compatible to heritage conservation and eventually strengthens local ability of municipality and community groups in managing urban growth.

The Urban Regeneration guideline is attached to **Annex 2, Volume IV**.

Based on the study of the policies and the proposed recommendations, different agencies identified for the policy updates is presented below.

| S. N. | Policies                                                       | Local Level                                                           | Provincial Level                                                 | DoA, MoCTCA                                         | MoLMCPA           | MoUD                                                                           | NDRRMA | KVDA                                                              | MoFAGA                 |
|-------|----------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|-------------------|--------------------------------------------------------------------------------|--------|-------------------------------------------------------------------|------------------------|
| 1     | Ancient Monument Preservation Act, 1956 (2013)                 | - Policies regarding Monuments at local level<br>- Implementation     | Policies regarding Monuments at province level<br>Implementation | Drafting Policies<br>Making local level responsible |                   |                                                                                |        |                                                                   |                        |
| 2     | Guthi Corporation Act, 1976 (2033)                             | Implementation                                                        |                                                                  |                                                     | Drafting Policies |                                                                                |        |                                                                   |                        |
| 3     | Town Development Act, 1988 (2045)                              | - Drafting policies for local level<br>- Implementation               |                                                                  |                                                     |                   | Drafting Policies<br>National level                                            |        |                                                                   | Support Municipalities |
| 4     | Kathmandu Valley Development Authority (KVDA) Act, 1988 (2045) | Implementation at KV municipalities                                   |                                                                  |                                                     |                   |                                                                                |        | Drafting Policies<br>Support KV municipalities for implementation | Support Municipalities |
| 5     | Ownership of Joint Housing Act 1997 (2054)                     | Implementation                                                        |                                                                  |                                                     |                   |                                                                                |        |                                                                   |                        |
| 6     | Building Act 2055 (1998)                                       | Implementation                                                        |                                                                  |                                                     |                   | -Update<br>-Drafting Amendment<br><br>-Assist in Implementation and Monitoring |        |                                                                   |                        |
| 7     | National Urban Policy (NUP) 2007                               | - Municipal development policy, act, guidelines and working procedure | Assist in Implementation for LL                                  |                                                     |                   | Drafting policies<br>Technical Support to Municipalities                       |        |                                                                   |                        |

| S. N. | Policies                                                                    | Local Level                                                                                                                                          | Provincial Level                           | DoA, MoCTCA | MoLMCPA                                                                                                                               | MoUD                                                                                                                              | NDRRMA | KVDA | MoFAGA |
|-------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|------|--------|
|       |                                                                             | shall be drafted by municipal government<br>- Implementa<br>tion                                                                                     |                                            |             |                                                                                                                                       |                                                                                                                                   |        |      |        |
| 8     | Building Regulations 2010 (2066)                                            | Implementa<br>tion                                                                                                                                   |                                            |             |                                                                                                                                       | -Update<br>-Drafin<br>g Amendme<br>nt<br><br>-Assist in<br>Implement<br>ation and<br>Monitoring                                   |        |      |        |
| 9     | National Adaptation Program of Action (NAPA, 2010)                          |                                                                                                                                                      |                                            |             |                                                                                                                                       | Update<br>and<br>Implement<br>ation                                                                                               |        |      |        |
| 10    | The National Shelter Policy, 2012 (2069)                                    | - Develop<br>local level<br>housing<br>policy<br>-<br>Implementa<br>tion                                                                             | Develop<br>provincial<br>housing<br>policy |             |                                                                                                                                       |                                                                                                                                   |        |      |        |
| 11    | National Land Use Policy, 2015 (2072)                                       | - Land use<br>policy, act,<br>guidelines<br>and working<br>procedure<br>shall be<br>drafted by<br>municipal<br>government<br>-<br>Implementa<br>tion | Implementa<br>tion                         |             | Drafting<br>policy<br>Defining<br>roles and<br>responsibili<br>ties of<br>different<br>tiers of<br>governmen<br>t<br>Coordinatio<br>n |                                                                                                                                   |        |      |        |
| 12    | Draft National Plan of Actions for Safer Building Construction, 2015 (2072) | Implementa<br>tion                                                                                                                                   |                                            |             |                                                                                                                                       | -Rewrite,<br><br>-Submit for<br>approval<br><br>- Mobilize<br>resources<br><br>-Assist in<br>Implement<br>ation and<br>Monitoring |        |      |        |
| 13    | National Building Code (NBC) 203, 2015 (2072) Guideline                     | Implementa<br>tion                                                                                                                                   |                                            |             |                                                                                                                                       | Update of<br>building<br>codes<br>Promotion<br>of use of<br>local<br>resources<br>(materials                                      |        |      |        |

| S. N. | Policies                                                                                                            | Local Level                                                          | Provincial Level                                                | DoA, MoCTCA | MoLMCPA | MoUD                                          | NDRRMA                                                         | KVDA | MoFAGA                    |
|-------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------|-------------|---------|-----------------------------------------------|----------------------------------------------------------------|------|---------------------------|
|       |                                                                                                                     |                                                                      |                                                                 |             |         | and technology )<br>Support to municipalities |                                                                |      |                           |
| 14    | National Reconstruction and Recovery Policy, 2015 (2072)                                                            | Implementation in coordination with NDRRMA and provincial government | Implementation in coordination with NDRRMA and local government |             |         |                                               | Drafting and Updating policies<br>Promotion of local resources |      |                           |
| 15    | Third United Nations Conference on Housing and Sustainable Urban Development (Habitat III), A National Report, 2016 | Implementation                                                       | Support to local level                                          |             |         |                                               |                                                                |      |                           |
| 16    | Post Disaster Recovery Framework (PDRF), 2016 (2073)                                                                |                                                                      |                                                                 |             |         |                                               | Drafting and Update of policies                                |      |                           |
| 17    | The Fundamental Construction Bylaws on Settlement Development, Urban Planning and Building Construction 2017 (2074) | Prepare guideline Implementation                                     | Support to local level                                          |             |         | Draft and Update policies                     |                                                                |      | Support local governments |
| 18    | National Urban Development Strategy, 2017 (2074)                                                                    |                                                                      |                                                                 |             |         | Implement                                     |                                                                |      |                           |
| 19    | Local Government                                                                                                    | Implementation                                                       |                                                                 |             |         |                                               |                                                                |      | Update Policies Support   |

| S. N. | Policies                                    | Local Level                                                                                                                                                                                | Provincial Level                | DoA, MoCTCA | MoLMCPA | MoUD                                                                                                                                                    | NDRRMA | KVDA | MoFAGA                                 |
|-------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|----------------------------------------|
|       | Operations Act (LGOA), 2017 (2074)          |                                                                                                                                                                                            |                                 |             |         |                                                                                                                                                         |        |      | Local government for implementation    |
| 20    | Right to Housing Act, 2018 (2075)           | Implementation                                                                                                                                                                             |                                 |             |         |                                                                                                                                                         |        |      |                                        |
| 21    | Land Use Act, 2019 (2076)                   | Shall be incorporated in municipal land use development plan                                                                                                                               | Assist Local Level              |             |         |                                                                                                                                                         |        |      |                                        |
| 22    | Proposed National Urban Policy, 2019 (2076) | <ul style="list-style-type: none"> <li>- Municipal development policy, act, guidelines and working procedure shall be drafted by municipal government</li> <li>- Implementation</li> </ul> | Assist in Implementation for LL |             |         | <ul style="list-style-type: none"> <li>- Drafting amendment or new urban policies</li> <li>- Technical Support to Municipalities</li> </ul>             |        |      | - Assist Local Level in implementation |
| 23    | House Pooling Procedure, 2020 (2077)        | - Assist in Implementation                                                                                                                                                                 |                                 |             |         | <ul style="list-style-type: none"> <li>- Drafting amendment</li> <li>- Technical Support to for implementation</li> </ul>                               |        |      | - Assist Local Level in implementation |
| 24    | Urban Development Bill, 2021 (2078)         | - Assist in Implementation                                                                                                                                                                 |                                 |             |         | <ul style="list-style-type: none"> <li>- Drafting amendment in policies, acts, procedures</li> <li>- Technical Support to for implementation</li> </ul> |        |      | - Assist Local Level in implementation |
| 25    | Draft Urban Regeneration Act 2021           | - Assist in Implementation                                                                                                                                                                 |                                 |             |         | <ul style="list-style-type: none"> <li>- Drafting amendment and new acts</li> <li>- Technical Support to for implementation</li> </ul>                  |        |      | - Assist Local Level in implementation |



| S. N. | Policies                                | Local Level                | Provincial Level | DoA, MoCTCA | MoLMCPA | MoUD                                                                                 | NDRRMA | KVDA | MoFAGA                                 |
|-------|-----------------------------------------|----------------------------|------------------|-------------|---------|--------------------------------------------------------------------------------------|--------|------|----------------------------------------|
| 26    | Draft Urban Regeneration Guideline 2021 | - Assist in Implementation |                  |             |         | - Drafting amendment and new guidelines<br>- Technical Support to for implementation |        |      | - Assist Local Level in implementation |

## 6.2 Institutional Framework

Urban recovery issues are multidimensional and hence engagement of multi-stakeholders is essential. Tenure of NRA, the leading agency for overall reconstruction, has been completed. Post disaster Recovery and reconstruction are the mandate of newly formed NDRRMA which handles disaster management of the country. MOUD/DUDBC is the lead agency for overall urban planning, development, redevelopment, reconstruction, renewal, regenerations, etc, while KVDA is the lead agency for urban planning and development in Kathmandu Valley. Furthermore, after the introduction of federal system in the county municipal planning and development as well as post disaster municipal (urban and rural) recovery and reconstruction comes under the authority of the provincial and local government, particularly Municipalities in their respective areas. Moreover, wards and the community itself also have a critical role in urban recovery and reconstruction at grass root level. In addition to these stakeholders there are also other stakeholders at all levels which need to be involved in this process. In this context, to address urban recovery and reconstruction issues an institutional mechanism on various tiers is needed. Further, institutional mechanism is needed to disseminate the policies developed at national level to the provincial and local level and provide input in national level policies formulation from implementation at the grass root level.

### Proposed Institutional Framework

Reconstruction led by NDRRMA  
through Recovery and  
Reconstruction Directorate

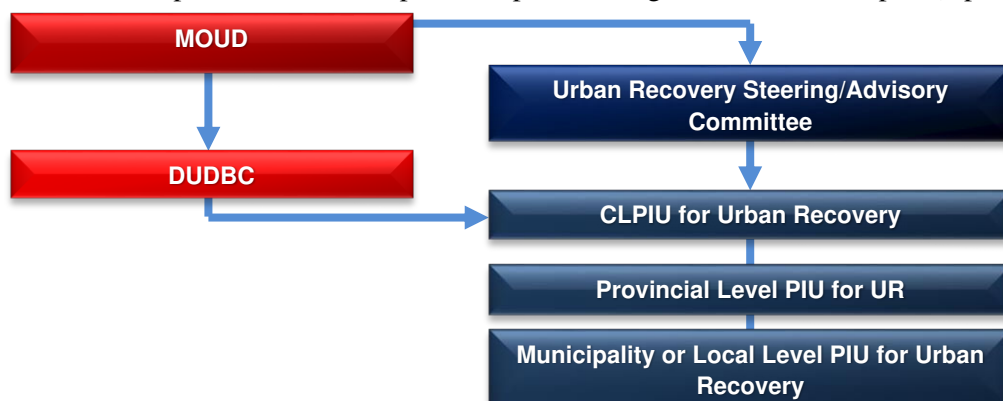
Various institutional frameworks of Urban Recovery can lead to the remaining reconstruction. An institutional framework is proposed hereby:

### 6.2.1 Proposed Structure for Leading/Guiding Urban Development/Recovery: MoUD as a lead

MOUD/DUDBC is the lead agency for nationwide overall urban planning, development, redevelopment, reconstruction, renewal, regenerations, etc,. Under MoUD, an urban recovery steering/advisory committee shall be formed which will direct urban recovery CLPIU in close coordination with Department of Urban Development and Building Construction (DUDBC). CLPIU will then direct provincial level project implementation units will eventually direct local level project implementation units as shown in figure 1.

The urban recovery steering committee will consist of:

- Chairman – Secretary, MoUD
- Members :
  - Secretary, MoFAGA
  - Secretary, MoLPCA
  - Secretary, MoTCA
  - DG, DUDBC
  - DG, DoA
  - Disaster Expert/ Urban Development Expert/ Heritage Conservation Expert (3 persons)



**Figure 1: Proposed Structure for Leading/Guiding Urban Recovery: MoUD as a lead**

The urban recovery steering committee will deliberate on the following matters:

1. Major policies for national urban development.
2. Policies to continue remaining reconstruction.
3. Strategic urban recovery and reconstruction plan jointly with NDRRMA, provinces and local government.
4. Oversight, guide, and support local government to formulate and implement local level urban development, recovery and reconstruction plan.
5. Monitor and evaluate implementation of local level urban recovery and reconstruction plan.

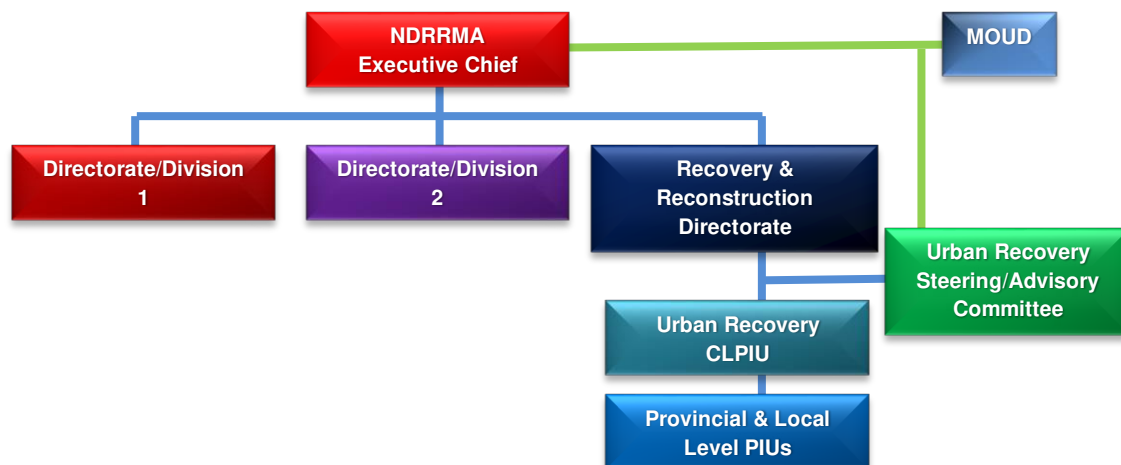
Similarly, Urban Recovery and Reconstruction Provincial Committees will be formed at relevant provinces under the Provincial Infrastructure Development Directorate as per the need. Provincial Level Project Implementation Units (PLPIU) will also be formed based on the need.

### 6.2.2 Proposed Structure for Leading Urban Recovery at Central Levels (NDRRMA as a lead)

After the tenure of NRA completed, based on the current Disaster Risk Reduction and Management Act, the recovery and reconstruction will have to be led by the newly established National Disaster Risk Reduction and Management Authority (NDRRMA). Among various Directorates/Divisions within NDRRMA, Recovery and Reconstruction Directorate should also be formed and this Directorate is proposed to lead the remaining recovery and reconstruction tasks. An Urban Recovery and Reconstruction Committee is proposed under the Recovery and Reconstruction Directorate which will direct urban recovery at central level. Central Level Project Implementation Unit (CLPIU) is to be formed with proper staffing which will eventually direct District Level Project Implementation Units (DLPIU) as shown in figure 2.

The following membership in the Urban Recovery and Reconstruction Steering Committee is proposed:

- Chairman – CEO, NDRRMA
- Members :
  - Joint Secretary, MoUD
  - Joint Secretary, MoFAGA
  - Joint Secretary, MoLPCA
  - Joint Secretary, MoTCA
  - DG, DUDBC
  - DG, DoA
  - Disaster Expert/ Urban Development Expert/ Heritage Conservation Expert (3 people)
- Member Secretary - Director, Recovery and Reconstruction Directorate of NDRRMA



**Figure 2: Proposed Structure for Leading Urban Recovery at Central Level (NDRRMA as a lead)**

The urban recovery steering committee will deliberate on the following matters:

1. Major policies for national urban recovery and reconstruction.
2. Strategic urban recovery and reconstruction plan jointly with provinces and local government.
3. Urban recovery and reconstruction plan including matters related to support local governments.
4. Oversight, guide, and support local government to implement local level urban recovery and reconstruction plan.
5. Monitor and evaluate implementation of local level urban recovery and reconstruction plan.

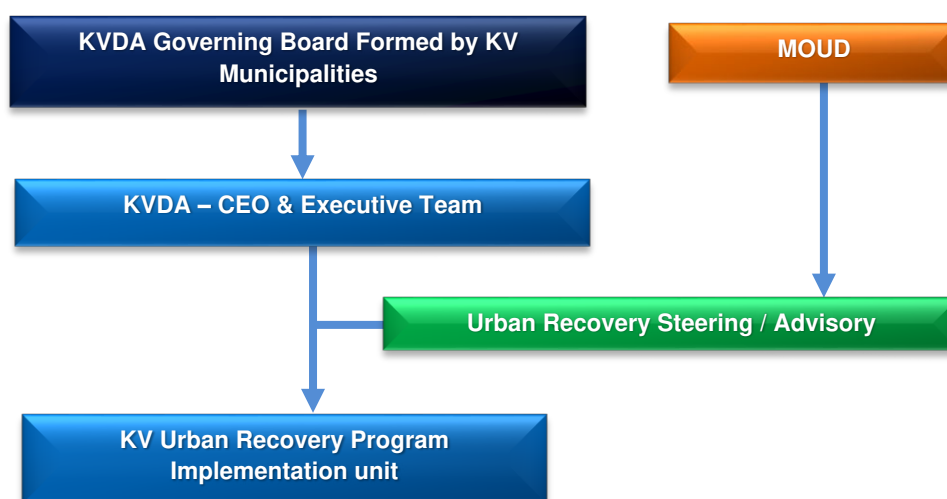
Similarly, Urban Recovery and Reconstruction Provincial Committees will be formed at relevant provinces under the Provincial Disaster Risk Reduction and Management Authority (PDRRMA) as per the need. Provincial Level Project Implementation Units (PLPIU) will also be formed based on the need.

### 6.2.3 Proposed Structure for Leading/Guiding Urban Recovery: KVDA as a lead

Kathmandu Valley Development Authority (KVDA) is mainly responsible for the development inside Kathmandu Valley. After NRA phases out, reconstruction will be led by KVDA. Under KVDA, a Governing Board formed by Kathmandu Valley Municipalities will direct KVDA's CEO and executive team. This team together with MoUD will guide an urban recovery steering/advisory committee which will eventually guide Kathmandu Valley urban recovery program implementation unit as shown in figure below.

The urban recovery steering committee will consist of:

- Chairman – Development Commissioner, KVDA
- Members :
  - Mayors, Kathmandu Valley
  - Joint Secretary, MoUD
  - Joint Secretary, MoFAGA
  - Joint Secretary, MoLPCA
  - Joint Secretary, MoTCA
  - DG, DUDBC
  - DG, DoA
  - Disaster Expert/ Urban Development Expert/ Heritage Conservation Expert (3 persons)



**Figure 3: Proposed Structure for Leading/Guiding Urban Recovery: KVDA as a lead**

The urban recovery steering committee will deliberate on the following matters:

1. Major policies for urban recovery and reconstruction at Kathmandu Valley.

2. Strategic urban development, recovery, and reconstruction plan jointly with Bagmati Province and local government.
3. Urban recovery and reconstruction plan including matters related to support local governments.
4. Oversight, guide, and support local government to implement local level urban recovery and reconstruction plan.
5. Monitor and evaluate implementation of local level urban development, recovery and reconstruction plan.

#### 6.2.4 Proposed Institutional Framework for Leading Post disaster Urban Recovery at Provincial Level

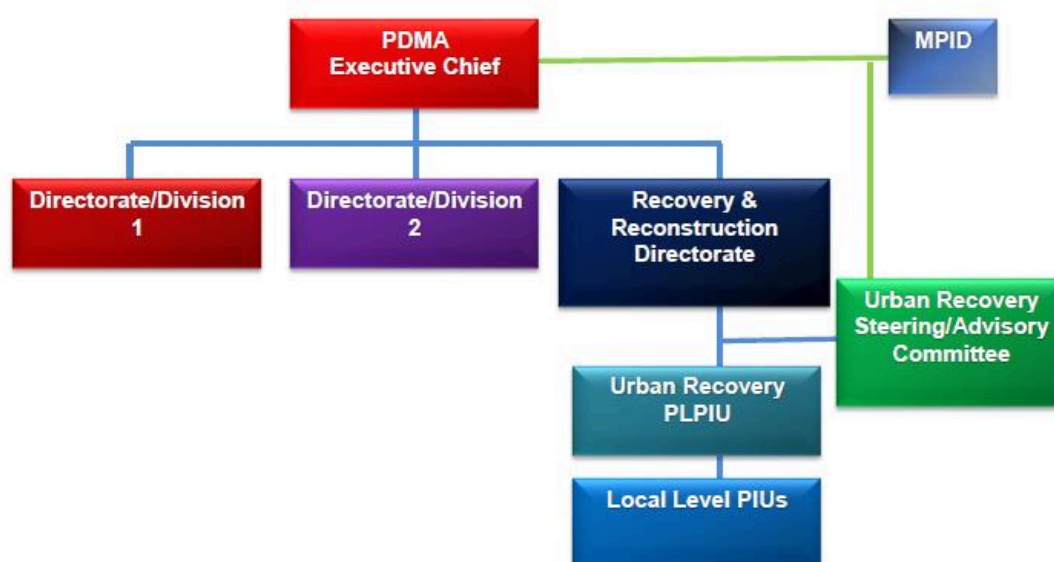


Figure 44: Proposed Structure for Leading/Guiding Urban Recovery: Provincial Government as lead as a lead

The Provincial level urban recovery steering committee will consist of:

- Chairman – CEO, PDMA
- Members :
  - Joint Secretary, PMoIPDD
  - Joint Secretary, PMoIAL
  - Joint Secretary, PMoLMCPA
  - Joint Secretary, PMoCT
  - Director, UDBC Directorate
  - Director, CTD directorate
  - Disaster Expert/ Urban Development Expert/ Heritage Conservation Expert (3 persons)

The Provincial level urban recovery steering committee will deliberate on the following matters:

6. Major policies for national urban recovery and reconstruction.
7. Strategic urban recovery and reconstruction plan jointly with local government.
8. Urban recovery and reconstruction plan including matters related to support local governments.
9. Oversight, guide, and support local government to implement local level urban recovery and reconstruction plan.
10. Monitor and evaluate implementation of local level urban recovery and reconstruction plan.

#### 6.2.5 Recovery and Reconstruction Committees at Local Levels

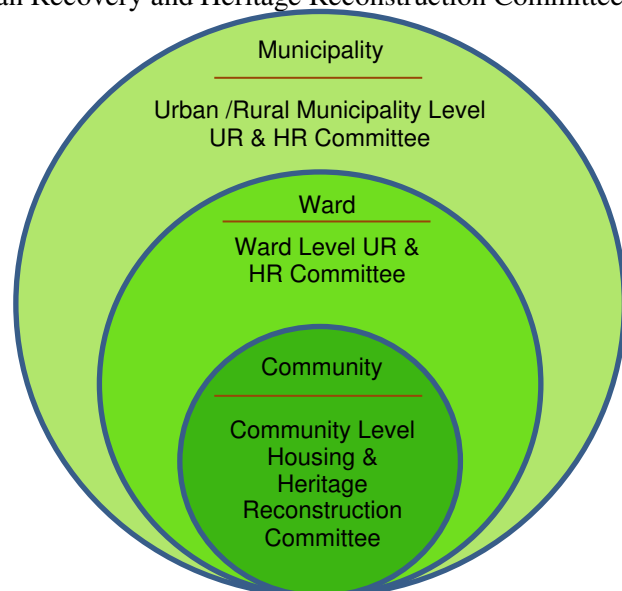
As per the Local Government Operation Act 2017, local level is authorized by GoN to formulate plans and implement it at a local level. In this context, a Municipal or Rural Municipal Level Urban Recovery

and Heritage Reconstruction Committee (ML-URHRC) will be formed in the municipalities where urban recovery and reconstruction tasks need to be implemented. Similarly, Ward Level Urban Recovery and Heritage Reconstruction Committee (WL-URHRC) will also be formed in relevant wards of the municipalities. Likewise, Community Level Housing and Heritage Reconstruction Committees (CL-HHRC) are proposed at relevant communities where urban recovery and reconstruction have to be implemented.

#### 6.2.5.1 Urban/Rural Municipal Level Urban Recovery and Heritage Reconstruction Committee (ML-URHRC)

The Municipality /Rural Municipality Level Urban Recovery and Heritage Reconstruction Committee will direct ward level urban recovery and reconstruction committee which will eventually direct community level housing and heritage reconstruction committee. The following structure of the committee is proposed:

- Chairperson - Mayor, Municipality /Rural Municipality
- Vice Chairperson - Deputy Mayor, Municipality /Rural Municipality
- Members :
  - Ward Chairpersons of the wards to be reconstructed
  - Ward members of respective wards
  - Related subject experts (3)
  - Union Representatives (2)
  - Private Sector Representatives (2)



**Figure 5: Recovery and Reconstruction Committees at Local Levels**

This committee will deliberate on the following matters or respond to a request for consultation:

1. Major policies about recovery and reconstruction at municipal level.
2. The urban recovery and reconstruction strategic plan and urban recovery and reconstruction plan.
3. Other matters necessary for urban recovery and reconstruction.

#### 6.2.5.2 Ward Level Urban Recovery and Heritage Reconstruction Committee (WL-URHRC)

Ward Level Urban Recovery and Heritage Reconstruction Committee will implement ward level urban recovery and reconstruction plan and direct/guide community level housing and heritage reconstruction committee. It will coordinate with local level and community to make local level plans. The following structure of the committee is proposed:

- Chairperson- Ward Chairperson
- Members :
  - Ward members of relevant ward
  - Representatives from reconstruction settlements (2)
  - Reconstruction experts from the same settlement (2)

#### 6.2.5.3 Community Level Housing and Heritage Reconstruction Committee (CL-HHRC)

Community Level Housing and Heritage Reconstruction Committee will represent/lead its community and coordinate with ward level to help ward level in making local level plans. This committee will consist of:



- Chairperson - Selected by the community
- Members :
  - Ward members of relevant ward (2)
  - Subject experts from the communities (3)
  - Women members from the community (2)

Four sub-committees are proposed under Community level housing and heritage reconstruction committee which will consist of:

- Housing Reconstruction Subcommittee
- Physical Infrastructure Reconstruction Sub-Committee
- Heritage Reconstruction and Rehabilitation Sub-Committee
- Livelihood and Economic Rehabilitation Sub-Committee

Each subcommittee will implement a respective field of recovery and reconstruction plan within their community.

### 6.3 Funding Mechanism

A dedicated funding mechanism for financing the remaining recovery and reconstruction work is needed. After the completion of NRA's tenure, the reconstruction fund is proposed to be lead and mobilized by NDRRMA. This fund should be regarded as the manifestation of the system of financial autonomy conferred by the law for effective financial performance. This reconstruction fund can obtain contribution from the following sources:

- Government of Nepal
- Prime Minister Relief Fund
- Organization or institution or individual contributor
- Grant aid in cash or kind or fund for the program from the foreign individual, government, or international organizations
- Any other source

The system of prior approval from the Ministry of Finance in coordination with the Ministry of Foreign Affairs before receiving the grant aid in cash or kind or fund for the program from the foreign (individual, government, or international organizations) should continue.

#### 6.3.1 Overall Funding Mechanism

National Urban Reconstruction Fund shall be set up funded by federal government of Nepal, Donors and Corporate Social Responsibility (CSR) fund to supply fund to local level urban development projects. These local level development projects shall also be funded by the respective local level and the province level in which it is situated.

#### Funding Sources for Recovery and Reconstruction

##### Existing funding sources

- a. NDRRMA grant
- b. NDRRMA Heritage subsidy grant
- c. NDRRMA Vulnerable population subsidy
- d. Subsidized loan by Banks
- e. Regular bank loans
- f. Grant from local government

##### Proposed Potential New Funding Sources

- a. Corporate Social Responsibility
- b. Subsidized Loans by Banks
- c. Housing Security Fund
- d. Engaging Cooperatives
- e. Insurance
- f. Housing loan repayment strategy through livelihood support

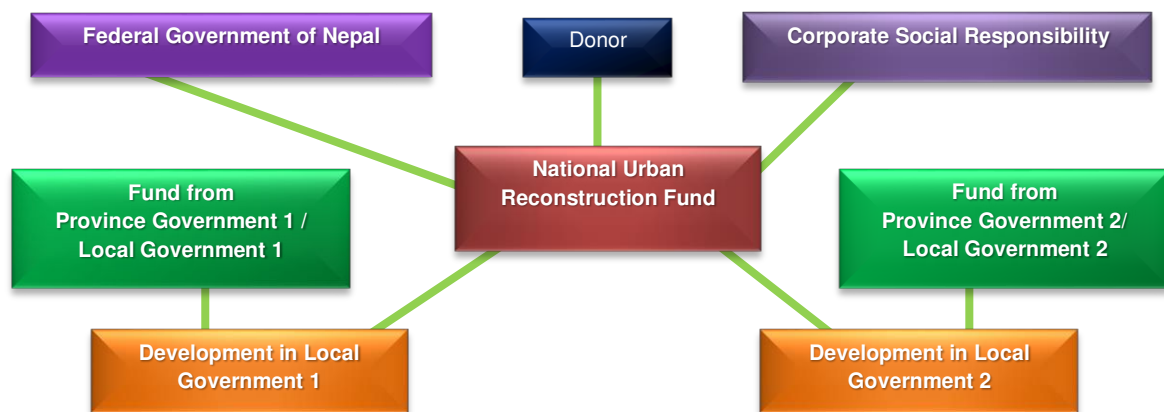


Figure 6: Proposed modality for National Urban Reconstruction Fund

### 6.3.2 Funding Mechanism for Housing

For housing reconstruction by individuals various funding modalities of NRA exists which are: NRA grant, NRA heritage subsidy grant, vulnerable subsidy, subsidized loan, grant from local government, and grant/loan from international agencies. These existing funding modalities should be continued by the successor of NRA and should be strengthened with Housing Security Fund, Corporate Social Responsibility (CSR) fund, engaging cooperatives, insurance, and various subsidized loans to perform better.

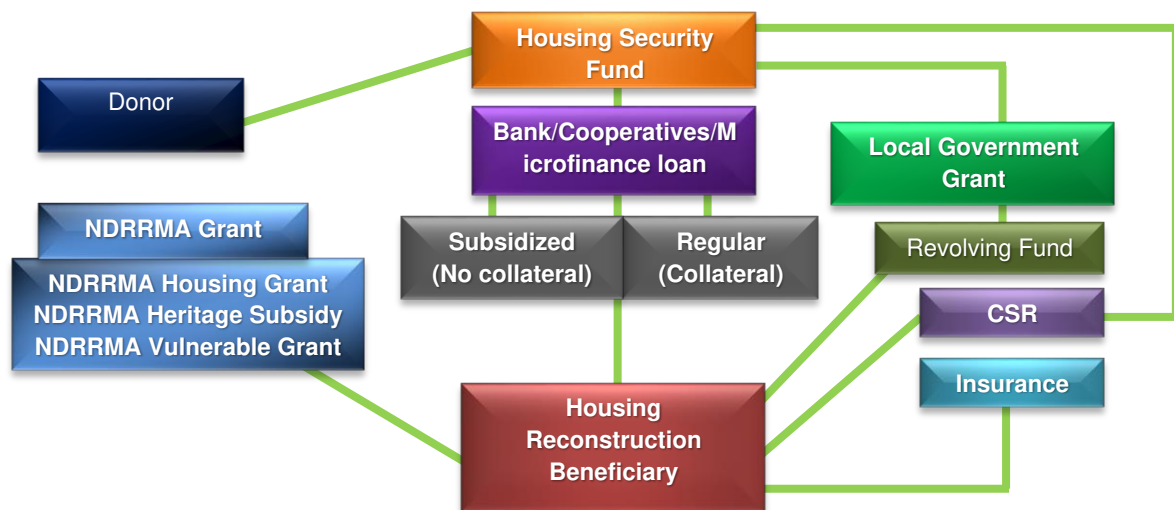


Figure 7: Proposed funding modality for Urban Housing Recovery

#### 6.3.2.1 Existing sources of fund

- a. **NDRRMA grant:** NDRRMA grant is the amount that will be segregated by for all disaster victims to reconstruct their houses damaged by the earthquake.
- b. **NDRRMA Heritage subsidy:** Heritage subsidy is the amount that will be segregated by NDRRMA for all disaster victims whose house lie in the heritage area.
- c. **NDRRMA Vulnerable population subsidy:** Vulnerable population subsidy is the amount that will be segregated by NDRRMA for all vulnerable population affected by the earthquake.
- d. **Bank Loans**
  - i. **Subsidized loan:** Subsidized loan is the loan provided by A, B and C class banks i.e., Commercial Bank, Development Bank and Finance Companies at an interest of 2%. This

- subsidized loan may come into existence after major disaster in close coordination of GoN and NDRRMA to help the disaster victims to reconstruct their house.
- ii. **Regular bank loans:** Commercial Bank, Development Bank and Finance Companies are also supplying the regular bank loans against the mortgage value.
  - e. **Grant from local government:** Many municipalities have been supplying heritage grant to conserve the heritage areas. This grant varies as per different municipalities as follows:
    - i. In Bhaktapur: 35% of reconstruction cost of building facade is provided as grant.
    - ii. In Sankhu: 1 lakhs NPR is supplied irrespective of the facade size.
    - iii. In KMC: NPR 500 per square feet based on the facade area is being planned which will be around 2 lakhs to 2lakhs 50 thousand per houses.
    - iv. In LMC: The grant calculation is in the process but is not yet decided.
  - f. **Donation from National and International Donors:** Various national and international donors have donated in the reconstruction after Gorkha Earthquake 2015.

### 6.3.2.2 Proposed Potential New Funding System

- a. **Housing Security Fund:** The existing funding modalities can be further strengthened to perform better through the establishment of “Housing Security Fund”. Housing security fund is a fund that accumulates fund from Government of Nepal, Prime Minister Relief Fund, national donors (organization, institution, and individual), international donors (government, organizations, individual) or any other source. This fund will remain under the operation of NDRRMA. Part of this fund will consist of grant money and part of it will consist of a loan. The benefit of obtaining loan through this fund is that this fund will be secured by the Nepal Rastra Bank and hence, the GoN can remain as a guarantee to obtain subsidized loan (at a rate of 2% interest) from Asian Development Bank or the World Bank. The housing security fund will act as a revolving fund to supply grant and collateral free subsidized loan to the disaster victims to reconstruct their houses. This fund may also consist of the CSR. The main functions of this loan are as follows:
  - i. This fund covers the subsidized interests of the housing loans.
  - ii. It also covers the loan if someone is unable to pay it back due to valid financial reasons.
  - iii. This fund is based on certification & recommendation of community, ward, and municipality.
  - iv. The payback period of this grant will be a minimum of 7 to 10 years.
- b. **Corporate Social Responsibility (CSR) fund:** CSR fund is an action of companies to keep positive impact on the society. Government of Nepal has mandated that, as a part of CSR, 1% profit of all business entities, including banks (A, B, C and D class), whose annual turnover exceeds 50 million NPR must be donated for a social cause (development works). This sums around NPR 1 Kharab (US\$ 1 billion) annually. However, CSR has terms, conditions, and limitation of its usage. GON and NRB fixes policy which prioritize the use of the CSR fund. Existing policy is in education, health, and financial literacy. If GON decides to use a part of this CSR fund for urban poor housing, it can be pooled in the housing security fund, then many of the urban poor will have access to a subsidized loan.
- c. **Engaging Cooperatives:** Till now, cooperatives have been supplying loan to support public who wish to reconstruct as a group through a company/cooperative of the beneficiaries at a high interest rate of up to 35%. If they are backed up by Nepal Rastra Bank, then they can supply loan at a minimum interest rate in following ways:
  - i. Cooperative takes the loan against each beneficiary given loan amount (up to 500,000 for each beneficiary) with no collateral, based on certification & recommendation of community, ward and municipality.
  - ii. “Housing Security Fund” supplies the subsidized interest and security of the loan amount.
  - iii. Cooperative gets remaining amount of loan from private/commercial banks with regular interest rates (with collateral). The amount of this loan will be based on project proposals.

- d. **Insurance:** GoN should create an insurance pool (like the Turkish Government after the Marmara earthquake) to transfer risk from the public to the private sector.
- e. **Housing loan repayment strategy through livelihood support:** We can have various livelihood support and economic recovery programs for needing people. They are:
  1. Specifically, designed projects
  2. Small and cottage industry development program
  3. Prime Minister self-employment fund
  4. Local micro-finance, cooperatives (government security/guarantee will enhance the local financial support)

To support these programs various aspects, need to be concerned. They are:

- Identification of local trade, industry, tourism
- Training and technical support
- Start-up – seed money

For instance, to encourage small and cottage industry development, in Khokana, reviving mustard oil mills or reviving handicrafts will revive its tradition, trade and tourism. After finding the local trade, we need to train the locals to flourish the trade. Finally, to encourage new entrepreneurs, they need to be supported financially. For this the seed money needs to be planned beforehand.

Technical and Financial Assistance for Housing and Livelihood Recovery can be obtained from Donor (Bilateral/Multilateral), INGOs and Internal Organizations, National / Local NGOs and community-based organizations, Private Sector (with the use of CSR fund), and government specially designed technical help programs.

- f. **Subsidized Loans:** Subsidized loans are provided by private commercial banks/local cooperatives/ microfinance (Without collateral or in group collateral) which is up to NPR 500,000. This kind of loan should be aided by GoN through Housing Security Fund.

### Total Subsidized Loan and Government Security Fund Needed

To estimate the amount of funds to be managed by GoN, subsidized funds or government security funds are calculated for a 12-year period. The calculation shows that an amount of 4.7 billion NPR must supply subsidized loan of NPR 500,000 at an interest rate of 2% for 30,000 vulnerable. If all 100,000 households are considered that amount sums to be 15.8 billion NPR. Similarly, the fund needed to be managed by GoN increases with the increase of loan amount and interest rate. For instance, subsidized loans of NPR. 1,000,000 at an interest rate of 2% up to 5 lakhs & at an interest rate of 4% from 5-10 lakh sums up to be 8.3 billion NPR for 30,000 vulnerable and it sums up to be 27.8 billion NPR for 100,000 households. Similarly, subsidized loans of NPR. 1,500,000 at an interest rate of 2% up to 5 lakhs and at an interest rate of 4% from 5-10 lakh and at an interest rate of 6% from 10-15 lakh sums up to be 10.7 billion NPR for 30,000 vulnerable and it sums up to be 35.9 billion NPR for 100,000 households. Furthermore, subsidized loans of NPR. 2,000,000 at an interest rate of 2% up to 5 lakhs, at an interest rate of 4% from 5-10 lakh, at an interest rate of 6% from 10-15 lakh and at an interest rate of 7% from 15-20 lakh sums up to be 12.6 billion NPR for 30,000 vulnerable and it sums up to be 42 billion NPR for 100,000 households. Additionally, subsidized loan of NPR. 2,500,000 at an interest rate of 2% up to 5 lakhs, at an interest rate of 4% from 5-10 lakh, at an interest rate of 6% from 10-15 lakh, at an interest rate of 7% from 15-20 lakh and at an interest rate of 8% from 20-25 lakh sums up to be 13.8 billion NPR for 30,000 vulnerable and it sums up to be 46 billion NPR for 100,000 households.

Details of the calculation of total subsidized loan and Government security fund needed is attached to **Annex 9, Volume II.**

#### 6.3.2.3 Cost recovery of subsidized loan

From the study of national urban regeneration projects carried out at Bhaktapur, Bandipur and Lalitpur (Pilanchhe), it is understood that the loan obtained for housing recovery can easily be paid back through

hospitality business (homestay). In this way, heritage value can be preserved, and tourism development will take place simultaneously.

From Pilanche it is seen that a building of a total built up area of 1440 square feet can be constructed at NPR 4,608,000.00 (at a cost of NPR 3,200.00/sq. ft). A reconstruction grant of NPR 3,00,000; a heritage grant of NPR 50,000 is provided by NRA and a construction material grant of NPR 4,50,000 is provided by Pilanche per household. This sums up to be NPR 8, 00,000. If an owner invests NPR 20,00,000 then he has to take a loan of remaining NPR 18,00,000 from bank. If this loan is at an interest rate of 2% up to 5 lakhs, at an interest rate of 4% from 5-10 lakh, at an interest rate of 6% from 10-15 lakh and at an interest rate of 8% from 15-18 lakhs for 7 years period, the house owner must pay an EMI of NPR 28,428.57 per month for 7 years. If 1 year is kept as a grace period, then the houseowner can rent rooms after this period. Considering the rent as NPR 1500 per night, and the occupancy rate as 55%, the house owner can earn NPR 50,000 per month earning a surplus of NPR 21,000 per month for renting 2 rooms.

Detail study of this capital and loan recovery is done with an example of Pilanche which is attached to **Annex 10, Volume II**.

### 6.3.3 Funding Mechanism for Infrastructure and Heritage

There are two categories of infrastructures / heritages structures: one that can generate revenue by sale of services and the other that cannot directly generate revenue in conventional way by sale of services.

#### 6.3.3.1 Revenue generating Infrastructure and service providing line agencies

Infrastructures/heritages structures that can generate revenue by sale of services are water supply, telephone, electricity, cable television, internet, tourism infrastructure etc. In this category of infrastructures / heritages structures, investment for development can be made in 4 separate ways. Firstly, investment can be done by concerned companies such as Kathmandu Upatyaka Khanepani Limited (KUKL), Nepal Telecom (NTC), Nepal Electricity Authority (NEA), cable TV operators, Internet Service Providers (ISPs), banks etc. as they are the profit bearers. However, Government enabling for loan or guarantee may be needed. Secondly, investment can be made through Government / Municipality grant contributions (such as the housing grant). Thirdly, grant/loan fund can be provided by Town Development Fund/Nepal Investment Board and other bilateral/multilateral funding agencies. Finally, investment can be made through Public Private Partnership.

#### 6.3.3.2 Nonrevenue generating Infrastructure providing line agencies

Infrastructures/heritages structures that cannot directly generate revenue in conventional way by sale of services are road, open spaces, community infrastructures, heritage structures etc. In this category of infrastructures / heritage structures, investment for development can be done in 4 separate ways. Firstly, investment can be made through a Government / Municipality grant. Secondly, investment can be made through grant funds from bilateral / multilateral funding institutions. Thirdly, investment can be made through private companies / private donations (CSR) and finally, investment can be made through Public Private Partnership which increases the potential for revenue generation.

#### Potential Funding for heritage structures and Infrastructures

##### Infrastructure generating revenue

1. Concerned companies through Government facilitation
2. Federal /Provincial/Local Government grant contributions
3. Grant/loan from Town Development Fund/Nepal Investment Board and other bilateral/multilateral funding agencies
4. Public Private Partnership.

##### Infrastructure with no revenue

1. Federal /Provincial/Local Government grant contributions
2. Grant/loan from bilateral/multilateral funding institutions
3. Private companies / private donations (CSR)
4. Public Private Partnership



## 6.4 Socio-Technical Support for Urban Recovery

Structure of urban area is composed of City Core Area, Inner Fringe area, Outer Fringe Area, and Rural Hinterland Area within urban premise. There are more than fifty traditional Newari settlement full of architectural, socio-cultural, and urban heritage in Kathmandu Valley and surrounding districts. Most of the buildings in these settlements are old and are not meticulously kept. Most of the urban houses and monuments of these settlements are damaged by the 2015 Gorkha earthquake. Large chunks of urban vulnerable population live in these settlements, and they are most affected by the earthquake. Depending upon the urban structure, the extent of the damage is different. The socio-technical help to households and communities should correspond to damage pattern, damaged structure, and extent of the damage together with their importance in the community. Proposed socio-technical support to earthquake affected households for City Core Area, Inner Fringe area, Outer Fringe Area, Rural Hinterland Area within urban premise and vulnerable family are as follows:

### **Orientations and Trainings:**

- Orientation of house owners/community and municipal staff on reconstruction and retrofitting of damaged houses
- Orientation of house owners and municipal staff on importance of traditional settlement and its heritages particularly for city core or traditional settlement.
- Training of municipal and free-lance consultants on design of traditional house using traditional construction materials and technology particularly for city core or traditional settlement.
- Training of local masons and carpenters on construction of traditional houses using local traditional construction materials and technology.
- Training of masons and engineers on reconstruction of earthquake safe houses.
- Training of local masons and engineers in designing and retrofitting of earthquake damaged houses.
- Training of people on small livelihood project implementation like, vegetable farming, Poultry, and goat rearing, etc. particularly for vulnerable population.

### **Formation of Community Reconstruction committee:**

- Form a community reconstruction committee for settlement mapping and inventory preparation.

### **Community Participation:**

- Engage community and local stakeholder on mapping verification and inventory preparation.
- Engage community reconstruction committee to prepare heritage settlement recovery plan.
- Engage community reconstruction committee for restoration and reconstruction of public places and buildings.
- Engage community and local stakeholder on mapping of vulnerable family.
- Form and Engage community reconstruction committee to assist vulnerable family on reconstruction and retrofitting of earthquake damaged houses.
- Engage community reconstruction committee to form group of vulnerable families for access to soft loan from microfinance bank for reconstruction and retrofitting of earthquake damaged houses and for livelihood project.

### **Technical Assistance:**

- Prepare design guidelines for reconstruction and restoration of traditional residential houses with the partner organisations, DoA, and DUDBC.
- Prepare design guidelines for reconstruction of fully damaged residential houses mostly prevalent in the inner and outer fringe area and rural surroundings area within urban premise with the aid from partner organisations, and DUDBC.
- Prepare design guidelines for retrofitting of partially damaged residential houses mostly prevalent in the inner and outer fringe area and rural surroundings area within urban premise with the aid from partner organisations, and DUDBC.



- Help desk and technical support centre for designing of the traditional and other houses and preparation of municipal drawings for building permit.
- Continuous door to door technical help in actual reconstruction and restoration of traditional and other earthquake damaged houses.
- Demonstration construction in public building or monuments restoration and reconstruction.
- Facilitate vulnerable families for access to soft loan from micro finance bank. Facilitate vulnerable families on application for building permit.

The cost of technical assistance is NPR 15270/HH (152 USD) for 5 years of reconstruction period which is about NPR 254/HH (USD 2.5/HH) per month for the same 5 years' period.

## 6.5 Implementation of Strategies

### 6.5.1 Major Preparatory Tasks

Preparation of urban recovery strategy took place from March to September 2021. This document will be reviewed by related ministries, CRS, NDRRMA, NHRP, CLPIU/Building, validation team and other experts. In the meantime, dissemination workshops and advocacy sessions with Nepal Rastra Bank, Ministry of Finance, MoUD, MoFAGA, MoLMCPA, MoCTCA and DUDBC will be organized to get the strategies endorsed. After all the reviews, Government of Nepal will endorse this urban recovery strategy. The endorsement will then be followed by the amendments in the acts and policies and implementation of the strategies.

#### 6.5.1.1 Preparation of Urban Recovery Strategy

Preparation of urban recovery strategy took place from 5<sup>th</sup> March to 15<sup>th</sup> September 2021. During this, a series of discussions with validation team and experts were organized. An essence of all this is summarized and presented in this report.

#### 6.5.1.2 Review of the Strategy

After the submission of this report, it will be reviewed by related ministries, CRS, NDRRMA, NHRP, CLPIU/Building, validation team and other experts and their feedback will be incorporated in the strategy.

#### 6.5.1.3 Discussions for endorsement

In the meantime, while the strategy is being reviewed, advocacy meetings, dissemination workshops with Nepal Rastra Bank, Ministry of Finance, MoUD, MoFAGA, MoLMCPA, MoCTCA, NDRRMA and DUDBC will be organized to get the strategies endorsed.

#### 6.5.1.4 Endorsement of strategy by GoN

After all the discussions and revisions, Government of Nepal will endorse this urban recovery strategy.

#### 6.5.1.5 Amendment in Acts and Policies

The endorsement will then be followed by the amendments in the acts and policies by the respective ministries as per the suggestions made in this report.

#### 6.5.1.6 Implementation of Strategies

International studies have shown that disaster recovery is a time-consuming task and Nepal is not an exception. Hence, implementation of the recovery strategy is assumed to be set up by the end of 2024.

## 6.5.2 Proposed Implementation Plan

The proposed implementation plan is as follows:

Table 25: Timeline for formulation and Implementation of Strategies

| Serial No. | Activities                                                                                                        | 2022 |  |  |  |  | 2023 |  |  |  |  | 2024 |  |  |  |  |
|------------|-------------------------------------------------------------------------------------------------------------------|------|--|--|--|--|------|--|--|--|--|------|--|--|--|--|
| 1          | Preparation of Integrated Urban Development and Recovery Strategy                                                 |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 2          | Review of the strategy by CRS, NHSRP and relevant ministries                                                      |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 3          | Submission of Integrated UDRS and Interactive discussion & Advocacy with Local provincial and national government |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 4          | Interactive discussion & Advocacy with MOUD, DUDBC,                                                               |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 5          | Interactive discussion & Advocacy with MOFAGA,                                                                    |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 6          | Interactive discussion & Advocacy with MOLMCPA,                                                                   |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 7          | Interactive discussion & Advocacy with MOCTCA, DOA                                                                |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 8          | Endorsement of the Strategy by Government of Nepal                                                                |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |
| 9          | Amendment in existing Policy, Acts, Regulations                                                                   |      |  |  |  |  |      |  |  |  |  |      |  |  |  |  |

[illegible]

### 6.5.3 Human Resource

In addition to the human resource discussed above, the human resource needed for Central Level Project Implementation Unit (CLPIU) and Provincial/Local Level Project Implementation Unit (LLPIU) are as following:

CLPIU and PLPIU will consist of 20 professionals including Project Director, Deputy Project Director, Structure Engineer, Engineer, Section Officer, Account Officer, Legal Advisor, Sub-engineer, Accountant, Nayab Subba, Computer Operator, Light vehicle driver and Office Helper. Similarly, LLPIU will consist of 12 professionals including all mentioned above except Section Officer, Account Officer, and Legal Advisor. The detail of human resource calculation for CLPIU, PLPIU and LLPIU is attached to **Annex 11, Volume II**.

#### 6.5.4 Monitoring and Evaluation Arrangement

A monitoring and evaluation unit under the guidance of NDRRMA must check and evaluate the implementation of the urban recovery strategies.

## 7 MODEL PROJECTS

Incorporating all the strategies discussed above, three potential projects have been designed, which are as follows:

### 7.1 Khokana Regeneration Project

| SN                                        | Features                                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Info                              |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Name of Project                                          | Khokana Regeneration Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Location                                  |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Province                                                 | Bagmati                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2                                         | District                                                 | Lalitpur District                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                         | Municipality                                             | Lalitpur Metropolitan City                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description of Project Area               |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Project Area                                             | 3.17 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                         | Population                                               | 4,927                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3                                         | Number of Household                                      | 1,056                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4                                         | Number of Heritage Structures                            | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description of Lalitpur Metropolitan City |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Area                                                     | 37.4 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                         | Population                                               | 226,728                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3                                         | Number of Household                                      | 70,256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project Components                        |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Monument Conservation                                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2                                         | Private Building Reconstruction                          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                         | Infrastructure Rehabilitation, Improvement and Extension | Improvement and extension of drinking water supply network with storage and treatment, rain water harvesting, surface water drainage and irrigation rehabilitation, underground electrification, street lights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation programs, public toilet improvement and extension |
| 4                                         | Social Development                                       | Intangible heritage conservation programs, awareness raising (conservation of heritage) and capacity development programs (skill transfer)                                                                                                                                                                                                                                                                                                                                                                 |
| 5                                         | Environment Management                                   | Solid waste management and sustainable green design promotion                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6                                         | Disaster Risk Management                                 | Minimize or mitigate impacts of Landslide, Fire, Earthquake and Epidemic                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7                                         | Economic Development                                     | Tourism development, local industry/ craft development, home stay and hotel/ resort development, café, restaurants and eateries, community center, vegetable market                                                                                                                                                                                                                                                                                                                                        |
| 8                                         | Institutional Development                                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for                                                                                                                                                                                                                                                                                                                                                     |

| SN                   | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                                                                    | traditional buildings and monument reconstruction, heritage bylaws, risk sensitive land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and brochure/ pamphlets on history and significance of the place and its' heritages                                                                                                                                                                                                                                                                                                                                                                                                   |
| Project Cost         |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1                    | Monument Conservation                                              | 217 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2                    | Private Building Reconstruction                                    | 1785 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3                    | Infrastructure Rehabilitation, Improvement and Extension           | 460 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4                    | Social Development                                                 | 240 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5                    | Environment Management                                             | 160 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6                    | Disaster Risk Management                                           | 81million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7                    | Economic Development                                               | 81 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 8                    | Institutional Development                                          | 10 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9                    | Project Management Cost                                            | 303 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 10                   | Total Project Cost                                                 | 3,337 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 11                   | Contingency (5%) +Contractor's profit (15%) + Vat (13%)            | 1,101million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 12                   | Grand Total Project cost                                           | 4,440 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Development Modality |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1                    | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:<br/>Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.<br/>Government / Municipality investment<br/>Grant/loan from Town Development Fund<br/>Public Private Partnership</p> <p>2<sup>nd</sup> Category:<br/>Government / Municipality investment<br/>Grant/loan from bilateral / multilateral funding institutions<br/>Private companies / private donations (CSR)<br/>Public Private Partnership</p>                                                                                                                                                                                                      |
| 2                    | Funding Modality for private buildings                             | <p>Construction of a small sized (600 sq. ft.), 3 storeyed brick in cement masonry which costs around NPR 8,460,000 (Cost Estimate of buildings is attached to Annex 12, Volume II.)<br/>Support from NRA and Local Level<br/>NRA reconstruction grant: <b>NPR 300,000.</b><br/>NRA vulnerable population grant: <b>NPR 50,000.</b><br/>NRA Heritage subsidy: <b>NPR 50,000.</b><br/>Local government heritage grant: NPR 100,000<br/>Total: NPR 500,000<br/>For Housing Reconstruction by Individuals obtaining loan<br/>Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b><br/>Regular bank loans against collateral: <b>NPR 7,460,00.</b></p> |

| SN                | Features                                                                                                                                                                        | Characteristics                                                                                                        |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|                   |                                                                                                                                                                                 | Or people can reconstruct as a group through a company/cooperative of the beneficiaries                                |
| 3                 | Livelihood Support                                                                                                                                                              | Promotion of local trade: handicrafts<br>Home stay<br>Capacity development in agriculture<br>Organic vegetable farming |
| 4                 | Institutional Framework (Federal, provincial, and local level)                                                                                                                  |                                                                                                                        |
| 5                 | Local Government                                                                                                                                                                |                                                                                                                        |
| Expected Outcomes |                                                                                                                                                                                 |                                                                                                                        |
| 1.                | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                 |                                                                                                                        |
| 2.                | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans. |                                                                                                                        |
| 3.                | Capacity enhancement of local government and community in planning towards green growth and heritage conservation.                                                              |                                                                                                                        |
| 4.                | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community enhancing livelihood.            |                                                                                                                        |
| 5.                | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                           |                                                                                                                        |



### 7.1.1 Khokana Regeneration plan and projects and prioritization of activities

#### Introduction to Khokana

Khokana is a former Village Development Committee (VDC) which has been merged with the neighboring VDC's Sainbu, Bungamati, Chhampi, Dukuchhap, and other 38 VDC's to form Lalitpur Metropolitan City in Lalitpur District, Province No. 3 of Nepal. It is a traditional Newari settlement which lies about 8 kilometers south of Kathmandu. It lies on the hills between Bagmati and Nakkhu rivers. Its traditional settlement of Khokana consists of an area of 3.17 sq.km. It consists of 4,927 population and 1,056 households. This traditional settlement consists of around 55 heritage structures.

#### History

Khokana was set up in the 16th century. It was the first historical place, set up by Newars when Kathmandu Valley was a lake. In that period, Kodey (Lower village) was the name of present Khokana, locally called 'Ku-Desh'. Khokana was claimed to be the first town lit by the Pharping Hydropower during the period of Prime Minister Chandra Shamsheer. Khokana is historically recognized for its oil mills.

Locals tell an interesting story about how 'Khokana' got its name. The word Khokana is derived from the Newari word 'khona' which means 'Telling while weeping.' According to the legend, A Maharjan priest of Pachali Bhairav of Kathmandu fainted due to the high cold fever. He was then taken to the bank of Bagmati River for cremation thinking of dead. He, along with his wife as 'Sati' was placed on the pyre. As the pyre burned, the man became conscious, and they were saved but were not allowed to live in their place. The couple fled to Kudesh; north-west of present Khokana. It is said that the goddess Shikali Devi blessed them, and a new settlement and they shifted towards high ground, east of Kudesh. This settlement started by a weeping (Khona in Newari) woman is said to be the origin of the name of the Khokana.

After the Gorkha Earthquake 2015, a total of 812 houses were fully destroyed, and 355 homes were partially destroyed in Khokana. A total of 28 people were injured, and 9 people were killed. 3 monuments were completely damaged while others had moderate to minor damage. Similarly, many other heritage structures such as stone spouts, ponds, and wells had minor damage.

#### Past efforts to conserve ancient settlement

Past efforts to conserve ancient settlement are as follows:

1. The historic settlement Khokana and its preservation requirement was initially documented in Physical Development Plan of Kathmandu Valley, 1969 (Department of Housing, and Physical Planning, 1969, then DHBPP now DUDBC).
2. In 1975, Kathmandu Valley Preservation of Physical Environment and Cultural Heritage-Protective Inventory Vol. 2s- inventory of monuments and building footprint and road network map of Khokana was published to preserve urban fabric of historic settlement of Khokana. For the first time it was documented as government's concept to preserve (Department of Housing, Building and Physical Planning, 1975, then DHBPP now DUDBC).
3. Khokana was enlisted in tentative list of World Heritage Centre or of the Secretariat of UNESCO, sent in 2008.

Besides the effort of making inventory of monuments, buildings, and road network map of Khokana in paper, no actual urban planning or conservation works has been carried out at Khokana.

#### Vision

Develop Khokana as a tourist destination and center for mustard oil production

#### Goal

The plan goal is to introduce new economic capabilities. It aims to create employment based on existing indigenous mustard oil productions and hospitality business to stimulate economic recovery to local and surrounding areas. The federal government, Bagmati province government and Lalitpur Metropolitan city links urban regeneration, urban development and urban reconstruction with key

facilities that have significant economic recovery. It targets areas where employment can be created through composite development projects (anchor businesses)

### **Main Objective**

To promote economic, social, and cultural activities by enhancing the overall ability of the area and using local resources.

### **Secondary Objectives**

- Use local characteristics to boost the economy of the depressed settlement
- Keep the community together by improving the living conditions of lagging residential neighborhoods

### **Target area**

Decaying main area, such as the main street and the areas behind it, and a residential neighborhood in poor condition.

### **Precondition/Assumption**

A special committee for urban regeneration in the MOUD/DUDBC enabled a nationwide implementation and to support local government in regeneration planning and execution.

### **Key Activities**

- Conservation of Cultural assets and traditional mustard oil pressing mill to promote tourism
- Expansion of infrastructure and projects such as culture, art, and tourism to boost the commercial market
- Improvement of roads, streets, lanes, and square
- Improvement of water supply, sanitation, and hygiene
- Reconstruction of damaged houses reaching traditional streetscape and outlook
- Creating community-centered incomes by expanding infrastructure for retirement and decaying residential areas and by creating a social economy (e.g., social enterprise cooperatives, local trade, and craft businesses)

### **Guiding Principles of Historic settlement/ town/city and settlement (Reconstruction & Recovery) Regeneration**

1. Acknowledging the historic town/city/settlement as a “Cultural Construct” where built structures and open spaces closely relate to social fabric “This principle fundamentally shifts culture to the forefront rather setting it aside”
2. Starting the reconciliation process with the (re)construction of culture landmarks and places of significance to local community
3. Prioritizing culture early in the planning process, starting with needs assessments and implementation of emergency interventions that reflect community priorities and needs
4. Engaging communities and local governments in every step of the reconstruction & recovery process
5. Using financial models that balance immediate/ short-term needs with the medium/long term development period in reconstruction plan
6. Ensuring effective management of the reconstruction process by striking a balance between people’s basic needs and the recovery of a city settlement’s historic character

| Urban regeneration funding and implementation plan of Khokana |                                                                                                                                  |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SN                                                            | Activities<br>(regeneration projects)                                                                                            | Total Amount<br>in NPR | Funding<br>agency                                                                                                                                          | Implementing<br>Organization                                                                                                    | Challenges                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1. Institutionalization Plan I                                |                                                                                                                                  |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| a                                                             | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management | 10 million             | <b>Main:</b><br>local Government<br><br><b>Others:</b><br>Central Government, State Government, local Government, INGOs, Donor Organizations               | <b>Main:</b><br>local Government<br><br><b>Others:</b><br>Private sector, ward office, tole development committees, INGOs, NGOs | <ul style="list-style-type: none"><li>There is no specific local policy on conservation and management of natural resources and manmade heritages and environment protection and management</li><li>There is no design guideline for traditional buildings and monument reconstruction</li><li>There is no RSLUP of LMC</li><li>There is no retrofitting guideline for vulnerable traditional houses</li><li>There is no guidelines for guides and brochures/ pamphlets on the history and significance of the place and its' heritages</li></ul> |
| b                                                             | Design guidelines for traditional buildings and monument reconstruction                                                          |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| c                                                             | Heritage bylaws                                                                                                                  |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| d                                                             | Risk sensitive land use plans (RSLUP)                                                                                            |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| e                                                             | Retrofitting guidelines for vulnerable houses                                                                                    |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| f                                                             | Guidelines for guides and brochures/ pamphlets on history and significance of the place and its' heritages                       |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2. Monument Conservation IIA                                  |                                                                                                                                  |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| a                                                             | Repair                                                                                                                           | 217 million            | <b>Main:</b><br>local Government                                                                                                                           | <b>Main:</b><br>local Government<br><br><b>Others:</b><br>ward office, tole development committees, INGOs, NGOs                 | There is no Policy for the promotion and support for regeneration of ancient human settlements, monument areas and traditional private buildings.                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                               | Restoration                                                                                                                      |                        | <b>Main:</b><br>Central Government, Donor Organizations<br><b>Others:</b><br>Central Government, State Government, local Government, INGOs, private sector |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | Reconstruction                                                                                                                   |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3. Infrastructure improvement and extension Plan IIB          |                                                                                                                                  |                        |                                                                                                                                                            |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| a                                                             | Improvement and extension of drinking water supply network with storage and treatment                                            | 460 million            | <b>Main:</b><br>local Government and concerned line agencies<br><b>Others:</b><br>Central Government, State Government, INGOs, Donor                       | KUKL/local Government                                                                                                           | <ul style="list-style-type: none"><li>Ambiguous responsibilities among line agencies</li><li>No policy to promote and support for installation of rainwater harvesting system</li><li>No design guidelines on Restoration and rehabilitation of public</li></ul>                                                                                                                                                                                                                                                                                  |
| b                                                             | Rainwater harvesting                                                                                                             |                        |                                                                                                                                                            | House Owners/local Government                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| c                                                             | Surface water drainage and irrigation rehabilitation                                                                             |                        |                                                                                                                                                            | KUKL/local Government                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Urban regeneration funding and implementation plan of Khokana |                                                                                                                                    |                     |                                                                                                                      |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SN                                                            | Activities (regeneration projects)                                                                                                 | Total Amount in NPR | Funding agency                                                                                                       | Implementing Organization       | Challenges                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| d                                                             | Underground electrification                                                                                                        |                     | Organizations, private sector                                                                                        | NEA/local Government            | <ul style="list-style-type: none"><li>spaces and heritage route</li><li>Municipality does not have special units to look after ancient heritage settlement</li></ul>                                                                                                                                                                                                                                                                                                      |
| e                                                             | Streetlights                                                                                                                       |                     |                                                                                                                      | NEA/local Government            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| f                                                             | Communication system                                                                                                               |                     |                                                                                                                      | NTC/local Government            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| g                                                             | Restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes)                                            |                     |                                                                                                                      | Private Sector/local Government |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| h                                                             | Road network rehabilitation and improvement                                                                                        |                     |                                                                                                                      | local Government                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| i                                                             | Parking space development for public buses and private vehicles                                                                    |                     |                                                                                                                      | local Government                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| j                                                             | Sanitation and hygiene improvement programs                                                                                        |                     |                                                                                                                      | KUKL/local Government           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| k                                                             | Public toilet improvement and extension                                                                                            |                     |                                                                                                                      | local Government                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4. Private Building Reconstruction III                        |                                                                                                                                    |                     |                                                                                                                      |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| a                                                             | Repair                                                                                                                             | 1785 million        | By house owners                                                                                                      | house owners                    | No detailed guidelines for repair and restoration of traditional buildings and monuments                                                                                                                                                                                                                                                                                                                                                                                  |
| b                                                             | Restoration                                                                                                                        |                     | By house owners                                                                                                      | house owners                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| c                                                             | Reconstruction – Construction of a small sized (600 sq. ft.), 3 storeyed bricks in cement masonry which costs around NPR 8,460,000 |                     | NRA reconstruction grant: NPR 300,000                                                                                | NRA+ House owners               | <ul style="list-style-type: none"><li>Inadequate grant amount for reconstruction of traditional houses inconsistency with the local streetscape.</li><li>National building code NBC 203 prohibits to construct traditional Newari house in mud mortar (three storey plus attic)</li><li>People could only construct their building as per the building bylaws which restrict number of floors.</li><li>Land cadastral and ownership certificate issues prevail.</li></ul> |
|                                                               |                                                                                                                                    |                     | NRA vulnerable population grant: NPR 50,000                                                                          | NRA+ House owners               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                               |                                                                                                                                    |                     | NRA Heritage subsidy: NPR 50,000                                                                                     | NRA+ House owners               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                               |                                                                                                                                    |                     | Local government heritage grant: NPR 100,000                                                                         | Local government+ House owners  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                               |                                                                                                                                    |                     | Subsidized Loans by banks (no collateral): NPR 500,000.                                                              | Bank+ House owners              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                               | Regular bank loans against collateral:                                                                                             | Bank+ House owners  | <ul style="list-style-type: none"><li>Too much paperwork makes the loan process cumbersome and complicated</li></ul> |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5. Social Development Plan (in NPR)IV                         |                                                                                                                                    |                     |                                                                                                                      |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| a                                                             | Intangible heritage conservation programs                                                                                          | 240 million         | Main: local Government, INGOs, NGOs                                                                                  | Main: local Government          | There is no specific policy for conservation of intangible heritage.                                                                                                                                                                                                                                                                                                                                                                                                      |
| b                                                             | Awareness raising for conservation of heritage                                                                                     |                     |                                                                                                                      | Others:                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Urban regeneration funding and implementation plan of Khokana |                                               |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
|---------------------------------------------------------------|-----------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SN                                                            | Activities<br>(regeneration projects)         | Total Amount<br>in NPR | Funding<br>agency                                                                                                                            | Implementing<br>Organization                                                                                                | Challenges                                                                                                                                                                    |
| c                                                             | Ability development programs (skill transfer) |                        | <b>Others:</b><br>Central Government, State Government                                                                                       | ward office, local tole development committees                                                                              |                                                                                                                                                                               |
| 6. Disaster Management Plan (in NPR) VA                       |                                               |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| a                                                             | General (formation of fund+ awareness)        | 81million              | <b>Main:</b><br>NDRRMA, local Government, INGOs, NGOs<br><b>Others:</b><br>Central Government, State Government, Donor Organizations         | <b>Main:</b><br>local Government<br><b>Others:</b><br>Private sector, ward office, tole development committees              | Lack of community-based disaster management committee for Disaster Risk Reduction and Management.                                                                             |
| b                                                             | Landslide                                     |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| c                                                             | Fire                                          |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| d                                                             | Earthquake                                    |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| 7. Environmental Management Plan (in NPR)VB                   |                                               |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| a                                                             | Solid waste management                        | 160 million            | <b>Main:</b><br>local Government, INGOs, NGOs<br><b>Others:</b><br>Central Government, State Government, Donor Organizations, Private sector | <b>Main:</b><br>local Government<br><b>Others:</b><br>Private sector, ward office, tole development committees, INGOs, NGOs | There is no policy to promote and support sustainable green design and construction.                                                                                          |
| b                                                             | Sustainable green design promotion            |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| 8. Economic Development Plan (in NPR)VC                       |                                               |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| a                                                             | Tourism development                           | 81 million             | <b>Main:</b><br>local Government, Private sector, INGOs, NGOs<br><b>Others:</b><br>Central Government, State Government, Donor Organizations | <b>Main:</b><br>local Government<br><b>Others:</b><br>Private sector, ward office, tole development committees, INGOs, NGOs | There is no local policy to promote and support the private sector for tourism development<br>There is no guidelines for proper development of home stay, local industry etc. |
| b                                                             | Oil Mill/Local industry/ craft development    |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| c                                                             | Home stay and hotel/ resort development       |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| d                                                             | Café, restaurants, and eateries               |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| e                                                             | Community center                              |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| f                                                             | Vegetable market                              |                        |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| Project Management Cost                                       |                                               | 303 million            |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| Total Project Cost                                            |                                               | 3,337 million          |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| Contingency (5%) +Contractor's profit (15%) + Vat (13%)       |                                               | 1,101 million          |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |
| Grand Total Project cost                                      |                                               | 4,440 million          |                                                                                                                                              |                                                                                                                             |                                                                                                                                                                               |

### 7.1.2 Total Construction Cost by Individual House Owners

It is estimated that Khokana needs a total budget of 1785 million to repair/retrofit/reconstruct the private buildings damaged by earthquake. Hence, an average amount per person is calculated to construct the damaged building individually.

To estimate the amount of funds to be managed by individual which sums to be around NPR. 16,80,000 per person, subsidized yearly instalment and regular instalment are calculated for a 7-year return period. The calculation shows that an instalment amount of NPR 278869 is required annually at a subsidized interest rate @2% up to 5 lakhs, @4% from 5-10 lakh, @6% from 10-15 lakh & @7% from 15-20 lakh. Similarly, the calculation shows that an instalment amount of NPR 334679.88 is required annually at a regular interest rate@10% (source: Everest Bank, August 2021).

Details of the calculation of total construction cost by individual is attached to **Annex 14, Volume II**.

### 7.1.3 Total cost of technical assistance

The cost of technical assistance is NPR 16,125,120/1056HH (161251 USD) for 5 years of reconstruction period or NPR 3,505,460/person (3505 USD) (household size: 4.6).

## 7.2 Sankhu Regeneration Project

| SN                                             | Features                                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Info</b>                            |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                                              | Name of Project                                          | Sankhu Regeneration Project                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Location</b>                                |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                                              | Province                                                 | Bagmati                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2                                              | District                                                 | Kathmandu District                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3                                              | Municipality                                             | Sankharapur Municipality                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Description of Project Area</b>             |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                                              | Project Area                                             | 0.15 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2                                              | Population                                               | 4,906                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3                                              | Number of Household                                      | 965                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4                                              | Number of Heritage Structures                            | 115                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description of Sankharapur Municipality</b> |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                                              | Area                                                     | 60.21 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2                                              | Population                                               | 25,338                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3                                              | Number of Household                                      | 5,607                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Project Components</b>                      |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1                                              | Monument Conservation                                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2                                              | Private Building Reconstruction                          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3                                              | Infrastructure Rehabilitation, Improvement and Extension | Improvement and extension of drinking supply network, rainwater harvesting and surface water drainage and irrigation rehabilitation, underground electrification, streetlights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation development programs, public toilet improvement and extension |
| 4                                              | Social Development                                       | Intangible heritage conservation programs, awareness (conservation of heritage) and capacity development programs (skill transfer)                                                                                                                                                                                                                                                                                                                                                     |
| 5                                              | Environment Management                                   | Solid waste management and sustainable green design                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6                                              | Disaster Risk Management                                 | Minimize and Mitigate impact of Landslide, Fire, Earthquake and Epidemic                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7                                              | Economic Development                                     | Tourism development, local industry/ handicraft development, home stay, hotel and resort development, café, restaurants, eateries, and agro-products' market development                                                                                                                                                                                                                                                                                                               |
| 8                                              | Institutional Development                                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for traditional buildings and monument reconstruction, heritage bylaws, risk sensitive                                                                                                                                                                                                                                              |



| SN                          | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                                                    | land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and awareness brochure/ pamphlets on history and significance of the place and it's heritages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Project Cost</b>         |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1                           | Monument Conservation                                              | 372 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2                           | Private Building Reconstruction                                    | 1,630 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3                           | Infrastructure Rehabilitation, Improvement and Extension           | 21 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4                           | Social Development                                                 | 235 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5                           | Environment Management                                             | 8 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6                           | Disaster Risk Management                                           | 4 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7                           | Economic Development                                               | 80 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8                           | Institutional Development                                          | 10 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 9                           | Project Management Cost                                            | 236 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10                          | Total Project Cost                                                 | 2,597 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 11                          | Contingency (5%) + Contractor's profit (15%) + Vat (13%)           | 857 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 12                          | Grand Total Project cost                                           | 3,454 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Development Modality</b> |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1                           | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:</p> <ol style="list-style-type: none"> <li>1. Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.</li> <li>2. Government / Municipality investment</li> <li>3. Grant/loan from Town Development Fund</li> <li>4. Public Private Partnership</li> </ol> <p>2<sup>nd</sup> Category:</p> <ol style="list-style-type: none"> <li>5. Government / Municipality investment</li> <li>6. Grant/loan from bilateral / multilateral funding institutions</li> <li>7. Private companies / private donations (CSR)</li> <li>8. Public Private Partnership</li> </ol>                                                                                                                                                                                                                                                                     |
| 2                           | Funding Modality for private buildings                             | <p><b>Construction of a small sized (600 sq. ft.), 3 storeyed brick in cement masonry which costs around NPR 8,460,000</b></p> <p><b>A. Support from NRA and Local Level</b></p> <ol style="list-style-type: none"> <li>1. NRA reconstruction grant: <b>NPR 300,000.</b></li> <li>2. NRA vulnerable population grant: <b>NPR 50,000.</b></li> <li>3. NRA Heritage subsidy: <b>NPR 50,000.</b></li> <li>4. Local government heritage grant: <b>NPR 100,000</b></li> </ol> <p><b>Total: NPR 500,000</b></p> <p><b>B. For Housing Reconstruction by Individuals obtaining loan</b></p> <ol style="list-style-type: none"> <li>1. Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b></li> <li>2. Regular bank loans against collateral: <b>NPR 7,460,00</b></li> </ol> <p>Or people can reconstruct as a group through a company/cooperative of the beneficiaries</p> |
| 3                           | Livelihood Support                                                 | <ul style="list-style-type: none"> <li>• Promotion of local trade: handicrafts</li> <li>• Home stay</li> <li>• Capacity development in agriculture</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| SN                       | Features                                                                                                                                                                        | Characteristics                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Organic vegetable farming</li> </ul>                                                                                                                                                                                                                                                                                                      |
| 4                        | Institutional Framework (Federal, provincial, and local level)                                                                                                                  | <pre> graph TD     NDRRMA[NDRRMA Executive Chief] --&gt; DD1[Directorate/Division]     NDRRMA --&gt; DD2[Directorate/Division]     NDRRMA --&gt; RRD[Recovery &amp; Reconstruction Directorate]     RRD --&gt; URSAC[Urban Recovery Steering/Advisory Committee]     RRD --&gt; URCPIU[Urban Recovery CLPIU]     URCPIU --&gt; PLLPIUs[Provincial &amp; Local Level PIUs] </pre> |
| 5                        | Local Government                                                                                                                                                                | <pre> graph TD     subgraph Municipality         MURH[Municipality /Rural Municipality Level UR &amp; HR Committee]     end     subgraph Ward         WRH[Ward Level UR &amp; HR Committee]     end     subgraph Community         CRH[Community Level Housing &amp; Heritage Reconstruction Committee]     end     MURH --- WRH --- CRH </pre>                                  |
| <b>Expected Outcomes</b> |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.                       | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.                       | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans. |                                                                                                                                                                                                                                                                                                                                                                                  |
| 3.                       | Capacity enhancement of local government and community in planning towards green growth and heritage conservation.                                                              |                                                                                                                                                                                                                                                                                                                                                                                  |
| 4.                       | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community enhancing livelihood.            |                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.                       | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                           |                                                                                                                                                                                                                                                                                                                                                                                  |

### 7.3 Bungamati Regeneration Project

| SN                                        | Features                                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Info                              |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Name of Project                                          | Bungamati Regeneration Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Location                                  |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Province                                                 | Bagmati                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2                                         | District                                                 | Lalitpur District                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                         | Municipality                                             | Lalitpur Metropolitan City                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description of Project Area               |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Project Area                                             | 0.09 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                         | Population                                               | 5,966                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3                                         | Number of Household                                      | 1,304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4                                         | Number of Heritage Structures                            | 78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description of Lalitpur Metropolitan City |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Area                                                     | 37.4 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                         | Population                                               | 226,728                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3                                         | Number of Household                                      | 70,256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project Components                        |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                         | Monument Conservation                                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2                                         | Private Building Reconstruction                          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                         | Infrastructure Rehabilitation, Improvement and Extension | Improvement and extension of drinking water supply network with storage and treatment, rain water harvesting, surface water drainage and irrigation rehabilitation, underground electrification, street lights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation programs, public toilet improvement and extension |
| 4                                         | Social Development                                       | Intangible heritage conservation programs, awareness raising (conservation of heritage) and capacity development programs (skill transfer)                                                                                                                                                                                                                                                                                                                                                                 |
| 5                                         | Environment Management                                   | Solid waste management and sustainable green design promotion                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6                                         | Disaster Risk Management                                 | Minimize or mitigate impacts of Fire, Earthquake and Epidemic                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7                                         | Economic Development                                     | Tourism development, local industry/ craft development, home stay and hotel/ resort development, café, restaurants and eateries, community center, vegetable market                                                                                                                                                                                                                                                                                                                                        |
| 8                                         | Institutional Development                                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for traditional buildings and monument reconstruction, heritage bylaws, risk sensitive land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and brochure/ pamphlets on history and significance of the place and its' heritages                                                                                          |
| Project Cost                              |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| SN                   | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | Monument Conservation                                              | 245 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                    | Private Building Reconstruction                                    | 2203 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3                    | Infrastructure Rehabilitation, Improvement and Extension           | 13 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4                    | Social Development                                                 | 286 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5                    | Environment Management                                             | 4.5 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6                    | Disaster Risk Management                                           | 2.3 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7                    | Economic Development                                               | 97.9 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8                    | Institutional Development                                          | 11.96 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9                    | Project Management Cost                                            | 286.5 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 10                   | Total Project Cost                                                 | 3,151 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11                   | Contingency (5%) + Contractor's profit (15%) + Vat (13%)           | 1,040 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 12                   | Grand Total Project cost                                           | 4,191 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Development Modality |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1                    | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:<br/>Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.<br/>Government / Municipality investment<br/>Grant/loan from Town Development Fund<br/>Public Private Partnership</p> <p>2<sup>nd</sup> Category:<br/>Government / Municipality investment<br/>Grant/loan from bilateral / multilateral funding institutions<br/>Private companies / private donations (CSR)<br/>Public Private Partnership</p>                                                                                                                                                                                                                                                                                                     |
| 2                    | Funding Modality for private buildings                             | <p>Construction of a small sized (600 sq. ft.), 3 storeyed brick in cement masonry which costs around NPR 8,460,000 (Cost Estimate of buildings is attached to Annex 12, Volume II.)<br/>Support from NRA and Local Level<br/>NRA reconstruction grant: <b>NPR 300,000.</b><br/>NRA vulnerable population grant: <b>NPR 50,000.</b><br/>NRA Heritage subsidy: <b>NPR 50,000.</b><br/>Local government heritage grant: NPR 100,000<br/>Total: NPR 500,000<br/>For Housing Reconstruction by Individuals obtaining loan<br/>Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b><br/>Regular bank loans against collateral: <b>NPR 7,460,00.</b></p> <p>Or people can reconstruct as a group through a company/cooperative of the beneficiaries</p> |
| 3                    | Livelihood Support                                                 | <p>Promotion of local trade: handicrafts<br/>Home stay<br/>Capacity development in agriculture<br/>Organic vegetable farming</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| SN                | Features                                                                                                                                                                        | Characteristics |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 4                 | Institutional Framework (Federal, provincial, and local level)                                                                                                                  |                 |
| 5                 | Local Government                                                                                                                                                                |                 |
| Expected Outcomes |                                                                                                                                                                                 |                 |
| 1.                | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                 |                 |
| 2.                | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans. |                 |
| 3.                | Capacity enhancement of local government and community in planning towards green growth and heritage conservation.                                                              |                 |
| 4.                | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community enhancing livelihood.            |                 |
| 5.                | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                           |                 |

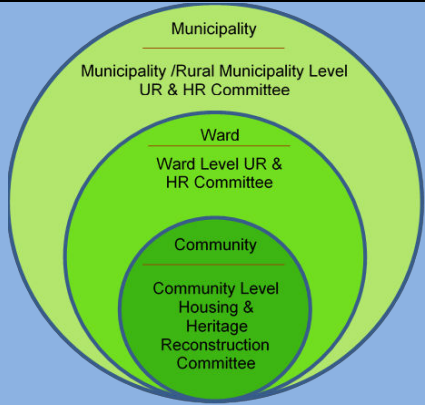
## 7.4 Dolakha Regeneration Project

| SN                                 | Features        | Characteristics              |
|------------------------------------|-----------------|------------------------------|
| <b>General Info</b>                |                 |                              |
| 1                                  | Name of Project | Dolakha Regeneration Project |
| <b>Location</b>                    |                 |                              |
| 1                                  | Province        | Bagmati                      |
| 2                                  | District        | Dolakha District             |
| 3                                  | Municipality    | Bhimeshwor Municipality      |
| <b>Description of Project Area</b> |                 |                              |
| 1                                  | Project Area    | 0.96 sq.km                   |
| 2                                  | Population      | 1,690                        |

| SN                                            | Features                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3                                             | Number of Household                      | 338                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4                                             | Number of Heritage Structures            | 64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description of Bhimeshwor Municipality</b> |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                             | Area                                     | 113 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2                                             | Population                               | 36,115                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3                                             | Number of Household                      | 10,255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project Components</b>                     |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                             | Monument Conservation                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2                                             | Private Building Reconstruction          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                             | Infrastructure Improvement and Extension | Improvement and extension of drinking water supply network, rainwater harvesting, surface water drainage and irrigation rehabilitation, underground electrification, street lights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation development programs, public toilet improvement and extension |
| 4                                             | Social Development                       | Intangible heritage conservation programs, awareness raising (conservation of heritage) and capacity development programs (skill transfer), programs to improve social life and bring back migrants                                                                                                                                                                                                                                                                                        |
| 5                                             | Environment Management                   | Solid waste management and sustainable green design promotion                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6                                             | Disaster Risk Management                 | Reduce and mitigate impacts of Landslide, Fire, Earthquake and Epidemics                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7                                             | Economic Development                     | Through tourism development, local industry/ handicraft revival and development, home stay, hotel and resort development, café, restaurants, and eateries development                                                                                                                                                                                                                                                                                                                      |
| 8                                             | Institutional Development                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for traditional buildings and monument reconstruction and restoration, heritage bylaws, risk sensitive land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and brochure/ pamphlets on history and significance of the place and heritages                                                               |
| <b>Project Cost</b>                           |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                             | Monument Conservation                    | 200 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2                                             | Private Building Reconstruction          | 572 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3                                             | Infrastructure Improvement and Extension | 140 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4                                             | Social Development                       | 81 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5                                             | Environment Management                   | 48 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6                                             | Disaster Risk Management                 | 25 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7                                             | Economic Development                     | 28 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8                                             | Institutional Development                | 3.4 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 9                                             | Project Management Cost                  | 109 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 10                                            | Total Project Cost                       | 1,206 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 11                                            | Contingency (5%) + Contractor's          | 398 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



| SN                          | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | profit (15%) + Vat (13%)                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 12                          | Grand Total Project cost                                           | 1,604 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Development Modality</b> |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1                           | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:</p> <ol style="list-style-type: none"> <li>1. Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.</li> <li>2. Government / Municipality investment</li> <li>3. Grant/loan from Town Development Fund</li> <li>4. Public Private Partnership</li> </ol> <p>2<sup>nd</sup> Category:</p> <ol style="list-style-type: none"> <li>5. Government / Municipality investment</li> <li>6. Grant/loan from bilateral / multilateral funding institutions</li> <li>7. Private companies / private donations (CSR)</li> <li>8. Public Private Partnership</li> </ol>                                                                                                                                                                                                                                                                       |
| 2                           | Funding Modality for private buildings                             | <p><b>Construction of a medium sized (850 sq. ft.), 3 storeyed stone in cement masonry which costs around NPR 3,700,000</b></p> <p><b>a. Support from NRA and Local Level</b></p> <ol style="list-style-type: none"> <li>1. NRA reconstruction grant: <b>NPR 300,000.</b></li> <li>2. NRA vulnerable population grant: <b>NPR 50,000.</b></li> <li>3. NRA Heritage subsidy: <b>NPR 50,000.</b></li> <li>4. Local government heritage grant: <b>NPR 100,000</b></li> </ol> <p><b>Total: NPR 500,000</b></p> <p><b>b. For Housing Reconstruction by Individuals obtaining loan</b></p> <ol style="list-style-type: none"> <li>1. Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b></li> <li>2. Regular bank loans against collateral: <b>NPR 2,700,000</b></li> </ol> <p>Or people can reconstruct as a group through a company/cooperative of the beneficiaries</p> |
| 3                           | Livelihood Support                                                 | <ul style="list-style-type: none"> <li>• Promotion of local trade: handicrafts</li> <li>• Home stay</li> <li>• Capacity development in agriculture</li> <li>• Organic vegetable farming</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4                           | Institutional Framework (Federal, provincial, and local level)     | <pre> graph TD     NDRRMA[NDRRMA Executive Chief] --&gt; D1[Directorate/Division]     NDRRMA --&gt; D2[Directorate/Division]     NDRRMA --&gt; RRD[Recovery &amp; Reconstruction Directorate]     RRD --&gt; URSAC[Urban Recovery Steering/Advisory Committee]     RRD --&gt; URCLPIU[Urban Recovery CLPIU]     URCLPIU --&gt; PLLPIUs[Provincial &amp; Local Level PIUs]     NDRRMA -.-&gt; URSAC   </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| SN                       | Features                                                                                                                                                                                                               | Characteristics                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 5                        | Local Government                                                                                                                                                                                                       |  |
| <b>Expected Outcomes</b> |                                                                                                                                                                                                                        |                                                                                   |
| 1.                       | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                                                        |                                                                                   |
| 2.                       | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans.                                        |                                                                                   |
| 3.                       | Ability enhancement of local government and community in planning towards green growth and heritage conservation.                                                                                                      |                                                                                   |
| 4.                       | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community. Ultimately enhancing livelihood and bringing back the migrated locals. |                                                                                   |
| 5.                       | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                                                                  |                                                                                   |

## 7.5 Nuwakot Regeneration Project

| SN                                | Features                                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Info                      |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                 | Name of Project                                          | Nuwakot Regeneration Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Location                          |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                 | Province                                                 | Bagmati                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2                                 | District                                                 | Nuwakot District                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3                                 | Municipality                                             | Bidur Municipality                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description of Project Area       |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                 | Project Area                                             | 8.44 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                 | Population                                               | 4,362                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3                                 | Number of Household                                      | 1,066                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4                                 | Number of Heritage Structures                            | 57                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description of Bidur Municipality |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                 | Area                                                     | 130.01 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2                                 | Population                                               | 54,351                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3                                 | Number of Household                                      | 12,505                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project Components                |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                 | Monument Conservation                                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2                                 | Private Building Reconstruction                          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                 | Infrastructure Rehabilitation, Improvement and Extension | Improvement and extension of drinking water supply network with storage and treatment, rain water harvesting, surface water drainage and irrigation rehabilitation, underground electrification, street lights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation programs, public toilet improvement and extension |
| 4                                 | Social Development                                       | Intangible heritage conservation programs, awareness raising (conservation of heritage) and capacity development programs (skill transfer)                                                                                                                                                                                                                                                                                                                                                                 |
| 5                                 | Environment Management                                   | Solid waste management and sustainable green design promotion                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6                                 | Disaster Risk Management                                 | Minimize or mitigate impacts of Landslide, Fire, Earthquake and Epidemic                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7                                 | Economic Development                                     | Tourism development, local industry/ craft development, home stay and hotel/ resort development, café, restaurants and eateries, community center, vegetable market                                                                                                                                                                                                                                                                                                                                        |
| 8                                 | Institutional Development                                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for traditional buildings and monument reconstruction, heritage bylaws, risk sensitive land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and brochure/ pamphlets on history and significance of the place and its' heritages                                                                                          |
| Project Cost                      |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| SN                   | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                    | Monument Conservation                                              | 179 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                    | Private Building Reconstruction                                    | 1802 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3                    | Infrastructure Rehabilitation, Improvement and Extension           | 1224 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4                    | Social Development                                                 | 209 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5                    | Environment Management                                             | 420 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6                    | Disaster Risk Management                                           | 216 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7                    | Economic Development                                               | 72 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8                    | Institutional Development                                          | 8.74 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 9                    | Project Management Cost                                            | 413.07 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10                   | Total Project Cost                                                 | 4,543 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11                   | Contingency (5%) + Contractor's profit (15%) + Vat (13%)           | 1,500 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 12                   | Grand Total Project cost                                           | 6,043 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Development Modality |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1                    | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:<br/>Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.<br/>Government / Municipality investment<br/>Grant/loan from Town Development Fund<br/>Public Private Partnership</p> <p>2<sup>nd</sup> Category:<br/>Government / Municipality investment<br/>Grant/loan from bilateral / multilateral funding institutions<br/>Private companies / private donations (CSR)<br/>Public Private Partnership</p>                                                                                                                                                                                                                                                                                                     |
| 2                    | Funding Modality for private buildings                             | <p>Construction of a small sized (600 sq. ft.), 3 storeyed brick in cement masonry which costs around NPR 8,460,000 (Cost Estimate of buildings is attached to Annex 12, Volume II.)<br/>Support from NRA and Local Level<br/>NRA reconstruction grant: <b>NPR 300,000.</b><br/>NRA vulnerable population grant: <b>NPR 50,000.</b><br/>NRA Heritage subsidy: <b>NPR 50,000.</b><br/>Local government heritage grant: NPR 100,000<br/>Total: NPR 500,000<br/>For Housing Reconstruction by Individuals obtaining loan<br/>Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b><br/>Regular bank loans against collateral: <b>NPR 7,460,00.</b></p> <p>Or people can reconstruct as a group through a company/cooperative of the beneficiaries</p> |
| 3                    | Livelihood Support                                                 | <p>Promotion of local trade: handicrafts<br/>Home stay<br/>Capacity development in agriculture<br/>Organic vegetable farming</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| SN                | Features                                                                                                                                                                        | Characteristics |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 4                 | Institutional Framework (Federal, provincial, and local level)                                                                                                                  |                 |
| 5                 | Local Government                                                                                                                                                                |                 |
| Expected Outcomes |                                                                                                                                                                                 |                 |
| 1.                | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                 |                 |
| 2.                | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans. |                 |
| 3.                | Capacity enhancement of local government and community in planning towards green growth and heritage conservation.                                                              |                 |
| 4.                | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community enhancing livelihood.            |                 |
| 5.                | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                           |                 |

## 7.6 Gorkha Regeneration Project

| SN                          | Features        | Characteristics             |
|-----------------------------|-----------------|-----------------------------|
| General Info                |                 |                             |
| 1                           | Name of Project | Gorkha Regeneration Project |
| Location                    |                 |                             |
| 1                           | Province        | Gandaki                     |
| 2                           | District        | Gorkha District             |
| 3                           | Municipality    | Gorkha Municipality         |
| Description of Project Area |                 |                             |

| SN                                 | Features                                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                  | Project Area                                             | 1.26 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2                                  | Population                                               | 3,942                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3                                  | Number of Household                                      | 1,024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4                                  | Number of Heritage Structures                            | 43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description of Gorkha Municipality |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                  | Area                                                     | 131.86 sq.km                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2                                  | Population                                               | 50,684                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3                                  | Number of Household                                      | 13,146                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project Components                 |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                  | Monument Conservation                                    | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2                                  | Private Building Reconstruction                          | Repair/ Restoration /Reconstruction (as required)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3                                  | Infrastructure Rehabilitation, Improvement and Extension | Improvement and extension of drinking water supply network with storage and treatment, rain water harvesting, surface water drainage and irrigation rehabilitation, underground electrification, street lights and communication system, restoration and rehabilitation of public spaces and heritage route (trail/Jatra routes), road network rehabilitation and improvement, parking space development for public bus and private vehicles, sanitation programs, public toilet improvement and extension |
| 4                                  | Social Development                                       | Intangible heritage conservation programs, awareness raising (conservation of heritage) and capacity development programs (skill transfer)                                                                                                                                                                                                                                                                                                                                                                 |
| 5                                  | Environment Management                                   | Solid waste management and sustainable green design promotion                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6                                  | Disaster Risk Management                                 | Minimize or mitigate impacts of Landslide, Fire, Earthquake and Epidemic                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7                                  | Economic Development                                     | Tourism development, local industry/ craft development, home stay and hotel/ resort development, café, restaurants and eateries, community center, vegetable market                                                                                                                                                                                                                                                                                                                                        |
| 8                                  | Institutional Development                                | Local policy on conservation and management of natural resources and manmade heritages and environment protection and management, design guideline for traditional buildings and monument reconstruction, heritage bylaws, risk sensitive land use plans, retrofitting guideline for vulnerable houses, guidelines for guides and brochure/ pamphlets on history and significance of the place and its' heritages                                                                                          |
| Project Cost                       |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1                                  | Monument Conservation                                    | 135 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2                                  | Private Building Reconstruction                          | 1730 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3                                  | Infrastructure Rehabilitation, Improvement and Extension | 183 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4                                  | Social Development                                       | 189 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5                                  | Environment Management                                   | 63 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| SN                   | Features                                                           | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6                    | Disaster Risk Management                                           | 32 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7                    | Economic Development                                               | 65 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8                    | Institutional Development                                          | 8 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9                    | Project Management Cost                                            | 240 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10                   | Total Project Cost                                                 | 2646 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 11                   | Contingency (5%)<br>+Contractor's profit (15%) + Vat (13%)         | 873 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 12                   | Grand Total Project cost                                           | 3519 million                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Development Modality |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1                    | Funding Modality for heritage buildings and public infrastructures | <p>1<sup>st</sup> Category:<br/>Water Supply Corporation Nepal, NTC, NEA, cable TV operators, ISPs etc.<br/>Government / Municipality investment<br/>Grant/loan from Town Development Fund<br/>Public Private Partnership</p> <p>2<sup>nd</sup> Category:<br/>Government / Municipality investment<br/>Grant/loan from bilateral / multilateral funding institutions<br/>Private companies / private donations (CSR)<br/>Public Private Partnership</p>                                                                                                                                                                                                                                                                                                     |
| 2                    | Funding Modality for private buildings                             | <p>Construction of a small sized (600 sq. ft.), 3 storeyed brick in cement masonry which costs around NPR 8,460,000 (Cost Estimate of buildings is attached to Annex 12, Volume II.)<br/>Support from NRA and Local Level<br/>NRA reconstruction grant: <b>NPR 300,000.</b><br/>NRA vulnerable population grant: <b>NPR 50,000.</b><br/>NRA Heritage subsidy: <b>NPR 50,000.</b><br/>Local government heritage grant: NPR 100,000<br/>Total: NPR 500,000<br/>For Housing Reconstruction by Individuals obtaining loan<br/>Subsidized Loans by banks (no collateral): <b>NPR 500,000.</b><br/>Regular bank loans against collateral: <b>NPR 7,460,00.</b></p> <p>Or people can reconstruct as a group through a company/cooperative of the beneficiaries</p> |
| 3                    | Livelihood Support                                                 | <p>Promotion of local trade: handicrafts<br/>Home stay<br/>Capacity development in agriculture<br/>Organic vegetable farming</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



| SN                | Features                                                                                                                                                                        | Characteristics                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4                 | Institutional Framework (Federal, provincial, and local level)                                                                                                                  | <pre> graph TD     NDRRMA[NDRRMA Executive Chief] --&gt; D1[Directorate/Division]     NDRRMA --&gt; D2[Directorate/Division]     NDRRMA --&gt; RRD[Recovery &amp; Reconstruction Directorate]     RRD --&gt; URC[Urban Recovery CLPIU]     RRD --&gt; URSC[Urban Recovery Steering/Advisory Committee]     URC --&gt; PLLP[Provincial &amp; Local Level PIUs] </pre> |
| 5                 | Local Government                                                                                                                                                                | <pre> graph TD     subgraph Municipality         MCHRC[Municipality /Rural Municipality Level UR &amp; HR Committee]     end     subgraph Ward         WCHRC[Ward Level UR &amp; HR Committee]     end     subgraph Community         CCHRC[Community Level Housing &amp; Heritage Reconstruction Committee]     end     MCHRC --- WCHRC --- CCHRC </pre>            |
| Expected Outcomes |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                      |
| 1.                | Awareness raised within community towards need for conservation of public heritage structures and intangible heritage elements.                                                 |                                                                                                                                                                                                                                                                                                                                                                      |
| 2.                | Active participation of community and local stakeholders in understanding the value of their space; collective input in documenting data and preparation of conservation plans. |                                                                                                                                                                                                                                                                                                                                                                      |
| 3.                | Capacity enhancement of local government and community in planning towards green growth and heritage conservation.                                                              |                                                                                                                                                                                                                                                                                                                                                                      |
| 4.                | Increased interest and support by local government in investing on development plans and entrepreneurship ventures proposed by local community enhancing livelihood.            |                                                                                                                                                                                                                                                                                                                                                                      |
| 5.                | Linkage with different heritage settlements set up; transformation of heritage settlement into a tourist destination.                                                           |                                                                                                                                                                                                                                                                                                                                                                      |

Cost Estimate of buildings is attached to **Annex 12, Volume II**.

Bill of quantities of Stone Masonry in Cement, Confined Masonry, RC and Brick in cement Masonry and drawings of these buildings is attached in **Annex 13 (Annex 13.1-Annex 13.6), Volume II**.

## **8 WHAT NEXT?**

This report has identified and analyzed the urban issues and challenges through available evidence and reports. It has proposed amendments to the existing Acts, Policies, Regulations and Working Procedures together with institutional framework to lead urban development/reconstruction and proposed the funding mechanism for urban recovery. Besides this report's scope, few other things still need to be considered such as:

- Economic Analysis of the model projects should be performed by a concerned agency
- An appropriate approach to involve private sector (not only in housing but in overall urban development) should be considered and developed with due concern to avoid gentrification
- Government regulations and guidelines should be developed to define the criteria for involvement of private sector

